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USA

December 19, 2024

Office of Food Additive Safety (HFS-200)
Center for Food Safety and Applied Nutrition
US Food and Drug Administration
5001 Campus Drive
College Park, MD 20740

RE: GRAS Determination for Onego Bio, Inc., for the use of ovalbumin from fermentation by *Trichoderma reesei* in baked goods and baking mixes, beverages and beverage bases (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections and frostings, dairy product analogs, egg products, fats and oils, frozen dairy desserts and mixes, fruit and water ices, gelatins, puddings, and fillings, grain products and pastas, gravies and sauces, milk products, plant protein products, snack foods (protein and snack bars), and soft candy.

Dear GRAS filing team,

Please find enclosed the Generally Recognized as Safe (GRAS) determination for the use of ovalbumin from fermentation by *Trichoderma reesei*. The substance is intended to be used as a general-purpose protein ingredient in a wide range of FDA-regulated food categories which include baked goods and baking mixes, beverages and beverage bases (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections and frostings, dairy product analogs, egg products, fats and oils, frozen dairy desserts and mixes, fruit and water ices, gelatins, puddings, and fillings, grain products and pastas, gravies and sauces, milk products, plant protein products, snack foods (protein and snack bars), and soft candy.

The protein ingredient is intended to replace fresh or powdered whole eggs and egg white protein powder in FDA-regulated food products, providing dietary protein along with functional properties such as foaming, gelling, and binding. These uses are not expected to increase the overall protein consumption of the general population. Information regarding the product identification, intended uses, the manufacturing process and safety is presented in the attached GRAS notice.

Onego Bio, Inc., has determined that its ovalbumin from fermentation by *Trichoderma reesei* is GRAS for its intended uses based on scientific procedures in accordance with 21 CFR § 170.30(a) and § 170.30(b), and in conformance with the guidance issued by the Food and Drug Administration (FDA) under 21 CFR § 170.36, 81 Fed. Reg. 54960 (Aug. 17, 2016). Therefore, the use ovalbumin from fermentation by *Trichoderma reesei*, as described in this GRAS Determination, is exempt from the requirement of premarket approval set forth in the Federal Food, Drug, and Cosmetic Act.

Analytical data, published studies, and information that has been used as the basis for this GRAS determination are available for FDA's review upon request.

We appreciate the FDA's consideration of this submission and are available to offer any further details required to support the assessment process. Thank you for your attention to this matter.

Sincerely,



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**GENERALLY RECOGNISED AS SAFE (GRAS) DETERMINATION FOR
OVALBUMIN FROM FERMENTATION BY *Trichoderma reesei***

Prepared by: Onego Bio, Inc.

9215 Brown Deer Road, Suite C
San Diego, CA 92121
USA

Date: December 19, 2024

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List of abbreviations

ACS	American Chemical Society
ADME	Absorption, distribution, metabolism and excretion
AFNOR	Association Française de Normalisation
AMDR	Acceptable Macronutrient Distribution Range
AMR	Antimicrobial resistance
AOAC	Association of Official Analytical Chemists
ATCC	American Type Culture Collection
BMBL	Biosafety in Microbiological and Biomedical Laboratories
BSL	Biosafety Level
bw	body weight
CD	Circular dichroism
CFR	Code of Federal Regulation
CFU	Colony Forming Unit
cGMP	Current Good Manufacturing Practice
CIAP	Calf Intestinal Alkaline Phosphatase
CoA	Certificate of Analysis
DIN	Deutsches Institut für Normung
DM	Dry Matter
DNA	Deoxyribonucleic acid
DO	Dissolved Oxygen
DRV	Daily Reference Value
EDI	Estimated Daily Intake
EFSA	European Food Safety Authority
EN	European Norm
EPA	US Environmental Protection Agency
ESI	Electrospray Ionization
FALCPA	Food Allergen Labeling and Consumer Protection Act
FAO	Food and Agriculture Organization of the United Nations
FARRP	Food Allergy Research and Resource Program
FASTER	Food Allergy Safety, Treatment, Education and Research
FCC	Food Chemical Codex
FD&C Act	Federal Food, Drug and Cosmetics Act
FDA	US Food and Drug Administration
FOIA	Freedom of Information Act
FNB	Food and Nutrition Board
FSIS	USDA Food Safety Inspection Service
GILSP	Good Industrial Large-Scale Practice
GlcNAc	N-acetylglucosamine
GRAS	Generally Recognized as Safe
GRN	GRAS Notice
HPLC	High-performance liquid chromatography
HPLC-UV	High-performance liquid chromatography coupled with ultraviolet detection
IgE	Immunoglobulin E
ISO	International Organization for Standardization

JGI	Joint Genome Institute
kDa	Kilo Daltons
LC-MS/MS	Liquid chromatography coupled with tandem mass spectrometry
LC-UV	Liquid chromatography coupled with ultraviolet detection
LOQ	Limit of Quantification
MCB	Master Cell Bank
Mdeg	Millidegree
MS	Mass Spectrometry
MS/MS	Tandem Mass Spectrometry
NCBI	US National Center for Biotechnology Information
NCHS	National Center for Health Statistics
ND	Not Detected
NF	National Formulary
NHANES	National Health and Nutrition Examination Surveys
NIH	National Institute for Health
NMKL	Nordisk Metodikkomité for Næringsmidler
NOVA	NON-ANIMAL OVA
NS	Not specified
OECD	Organisation for Economic Co-operation and Development
OVA	Ovalbumin
PCR	Polymerase Chain Reaction
PDCAAS	Protein Digestibility Corrected Amino Acid Score
pI	Isoelectric point
PNGase F	Peptide-N-Glycosidase F
PPM	Parts Per Million
PTM	Post-translational modifications
qPCR	Quantitative Polymerase Chain Reaction
QTOF	Quadrupole Time-of-Flight
RT-PCR	Real-Time Polymerase Chain Reaction
SDS-PAGE	Sodium Dodecyl Sulphate–Polyacrylamide Gel Electrophoresis
SLC	Solute Carriers
SOP	Standard Operating Procedure
T _m	Melting temperature
TSA	Thermal Shift Assay
TSCA	Toxic Substances Control Act
TSS	Total Suspended Solids
UPW	Ultrapure water
US	United States
UPLC	Ultra Performance Liquid Chromatography
USDA	United States Department of Agriculture
USP	United States Pharmacopeia
UV	Ultraviolet
WGS	Whole Genome Sequencing
WHO	World Health Organization
WWEIA	What We Eat in America

1.1. 21 CFR §170.225 SIGNED STATEMENT OF THE CONCLUSION OF GENERALLY RECOGNIZED AS SAFE (GRAS) AND CERTIFICATION OF CONFORMITY TO 21 CFR §170.205-170.260

In accordance with Title 21 of the Code of Federal Regulations (CFR) § 170 Subpart E (§170.203-170.285), Onego Bio, Inc. ("Onego Bio", "Onego") hereby informs the United States (US) Food and Drug Administration (FDA) that Ovalbumin from fermentation by *Trichoderma reesei* is not subject to the premarket approval requirements outlined in the Federal Food, Drug and Cosmetics Act (FD&C Act) based on Onego Bio's conclusion that the notified substance is Generally Recognized as Safe (GRAS) under the proposed conditions of use described herein.

1.1. Submission of GRAS notice

Onego Bio, Inc. is hereby submitting a GRAS notice in accordance with Subpart E of Part 170.

1.2. Name and address of the notifier

Onego Bio, Inc.
9215 Brown Deer Road, Suite C
San Diego, CA 92121
USA

All communications on this matter are to be sent to Atova Regulatory Consulting, SLU:

Dr. Hannah Lester
Atova Regulatory Consulting, SLU
Passeig de Gracia, 50
08007 Barcelona
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Email: hannah@atovaconsulting.com

1.3. Common or usual name of the notified substance

Animal-free **or** non-animal egg protein (ovalbumin) from fermentation **or** precision fermentation
Animal-free **or** non-animal egg white protein from fermentation **or** precision fermentation
Animal-free **or** non-animal egg protein from fermentation **or** precision fermentation
Animal-free **or** non-animal egg protein (ovalbumin)
Animal-free **or** non-animal egg white protein
Animal-free **or** non-animal egg protein

1.4. Conditions of use

Ovalbumin from fermentation by *T. reesei* ("NON-ANIMAL OVA" or "NOVA", hereinafter) is intended to be used as a general-purpose protein ingredient, at a maximum incorporation rate of 25%, in a wide range of food categories. These categories, as defined in 21 CFR part 170.3(n), include baked goods and baking mixes, beverages and beverage bases (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections and frostings, dairy product analogs, egg products, fats and oils, frozen dairy desserts and mixes, fruit and water ices, gelatins, puddings, and fillings, grain products and pastas, gravies and sauces, milk products, plant protein products, snack foods (protein and snack bars), and soft candy. For details, please see **Table 10**.

The protein ingredient is manufactured following cGMP and a Food Safety Plan, in accordance with 21 CFR Part 117. NOVA is intended to replace fresh or powdered whole eggs and egg white protein powder in food products, providing dietary protein along with functional properties such as foaming, gelling, and binding. These uses are not expected to increase the overall protein consumption of the general population.

Ovalbumin from fermentation by *T. reesei* is not intended for use in infant formula or in meat, poultry, egg products, catfish, or any products subject to the jurisdiction of the United States Department of Agriculture (USDA) and the Food Safety Inspection Service (FSIS).

1.5. Basis for GRAS determination

Onego Bio notifies the Agency of its determination that its NOVA from fermentation by *T. reesei* is GRAS for its intended use, consistent with Section 201(s) of the FD&C Act. This GRAS conclusion is based on scientific procedures in accordance with 21 CFR § 170.30(a) and § 170.30(b) and has followed the guidance issued by the FDA under 21 CFR § 170.36, 81 Federal Register 54960 (Aug. 17, 2016).

The GRAS status of ovalbumin from fermentation by *T. reesei* is based on data generally available in the public domain relating to the safety of ovalbumin produced from fermentation and the safety of the production organism, *T. reesei*.

1.6. Exemption from Premarket approval

The notified substance is not subject to the premarket approval requirements of the FD&C Act based on our conclusion that the substance is GRAS under the conditions of intended use and when meeting the specifications described herein, when used as a food ingredient and source of protein in selected conventional food products.

1.7. Availability of information

Consistent with 21 CFR § 170.225(c)(7), the data and information that serve as a basis for this GRAS determination are available for FDA review and copying, by contacting Atova Regulatory Consulting, SLU during conventional office hours at:

Dr. Hannah Lester
Atova Regulatory Consulting, SLU
Passeig de Gracia, 50
08007 Barcelona
Spain

Phone number: +34 686 999 247
Email: hannah@atovaconsulting.com

Alternatively, a complete copy of the data and information that serve as a basis for this GRAS determination will be sent to FDA upon request, either electronically or on paper.

1.8. Freedom of information act (FOIA) – Data exempt from disclosure

It is Onego Bio's view that no data and information presented in this document contains any trade secret, confidential or financial information that is privileged or confidential. Thus, all information and data in this submission are not exempt from the Freedom of Information Act (FOIA), 5 U.S.C. Section 552.

1.9. Certification of Information Included In the GRAS notification

As an authorized representative of Onego Bio, Inc., the undersigned hereby certifies that to the best of our knowledge, the data and information provided in this GRAS notice constitutes a complete, representative, and balanced submission. This includes all relevant information, both favorable and unfavorable, known to Onego Bio, Inc. and is pertinent to assessing the safety and GRAS status of the ovalbumin from fermentation by *T. reesei* intended for use as a food ingredient, as detailed in this notice.

Date:

December 19, 2024



Maija Itkonen
CEO
Onego Bio, Inc.
Notifier

2. 21 CFR §170.230 IDENTITY, METHOD OF MANUFACTURE, SPECIFICATIONS, AND PHYSICAL OR TECHNICAL EFFECT OF THE NOTIFIED SUBSTANCE

2.1. Identity of the notified substance

2.1.1. Introduction

Onego Bio produces a highly purified ovalbumin by precision fermentation using a genetically modified strain of *T. reesei* expressing the ovalbumin gene from *Gallus gallus*. *T. reesei* is a filamentous fungus with a long history of safe use in the production of food enzymes by precision fermentation. The notified substance is referred to as ovalbumin from fermentation by *T. reesei* or NON-ANIMAL OVA (NOVA) throughout this GRAS determination, although other common names may be used as detailed in **Section 1.3**.

NOVA is a water-soluble, white to off-white powder with a moisture content of maximum 10% and protein content of minimum 75%, out of which at least 80% is ovalbumin. Onego's NOVA is substantially equivalent to its native counterpart and can be used as an ingredient in a range of food products as a replacement for fresh or powdered eggs and egg white protein powder.

The levels of impurities, namely heavy metals and microbial contaminants are all below limits of safety concern. NOVA contains no residual recombinant DNA or viable cells from the production organism. No mycotoxins of food safety concern were detected.

2.1.2. Chemical name and physicochemical properties

Egg albumin, or ovalbumin (OVA), is the predominant protein in the egg white of *Gallus gallus* (chicken), accounting for about 54% of the total albumen by weight. It is also known as "Gal d 2", which corresponds to its allergen designation (Mathew & Pflughaar, 2023).

Native OVA is a water-soluble globulin. It is a monomeric, phosphorylated glycoprotein with a polypeptide chain consisting of 385 amino acids, including six cysteines with a disulfide bond. N-terminal and C-terminal of the protein are acetylated glycine and proline, respectively, with a carbohydrate group attached to the N-terminal. The ovalbumin's crystal structure consists of a three-turn α -helix loop, a reactive center, and six strands of β -sheet A (Aro, 2023; Razi, 2023; Guha, 2019). Hen egg ovalbumin is a mixture of three kinds of proteins, A1, A2, and A3 with two, one, and no phosphoryl residues, respectively, with isoelectric points (pI) ranging from 4.75 to 4.94 (Kitabatake, 1988).

2.1.3. Substantial equivalence to native protein

Onego Bio has conducted a comprehensive characterization of ovalbumin from fermentation by *T. reesei* to conclusively determine substantial equivalence to its native counterpart. The amino acid sequence, glycosylation pattern, molecular weight, isoelectric point, secondary structure, thermostability, and *in vitro* digestibility were evaluated and compared to native OVA, as detailed below. The results from all the analyses confirmed that ovalbumin from fermentation by *T. reesei* is substantially equivalent to ovalbumin from chicken egg white.

For the analyses described below, two commercial reference OVA samples were used: 1) Worthington OVA (Worthington Biochemical Corporation, LS003059) and 2) Sigma OVA (Sigma Aldrich, A5503). Both Worthington and Sigma OVA samples were purchased as lyophilized powders with a purity higher than 95% and 98%, respectively. Both originate from chicken egg white.

2.1.3.1. Amino acid sequence: LC-MS/MS peptide mapping

A high-resolution peptide mapping (PepMap) analysis using liquid chromatography coupled with tandem mass spectrometry (LC-MS/MS) was performed on samples of NOVA and native OVA from chicken egg white to compare their protein sequences. LC-MS/MS is a widely used method in proteomics, favored for its high sensitivity, precision, and ability to provide detailed protein characterization (Neagu, 2022).

For this analysis, Worthington OVA and NOVA samples from three non-consecutive production lots (ID-7695, ID-6309, and ID-8711) were denatured and reduced using a urea-DTT solution, alkylated with iodoacetamide, and digested with trypsin into peptides. LC-MS/MS analysis was carried out using an ultra-performance liquid chromatography (UPLC) system coupled with a quadrupole time-of-flight (QTOF) mass spectrometer, employing positive electrospray ionization (ESI). The mass range for data acquisition was m/z 67–2490, and results were filtered to retain only high-confidence peptide identifications.

As recorded in the UniProt database (accession number P01012), native ovalbumin consists of 386 amino acid residues, representing the full-length precursor that includes an N-terminal methionine. In *Gallus gallus*, this methionine is removed post-translationally in the mature protein, resulting in a sequence of 385 amino acids, as detailed in **Table 1**. The PepMap results for the commercial OVA sample were consistent with the reported sequence, with a sequence coverage of 100%.

For Onego's NOVA, PepMap analysis confirmed an amino acid sequence of 385 residues, identical to the native protein (UniProt accession number P01012), with 100% sequence coverage (Figure 1). Notably, the expression cassette designed by Onego Bio omits the N-terminal methionine residue, as *T. reesei* does not remove it as efficiently as *Gallus gallus*. This modification ensures that ovalbumin produced by *T. reesei* corresponds exactly to the mature native OVA commonly consumed from chicken eggs and egg-containing products. While the sequence of Onego's ovalbumin expressed in *T. reesei* has not been submitted to the NCBI or UniProt, it is fully detailed in **Table 2** as determined experimentally via LC-MS/MS in at least three non-consecutive production batches. To further support the sequence equivalence of NOVA to its native counterpart, a comparison of the percentages of each amino acid in OVA and NOVA is presented in **Section 2.5 - Batch analysis**, Table 8, where the mean levels of each amino acid in NOVA are compared to theoretical values for native ovalbumin, demonstrating a similar amino acid composition.

Table 1. Amino acid sequence of mature native OVA, determined by PepMap
GSIGAASMEFCFDVFKELKVHHANENIFYCPIAIMSALAMVYLGAKDSTRQINKVVRFDKLPFGGDSIEAQCG TSVNVHSSLRDILNQITKPNVDVYSFSLASRLYAEERYPILPYLCVKELYRGGLEPINFQTAADQARELINSWV ESQTNGIIRNVLPSSVDSQTAMVLVNAIVFKGLWEKAFKDEDTQAMPFRVTEQESKPVQMMYQIGLFRVAS MASEKMKILELPFASGTMSMLVLLPDEVSGLEQLESIIINFEKLTWETSSNVMEERKIKVYLPRMKMEEKYNLTS VLMAMGITDVFSSSANLSGISSAESLKISQAVHAAHAEINEAGREVVGSAEAGVDAASVSEEFRAHPFLFCIK HIATNAVLFFGRCVSP

Table 2. Amino acid sequence of NOVA, determined by PepMap in three non-consecutive production lots
GSIGAASMEFCFDVFKELKVHHANENIFYCPIAIMSALAMVYLGAKDSTRQINKVVRFDKLPFGGDSIEAQCG TSVNVHSSLRDILNQITKPNVDVYSFSLASRLYAEERYPILPYLCVKELYRGGLEPINFQTAADQARELINSWV ESQTNGIIRNVLPSSVDSQTAMVLVNAIVFKGLWEKAFKDEDTQAMPFRVTEQESKPVQMMYQIGLFRVAS MASEKMKILELPFASGTMSMLVLLPDEVSGLEQLESIIINFEKLTWETSSNVMEERKIKVYLPRMKMEEKYNLTS VLMAMGITDVFSSSANLSGISSAESLKISQAVHAAHAEINEAGREVVGSAEAGVDAASVSEEFRAHPFLFCIK HIATNAVLFFGRCVSP

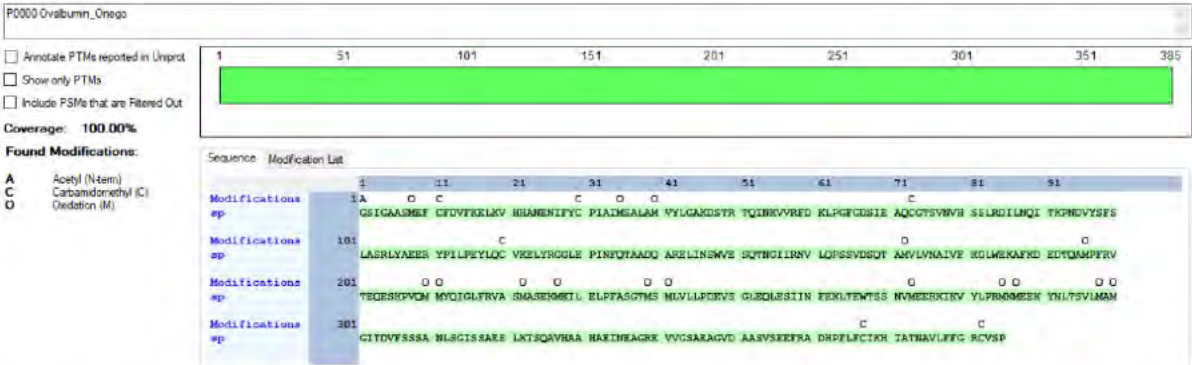


Figure 1. Sequence coverage for a sample of NOVA, determined by high-resolution peptide mapping (LC-MS/MS)

2.1.3.2. Glycan analysis: LC-MS

As a recombinant protein, NOVA is expected to exhibit post-translational modifications (PTMs) characteristic of its production organism, *T. reesei*, which is known to predominantly introduce N-linked glycosylations of the high-mannose type in alignment with the glycosylation patterns typically observed in filamentous fungi (Van Eerde, 2020, Nevalainen, 2014).

Notably, native chicken OVA is similarly N-glycosylated with mannose-based glycans, featuring a single N-linked glycosylation site and glycans connected to the polypeptide chain by N-acetylglucosamine (GlcNAc) - primarily (Man)₅(GlcNAc)₂ and (Man)₆(GlcNAc)₂ (Aro, 2023). OVA's glycans exhibit bi- to penta-antennary structures, with little evidence of larger, more complex structures. As is common in many avian N-linked sugars, OVA's glycans generally lack the terminal galactose found in complex mammalian glycans, although some reports describe (Man)₅(GlcNAc)₂(Gal)₁ in chicken OVA. Native chicken OVA is also phosphorylated (Harvey, 2000).

Onego Bio performed an LC-MS analysis to characterize NOVA's glycan pattern, confirming that its glycosylation profile aligns with those commonly observed in proteins expressed by *T. reesei*. The N-glycans identified in NOVA consist of mannose chains ranging from 5 to 11 units, linked to the polypeptide chain by GlcNAc residues. These results are summarized in Table 3, which provides a comparison of the glycans in NOVA and OVA.

Table 3. Comparison of glycans in native OVA and Onego's NOVA		
Parameter	OVA ^{a,b}	NOVA ^b
Glycan type	N-glycans	N-glycans
Number of residues glycosylated	1	0-2
Link to polypeptide chain	GlcNAc	GlcNAc
Identities of the carbohydrates	GlcNAc, Mannose, Galactose	GlcNAc and Mannose
Number of mannose units	3 to 7 ^b	5 to 11 ^b
Predominant glycoform	(Man) ₅ (GlcNAc) ₂ (Man) ₆ (GlcNAc) ₂	(Man) ₅ (GlcNAc) ₂
Isoelectric point	4.75 to 4.94 ^c 4.93 ^d	4.93 ^d

^a: Aro et al., 2023; Harvey et al., 2000.
^b: Determined experimentally by LC-MS glycan analysis. Worthington OVA was used as the reference sample.
^c: Kitabatake, 1988. The pI values for OVA with 2, 1, and 0 phosphoryl residues are 4.75, 4.89, and 4.94, respectively.
^d: Calculated using Geneious Prime software, on non-glycosylated and non-phosphorylated proteins.

Although NOVA exhibits slight differences in glycosylation compared to its native counterpart, the overall similarity remains high. In fact, one of the predominant glycoforms (Man)₅(GlcNAc)₂ is the same in both proteins and is the main N-glycan on NOVA. To assess the impact of these glycosylation differences, the isoelectric points (pI) of non-glycosylated and non-phosphorylated NOVA and OVA

were calculated using Geneious Prime. As shown in **Table 3**, the results of this calculation align with the value of 4.94 reported in the scientific literature for non-phosphorylated OVA (Kitabatake, 1988).

2.1.3.3. Molecular weight: SDS-PAGE

The molecular weight of NOVA was compared to that of its natural counterpart by means of Sodium Dodecyl Sulfate–Polyacrylamide Gel Electrophoresis (SDS-PAGE). Results are displayed in **Figure 2**.

For this analysis, NOVA samples from three non-consecutive production batches and Worthington OVA, all in powder form, were dissolved in ultrapure water to a concentration of 1 mg/mL. The protein samples were then denatured by boiling in denaturing buffer for 5-10 minutes.

Both OVA and NOVA were analyzed with and without PTMs. PTM removal was essential for this comparison, as differing PTM patterns would otherwise result in distinct electrophoretic profiles. To this end, CIAP (calf intestinal alkaline phosphatase, Invitrogen) was used to remove OVA's phosphate groups, while PNGase F (Peptide-N-Glycosidase F, New England Biolabs) was used to remove the N-glycans in both NOVA and OVA. Given that NOVA lacks the phosphate groups present in OVA, CIAP treatment was not required for its analysis. Denatured samples designated for PTM removal were treated with PNGase F and CIAP according to manufacturer's protocols. NOVA and OVA samples from which PTMs have been removed are referred to as "treated".

All samples - treated and untreated NOVA and OVA - were loaded onto a Criterion TGX Precast Gel (BioRad) which was run at constant 250 V for 30 minutes. The gel was stained with PageBlue™ staining solution. In all cases, the sample amounts loaded on each gel corresponded to 600 ng of ovalbumin (8 to 9 µL).

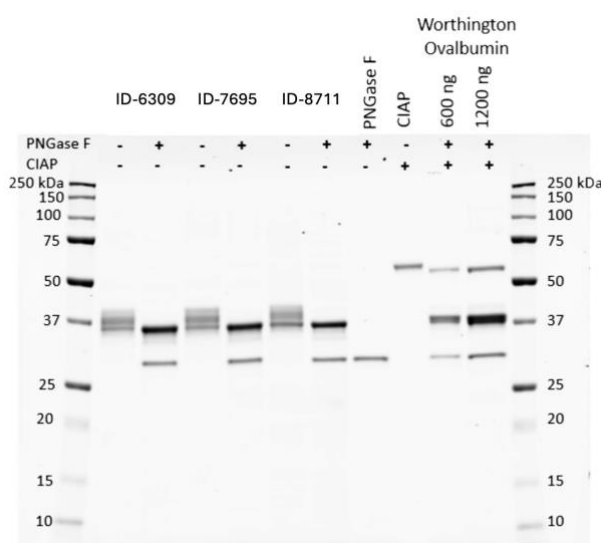


Figure 2. SDS-PAGE profile of commercial OVA and three non-consecutive production lots of NOVA, with and without PTMs.

Lane 1: molecular weight marker in kDa (Precision Plus Protein™ All Blue Standard, BioRad, 8 µL). Lane 2: NOVA lot ID-6309 untreated (600 ng). Lane 3: NOVA lot ID-6309 treated with PNGase F (600 ng). Lane 4: NOVA lot ID-7695 untreated (600 ng). Lane 5: NOVA lot ID-7695 treated with PNGase F (600 ng). Lane 6: NOVA lot ID-8711 untreated (600 ng). Lane 7: NOVA lot ID-8711 treated with PNGase F (600 ng). Lane 8: PNGase F. Lane 9: CIAP. Lanes 10 and 11: control Worthington OVA (600 ng and 1200 ng), treated with CIAP and PNGase F. Lane 12: molecular weight marker in kDa (Precision Plus Protein™ All Blue Standard, BioRad, 8 µL).

As expected, untreated NOVA samples displayed heterogeneous profiles with diffuse bands, reflecting the presence of PTMs that affect protein electrophoretic migration. Enzymatic treatment with PNGase F to remove N-glycans produced single, sharp bands for all three NOVA batches, facilitating direct comparison with native OVA. The migration patterns and molecular weights of deglycosylated NOVA samples were comparable to those of deglycosylated and dephosphorylated OVA, confirming the molecular weight equivalence of their core amino acid sequences.

2.1.3.4. Secondary structure: circular dichroism

The secondary structural motifs of NOVA and its native counterpart were analyzed by circular dichroism (CD) spectroscopy. Far-UV (190 to 260 nm) CD measurements were performed at 20°C, using a 0.1 mg/mL protein concentration in 10 mM sodium phosphate buffer (pH 7.2). For each sample, CD spectra were recorded in duplicate and averaged using a bandwidth of 1 nm, a wavelength step of 0.5 nm, and a time-per-point setting of 0.5 s. The CD spectra are presented in measured ellipticities (mdeg) without smoothing and with baseline correction for buffer contribution. Three non-consecutive production batches of NOVA (ID-7695, ID-6309, and ID-8711) were analyzed against a control OVA sample (Sigma).

As shown in Figure 3, both NOVA and native OVA displayed typical features of an alpha-helical protein, including negative bands at 208 and 222 nm and a positive band at 193 nm. The CD spectra of both proteins exhibited similar conformational motifs, confirming their equivalence in secondary structure.

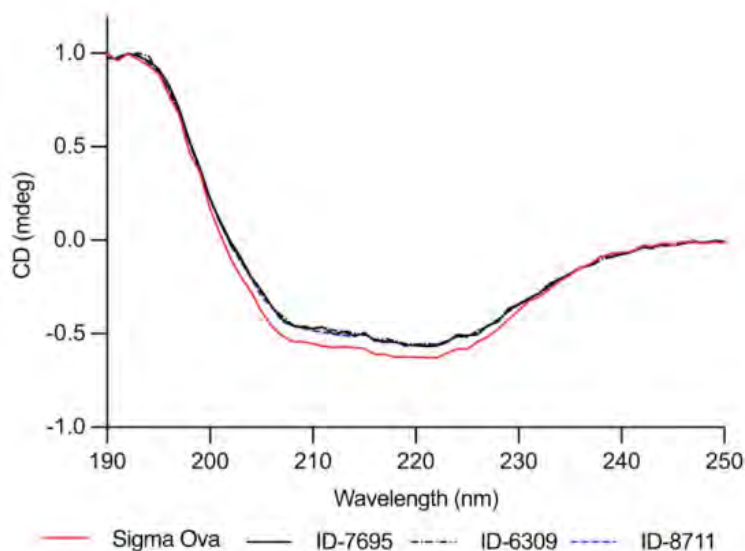


Figure 3. Circular dichroism (CD) spectra of native OVA and three non-consecutive production batches of NOVA.

The native chicken ovalbumin (Sigma OVA) is represented by the solid red line, while three production batches of NOVA are shown as: ID-7695 with a solid black line, ID-6309 with a dashed black line, and ID-8711 with a dashed blue line. The spectra were recorded from 260 to 190 nm in 10 mM Na-phosphate buffer, pH 7.2, at 20°C.

2.1.3.5. Thermostability and digestibility

Considering that the glycosylation pattern of NOVA differs slightly from that of its native counterpart, Onego Bio compared the thermostability and protein digestibility of the two proteins to verify that these variations do not impact the stability or functionality of ovalbumin from fermentation by *T. reesei*.

The melting temperatures (T_m) of NOVA from three non-consecutive production batches (ID-7695, ID-6309, and ID-8711) and a commercial OVA reference (Sigma) were determined using the Thermal Shift Assay (TSA), also known as differential scanning fluorimetry (ThermoFluor™). The assay was conducted over a temperature range of 25°C to 95°C with a gradient of 1°C per minute, following the method described by Huynh and Partch (2015). Each sample, prepared at a concentration of 1 mg/mL, was analyzed in triplicate. Data was processed and plotted using GraphPad Prism 10. Results, presented in **Table 4**, confirm that NOVA exhibits structural stability under thermal stress conditions, closely resembling the behavior of native chicken OVA despite minor differences in glycosylation.

The protein digestibility of NOVA was calculated using the Protein Digestibility Corrected Amino Acid Score (PDCAAS) method as referenced in [US Federal Register, Vol 58, Nr 3](#). The amino acid content was measured in three non-consecutive production lots of NOVA (**Section 2.5 - Batch analysis**) and a

digestibility factor of 1.0 was used for calculation, based on the reported digestibility of egg and egg white protein (Hoffmann & Falvo, 2004; Matsuoka, 2019).

The results confirmed the high digestibility of ovalbumin produced by *T. reesei*, with a PDCAAS of 1.0 across all NOVA batches analyzed, comparable to that of conventional egg white protein.

Table 4. Melting temperatures (T _m) of native OVA and three non-consecutive production batches of NOVA	
Sample	T _m (°C) ^a
NOVA ID-7695	73.28 ± 0.07
NOVA ID-6309	72.28 ± 0.96
NOVA ID-8711	72.56 ± 0.48
Native chicken OVA (Sigma)	74.93 ± 0.47
^a Each sample was measured in triplicate	

2.2. Source organism

2.2.1. Host strain

To manufacture NOVA, Onego Bio utilizes the filamentous fungus *T. reesei* TR112 as the production strain, which was derived from the well-known wild type strain QM6a.

QM6a is the parental strain to nearly all *T. reesei* industrial strains (Nevalainen, 1994) and is auxotrophic for uracil production due to the deletion of endogenous *PYR4* (orotidine 5'-phosphate carboxylase) gene – analogous to the *URA3* gene in *Saccharomyces cerevisiae*.

T. reesei strain QM6a is deposited at the American Type Culture Collection (ATCC) under [13631™](#). ATCC has classified the microorganism as Biosafety Level 1 (BSL-1). Moreover, *T. reesei* complies with the Organization for Economic Co-operation and Development (OECD) criteria for Good Industrial Large-Scale Practice (GILSP) microorganisms (OECD, 1993). It also meets the criteria for a safe production microorganism (Pariza and Foster, 1983; Pariza and Johnson, 2001).

The taxonomic identity of *T. reesei* is as follows:

Kingdom: Fungi
 Phylum: Ascomycota
 Class: Sordariomycetes
 Order: Hypocreales
 Family: Hypocreaceae
 Genus: *Trichoderma*
 Species: *Trichoderma reesei*
 Strain: QM6a
 Synonyms: *Hypocrea jecorina*

2.2.2. Production strain

The production strain used in the manufacture of recombinant fermentation-derived OVA is designated TR112 and was constructed from *T. reesei* strain QM6a by means of well-known genetic engineering techniques.

The expression cassette incorporated into the host strain contained the gene sequence for ovalbumin from *Gallus gallus*, codon-optimized for efficient *T. reesei* expression while preserving the original amino acid sequence, along with DNA homologous to the desired integration site. The gene sequence was synthesized *de novo* by GenScript. The cassette also included the endogenous *PYR4* (orotidine 5'-phosphate carboxylase) gene as a selection marker, which had been removed from the parental strain. OVA expression was driven under the control of either an endogenous *T. reesei* promoter or orthologous promoters, along with necessary signal and terminator sequences, and induction facilitated

by the fermentation medium. The expression cassette does not contain any antimicrobial resistance genes or mobile genetic elements.

The approach included the selective modulation of endogenous transcription factors and the integration of exogenous transcription factors from non-pathogenic and non-toxigenic sources. No antibiotic markers were used in the construction of the production strain. The absence of antimicrobial resistance (AMR) genes was confirmed via qPCR and whole genome sequencing (WGS). Additionally, WGS was used to confirm the identity of the production strain and the correct insertion of the expression cassette, while the insertion copy number was confirmed by qPCR. While the genome of *T. reesei* strain TR1112 is well characterized to ensure the safety of the production platform, Onego Bio has not made it publicly available as it is considered confidential business information.

Likewise, due to the highly proprietary nature of the production strain, Onego Bio has not deposited the production strain in a public culture collection. However, the master cell bank (MCB) of the production strain is deposited in a third-party facility for safe storage under appropriate conditions.

The production strain contains a single cassette insert. To improve protein yield and purity in the future, Onego Bio may optimize the strain through targeted host gene knockouts, deleting and replacing host genes with ovalbumin expression cassettes as described above. Potential target genes include enzymes involved in the metabolism of carbohydrates that are not utilized in our process. These targeted replacements would enhance ovalbumin production while decreasing the expression of host protein, thereby increasing the purity of the protein. Additionally, random expression cassette integrations may be considered. All integrations will be confirmed by WGS, and qPCR as described above. Product specifications will remain consistent throughout these strain optimizations and all genetic modifications to the production strain will be carefully evaluated to ensure that they do not compromise the safety and/or quality of the final product.

2.2.3. Genetic stability of the production organism

Multi-generational studies demonstrated that the expression cassette is stably inserted into the host strain genome, as evidenced by consistent ovalbumin expression levels. HPLC-UV analysis confirmed the stability of ovalbumin production across early (~25 generations) and extended (~75 generations) fermentation runs.

2.3. Production process

Ovalbumin from precision fermentation with *T. reesei* is produced by submerged fermentation of a *T. reesei* strain that has been constructed as described in **Section 2.2.2**. The fermentation process takes place in a liquid culture medium composed of standard ingredients used in the food/enzyme industry in facilities that adhere to current cGMPs and have a Food Safety Plan, in accordance with 21 CFR Part 117. All raw materials, including processing aids, are obtained from qualified suppliers and are safe and suitable for use in food products, conforming to either the specifications set out in the *Food Chemicals Codex* (14th edition) or to other applicable regulatory standards. These raw materials are of high purity and quality, regulated and suitable for their intended use and are food-grade, in accordance with 21 CFR § 170.230(c).

2.3.1. Description of the manufacturing process

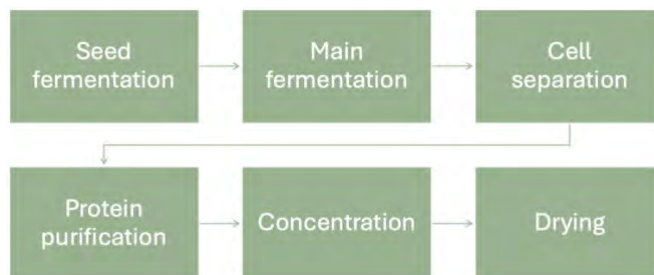


Figure 4. Overview of the production process of NOVA

Fermentation

The production of each batch of ovalbumin via precision fermentation with *T. reesei* begins with the preparation of an inoculum of the *T. reesei* production strain. This process starts by thawing a cryopreserved vial of the production strain and expanding it into a seed fermenter, in liquid culture medium, under controlled conditions.

The inoculum from the seed expansion is introduced into the main stirred-tank fermenter, where ovalbumin production occurs. The main fermentation is conducted under tightly regulated conditions. During all steps of fermentation, physical and chemical control methods are used, and microbiological analysis is done periodically to ensure the absence of foreign microorganisms.

Protein purification

During fermentation, *T. reesei* secretes ovalbumin into the liquid culture medium. For protein purification, the fermentation broth is harvested, the cell biomass is removed, and the resulting culture liquid is clarified through a multi-step filtration process. NOVA is then concentrated via ultrafiltration and diafiltration. The resulting NOVA material is sterile filtered before drying. The finished product is a white to off-white powder with a protein content of minimum 75%, out of which at least 80% is ovalbumin.

2.3.2. 2.4Raw materials

All raw materials and processing aids used in the manufacturing process of ovalbumin from fermentation by *T. reesei* are of high purity and quality, regulated and suitable for their intended use. They are food-grade in accordance with 21 CFR § 170.230(c).

All raw materials are used in the fermentation, including media components and processing aids, are standard substances used in the food enzyme industry. They are of suitable purity and quality for use in food, complying with quality specifications that include limits on heavy metals and microbiological contaminants. None of the raw materials are (or are derived from) major allergens and no antibiotics are used in the manufacturing process of ovalbumin from fermentation by *T. reesei*.

2.3.3. Quality control

Ovalbumin from fermentation by *T. reesei* is manufactured following cGMPs with the implementation of a Food Safety Plan in compliance with 21 CFR Part 117.

Each ovalbumin production lot is monitored for compliance with quality specifications as established by Onego Bio (**Section 2.4**). SOPs are followed throughout the entire manufacturing process to ensure consistent production and a safe final product.

2.4. Product specifications

The batch release specifications for NOVA are derived from the *Enzyme Preparations* monograph in the Food Chemicals Codex (FCC) 14th edition, in accordance with the FDA's guidance for industry on GRAS notices for enzyme preparations ([FDA 2010](#)). These specifications also consider those for other proteins produced through precision fermentation with GRAS status.

Table 5 below provides a detailed list of the product specifications along with the corresponding analytical methods used for each analysis. All analytical methods are validated for their intended purpose. All tests are conducted in ISO-17025-accredited testing facilities.

Table 5. Specifications for ovalbumin from fermentation by <i>T. reesei</i>			
Parameter	Specification	Method	LOQ
Physicochemical properties			
Appearance	White to off-white powder	Visual inspection	-
Solubility	Soluble in water	Visual inspection	-
pH	4.5 to 7.0	AOAC 981.12 ¹ Potentiometry	-
Composition			
Ovalbumin as % of protein	≥ 80%	HPLC-UV ² Internal method, validated	-
Moisture	≤ 10%	NMKL 23:1991 ³ , mod. Gravimetry	0.3%
Ash	≤ 5%	NMKL 173:2005 ⁴ , mod. Gravimetry	0.1%
Protein Dumas	≥ 75%	AOAC 968.06 ⁵ , mod. AOAC 992.15 ⁶ , mod. Dumas (TCD), N x 6.25	0.1%
Total fat	≤ 1.5%	NMKL160 ⁷ mod. Gravimetry	0.2%
Carbohydrates	≤ 20%	Calculated	-
Heavy metals			
Arsenic (As)	≤ 0.1 mg/kg	DIN EN 15763:2010 (2010-04) ⁸ , mod. ICP-MS	0.1 mg/kg
Cadmium (Cd)	≤ 0.1 mg/kg		0.01 mg/kg
Lead (Pb)	≤ 0.1 mg/kg		0.05 mg/kg
Mercury (Hg)	≤ 0.1 mg/kg		0.005 mg/kg
Microbiological contaminants			
Total Plate Count	< 10,000 CFU/g	NMKL 86:2013 ⁹ Culture, non-chromogenic media	1,000 CFU/g
Yeast & Molds	< 100 CFU/g	NMKL 98:2005 ¹⁰ Culture, non-chromogenic media	10 CFU/g
Coliforms	< 10 CFU/g	NMKL 44:2004 ¹¹ Culture, non-chromogenic media	10 CFU/g
<i>E. coli</i>	< 10 CFU/g	ISO 16649-2:2001 ¹² Culture, non-chromogenic media	10 CFU/g
<i>Salmonella</i> (/25 g)	Not Detected	BACGene <i>Salmonella</i> spp ¹³ RT-PCR	0.25 g
<i>Listeria monocytogenes</i> (/25 g)	Not Detected	AFNOR BRD 07/04-09/98 ¹⁴ Culture, chromogenic media	0.25 g

AFNOR = French Association for Standardization (Association Française de Normalisation); AOAC = Association of Official Analytical Chemists; BRD= Biological Reference Document; CFU = colony-forming units; DIN = Deutsches Institut für Normung (German Institute for Standardization); EN= European Norm; HPLC-UV = high-performance liquid chromatography coupled with ultraviolet detection; ICP-MS = inductively coupled plasma mass spectrometry; ISO = International Organization for Standardization; LOQ = limit of quantification; NMKL= Nordisk Metodikkomité for Næringsmidler (Nordic-Baltic Committee on Food Analysis); RT-PCR = real-time polymerase chain reaction.

¹ AOAC 1982. Method 981.12: *pH of acidified foods*. In: Official methods of analysis of AOAC International. 22nd ed. Gaithersburg, MD.

² The HPLC-UV method for OVA detection was developed at Onego Bio and validated in an accredited facility.

³ NMKL 1991. Method 23: *Moisture and ash. Gravimetric determination in meat and meat products*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.

⁴ NMKL 2005. Method 173, 2nd Ed.: *Ash, gravimetric determination in foods*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.

- ⁵ AOAC 1969. Method 968.06: *Protein (crude) in animal feed*. In: Official methods of analysis of AOAC International. 22nd ed. Gaithersburg, MD.
- ⁶ AOAC 1996. Method 992.15: *Crude protein in meat and meat products*. In: Official methods of analysis of AOAC International. 22nd ed. Gaithersburg, MD.
- ⁷ NMKL 1998. Method 160: *Fat, determination in foods*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.
- ⁸ DIN 2010. Method DIN EN 15763: *Foodstuffs - Determination of trace elements - Determination of arsenic, cadmium, mercury and lead in foodstuffs by inductively coupled plasma mass spectrometry (ICP-MS) after pressure digestion*. German Institute for Standardization (Deutsches Institut für Normung), Berlin, Germany.
- ⁹ NMKL 2013. Method 86, 5th Ed.: *Aerobic microorganisms. Determination in foods at 37 °C, 30 °C, 25 °C, 20 °C, 17/7 °C or 6.5 °C by the colony count method*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.
- ¹⁰ NMKL 2005. Method 98, 4th Ed.: *Mould and yeasts. Detection in foods and feeds*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.
- ¹¹ NMKL 2004. Method 44, 6th Ed.: *Coliform bacteria. Detection in foods and feeds*. Nordic-Baltic Committee on Food Analysis (Nordisk Metodikkomité for Næringsmidler), Oslo, Norway.
- ¹² ISO 2001. Method 16649-2:2001. Ed. 1: *Microbiology of food and animal feeding stuffs — Horizontal method for the enumeration of beta-glucuronidase-positive Escherichia coli — Part 2: Colony-count technique at 44 degrees C using 5-bromo-4-chloro-3-indolyl beta-D-glucuronide*. International Organization for Standardization.
- ¹³ Eurofins BACgene real-time PCR detection kit. Validated against the AOAC Performance Tested Methods criteria, demonstrating equivalence to FDA, USDA, and ISO reference methods as relevant to specific matrices. <https://www.eurofinsus.com/food-testing/resources/pathogen-testing-with-bacgene-pcr/>
- ¹⁴ AFNOR 2023. Certificate 07/04-09/98 for the analysis method RAPID'L.mono (detection) - Validated for the detection of *Listeria* spp. and *Listeria monocytogenes*, manufactured by BIO-RAD. Method validated in accordance with the NF VALIDATION general rules and the certification rules NF102 - *Validation of alternative analysis methods (Application to the food industry)*. French Association for Standardization (Association Française de Normalisation). https://nf-validation.afnor.org/wp-content/uploads/sites/2/2014/03/BRD-07-04-09-98_en.pdf

2.5. Batch analysis

Table 6 presents data from the analysis of three non-consecutive production batches of NOVA (ID-7695, ID-6309, and ID-8711). The results show compliance with product specifications across all lots, thus demonstrating the consistency of the manufacturing process. All CoAs are provided in **Section 7** as supporting information.

Table 6. Batch analysis of ovalbumin from fermentation by <i>T. reesei</i>					
Parameter	Unit	Specification	Batch 1 ID-7695	Batch 2 ID-6309	Batch 3 ID-8711
Appearance	-	White to off-white powder	Complies	Complies	Complies
pH	-	4.5 to 7.0	6.4	6.8	6.7
Composition					
Ovalbumin as % of protein	g/100g	≥ 80	85.8	86.4	90.9
Moisture	g/100g	≤ 10	8.0	6.9	4.9
Ash	g/100g	≤ 5	3.0	3.2	4.4
Protein Dumas	g/100g	≥ 75	88.2	85.7	84.6
Total fat	g/100g	≤ 1.5	< LOQ	< LOQ	< LOQ
Carbohydrates	g/100g	≤ 20	3.9	4.1	6.1
Heavy metals					
Arsenic (As)	mg/kg	≤ 0.1	< 0.1	< 0.1	< 0.1
Cadmium (Cd)	mg/kg	≤ 0.1	0.01	< 0.01	< 0.01
Lead (Pb)	mg/kg	≤ 0.1	0.05	0.06	0.05
Mercury (Hg)	mg/kg	≤ 0.1	< 0.005	< 0.005	< 0.005
Microbiological impurities					
Total Plate Count	CFU/g	< 10,000	< 1,000	< 1,000	< 1,000
Yeast	CFU/g	< 100	< 10	< 10	< 10
Molds	CFU/g	< 100	Est 20	< 10	< 10
Coliforms	CFU/g	< 10	< 10	< 10	< 10
<i>E. coli</i>	CFU/g	< 10	< 10	< 10	< 10
<i>Salmonella</i>	/25 g	Not Detected	Not Detected	Not Detected	Not Detected
<i>Listeria monocytogenes</i>	/25 g	Not Detected	Not Detected	Not Detected	Not Detected

CFU= colony forming units; Est=estimated; LOQ=limit of quantification.

The batch analysis of NOVA shows protein content ranging from 84.6% to 88.2% across the non-consecutive production batches, with NOVA content between 85.8% and 90.9%. The purity of ovalbumin from fermentation by *T. reesei* was determined using a validated HPLC-UV method developed by Onego Bio for the determination of ovalbumin in lyophilized powder samples. A representative chromatogram for NOVA lot ID-8711 is shown in **Figure 5**. The chromatographic results confirmed the identity and purity of ovalbumin from fermentation by *T. reesei*, displaying a single major peak with a retention time matching that of native OVA. Purity is calculated based on the percentage of the peak area.

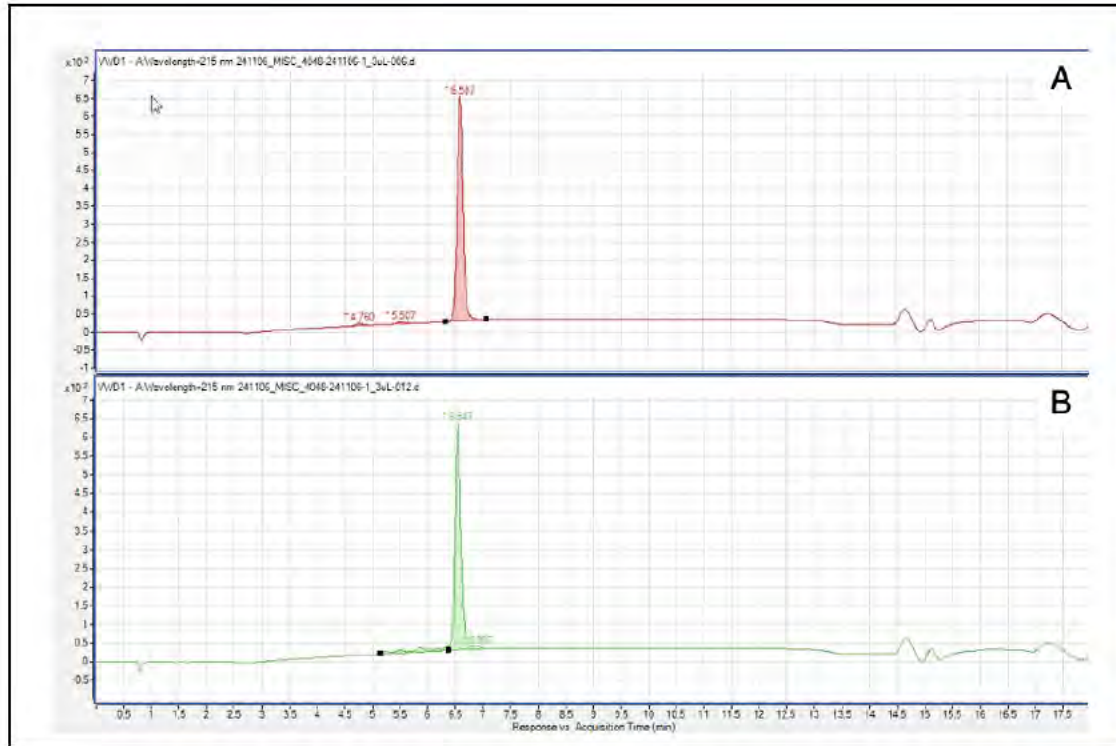


Figure 5. HPLC-UV chromatograms of native OVA and NOVA at 215 nm.

A: native chicken ovalbumin (Worthington LS003059). B: Onego Bio's NOVA (lot ID-8711). The HPLC-UV method used has been developed by Onego for the determination of ovalbumin purity in lyophilized powder samples and validated at a qualified third-party facility. Purity is determined based on peak area percentage. OVA and NOVA samples were dissolved in ultra-pure water (UPW), subjected to the same chromatographic conditions, and analyzed at 215 nm and 280 nm.

With respect to the analysis of contaminants in NOVA, the batch analysis presented in **Table 6** further indicates that heavy metal levels in NOVA are below the limit of quantification (LOQ) of the employed analytical methods (**Table 5**). All microbiological contaminants are within specifications and below the LOQ, except for mold levels in NOVA lot ID-7695, which were slightly elevated (est. 20 CFU/g) but still below the specified limit of acceptance. In addition, while *T. reesei* is not associated with mycotoxin production under standard industrial submerged fermentation conditions ([EPA, 2020, 40 CFR 725.420](#)), Onego Bio conducted testing on NOVA for common food-related mycotoxins, as outlined by the FDA ([2022](#)). This was done as a precaution due to the fungal origin of the production organism. All tested mycotoxins were below the limit of detection (**Table 7**), confirming that *T. reesei* strain TR112 does not produce mycotoxins under NOVA's manufacturing conditions. CoAs are provided in **Section 7** as supporting information.

Parameter	Unit	Specification	Batch 1 ID-7695	Batch 2 ID-6309	Batch 3 ID-8711
Aflatoxin B1	µg/kg	NS	< 1.0	< 1.0	< 1.0
Aflatoxin B2	µg/kg	NS	< 1.0	< 1.0	< 1.0
Aflatoxin G1	µg/kg	NS	< 1.0	< 1.0	< 1.0
Aflatoxin G2	µg/kg	NS	< 1.0	< 1.0	< 1.0

Table 7. Analysis of mycotoxins in ovalbumin from fermentation by <i>T. reesei</i>					
Parameter	Unit	Specification	Batch 1 ID-7695	Batch 2 ID-6309	Batch 3 ID-8711
Fumonisin B1	µg/kg	NS	< 20	< 20	< 20
Fumonisin B2	µg/kg	NS	< 20	< 20	< 20
Deoxynivalenol	µg/kg	NS	< 20	< 20	< 20
HT2-toxin	µg/kg	NS	< 10	< 10	< 10
T2-toxin	µg/kg	NS	< 10	< 10	< 10
Zearalenone	µg/kg	NS	< 10	< 10	< 10
Ochratoxin A	µg/kg	NS	< 2.0	< 2.0	< 2.0
NS = not specified					

To further substantiate the equivalence of NOVA to its native counterpart, Onego Bio has characterized NOVA's amino acid composition as detailed in **Table 8**. The mean values (in mol. % of the total protein fraction) of each amino acid in three production batches of NOVA are compared to theoretical values for native ovalbumin, demonstrating a similar amino acid composition. The slight deviations from the theoretical values can be attributed to the purity of ovalbumin in the product (85.5% to 90.9%) and the presence of host strain proteins as discussed in **Section 2.5.2** below.

Table 8. Amino acid analysis of NOVA compared to values for native OVA					
Test item	Batch 1 ID-7695 (g/100g)	Batch 2 ID-6309 (g/100g)	Batch 3 ID-8711 (g/100g)	NOVA Mean values (mol. %) ¹	Theoretical values for native OVA (mol. %) ²
Alanine	5.96	5.72	5.86	6.3%	9.1 %
Arginine	5.01	5.07	5.01	5.4%	3.9 %
Asparagine + aspartic acid	8.57	8.33	8.10	8.9%	8.0 %
Cystine	1.20	1.21	1.24	1.3%	1.6%
Glutamine + glutamic acid	13.1	13.0	13.1	14.1%	12.4%
Glycine	3.14	3.00	2.90	3.2%	4.9%
Histidine	1.96	1.94	1.93	2.1%	1.8%
Isoleucine	5.64	5.69	5.67	6.1%	6.5%
Leucine	8.07	8.04	7.96	8.6%	8.3%
Lysine	6.21	6.23	6.12	6.7%	5.2%
Methionine	3.78	3.80	3.91	4.1%	4.4%
Phenylalanine	6.15	6.10	6.07	6.6%	5.2%
Proline	3.43	3.29	3.25	3.6%	3.6%
Serine	6.64	6.57	6.59	7.1%	9.8%
Threonine	3.83	3.74	3.52	4.0%	3.9%
Tryptophane	1.38	1.29	1.27	1.4%	0.8%
Tyrosine	3.76	3.62	3.53	3.9%	2.6%
Valine	6.09	6.03	6.06	6.5%	8.0%
Sum of amino acids	93.92	92.67	92.09	100%	100%
¹ Mean of the measured amino acid composition of three NOVA batches (in mol. % of total protein fraction)					
² Theoretical values of native chicken OVA. The amino acid sequence of the native mature protein was derived from the UniProt database, accession No. P01012 . Theoretical values of each amino acid were calculated with the ProtParam tool of the Swiss Institute of Bioinformatics (SIB - ExPASy: https://web.expasy.org/cgi-bin/protparam/protparam)					

From the amino acid composition overview in **Table 8**, the average total amino acid content in NOVA is 92.89%, while the mean crude protein content, determined by the Dumas method, is 86.17% (**Table 6**). The Dumas method estimates protein content by measuring nitrogen (N-content x 6.25), based on the assumption that proteins contain 16% nitrogen (i.e., $1/0.16 = 6.25$). This method accounts for both proteins and other nitrogen-containing compounds but may be less precise than amino acid analysis, as the nitrogen content can vary among different proteins. While amino acid analysis provides a more direct and accurate measure of protein content, the similar results obtained through both methods suggest that the Dumas method is appropriate for estimating protein content in NOVA produced via *T. reesei* fermentation.

2.5.1. Absence of production organism in the final product

The absence of the *T. reesei* production strain in fermentation-derived NOVA was confirmed by means of a combined approach that includes the detection of viable cells through a culture method, and detection of DNA from the production organism by PCR. This approach is based on the guidelines provided by the European Food Safety Authority (EFSA) in their Guidance on the Characterisation of Microorganisms Used as Feed Additives or as Production Organisms (EFSA, 2018), regarded as an international reference for the safety assessment of protein production platforms of microbial origin. Samples from three representative, non-consecutive production lots of NOVA (ID-7695, ID-6309, and ID-8711) were evaluated.

For PCR analysis, DNA from the production strain (*T. reesei* TR112) was amplified on 96-well plates using a standard PCR program. The products were mixed with loading dye (Thermo Scientific), separated on an agarose gel, and visualized to detect the target amplification product. For each NOVA sample, 11 independent PCR reactions were prepared, each with equivalent volumes of the following templates:

1. Positive control: total genomic DNA from the production strain (1 reaction).
2. Negative control: pure solvent (1 reaction).
3. DNA dilution series: NOVA solutions of the same concentration spiked with DNA from the production strain in a range of dilutions, to determine the limit of detection (7 reactions).
4. NOVA test sample: NOVA solution without DNA from production strain, run in duplicate (2 reactions).

Results of the PCR analysis, shown in representative **Figure 6**, confirm the absence of recombinant DNA in Onego's ovalbumin from fermentation by *T. reesei*, above the assay detection limit. The sensitivity of the assay was determined to be 400 fg/μL (0.4 ng/g), which is sufficient to ensure reliable detection of any residual DNA from the production organism in NOVA samples.

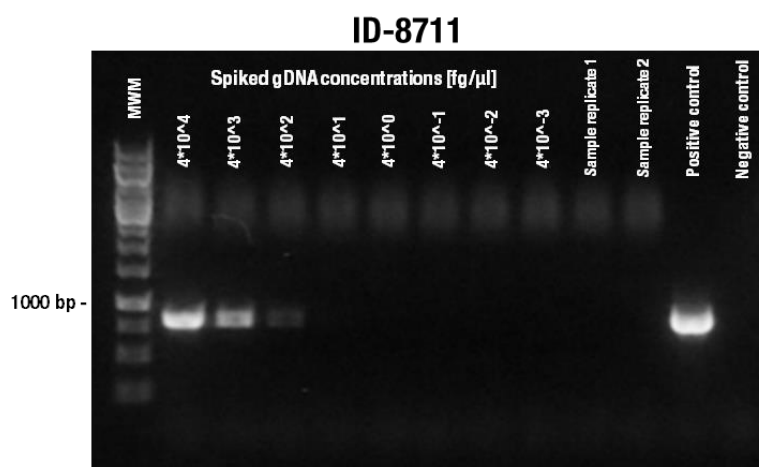


Figure 6. PCR analysis of NOVA lot ID-7695 to assess the presence of recombinant DNA from the production organism.

Lane 1: molecular weight marker (MWM). Lanes 2 to 9: DNA dilution series - NOVA sample spiked with host genomic DNA at concentrations of 4×10^4 fg/μL to 4×10^{-3} fg/μL. Lanes 10 and 11: NOVA test sample, replicates 1 and 2. Lane 12: positive control (host genomic DNA, 1 ng). Lane 13: negative control (25 mM NaOH). All other NOVA lots analyzed displayed comparable results.

The absence of viable cells from the production strain was evaluated using an internal culture method. NOVA samples from three non-consecutive production lots were inoculated onto potato dextrose sorbitol (PDS) agar plates and incubated at an appropriate temperature for a specified duration, in accordance with internal SOPs. The following controls were used:

1. Positive control 1: NOVA samples in sterile dilution solution, spiked with pre-determined low counts of viable *T. reesei* TR112 spores.

2. Positive control 2: pre-determined low counts of viable *T. reesei* TR112 spores in sterile dilution solution.
3. Negative control: sterile dilution solution was used as negative control.

All samples were assessed in triplicate. All NOVA samples evaluated were confirmed free from detectable, viable production organism, as no colony formation was observed in the plating test. This result is consistent with expectations, as all viable cells from the production organism are removed during downstream processing.

2.5.2. Proteins from the production organism in the final product

Onego Bio characterized the protein fraction in NOVA by quantitative proteomic analysis (LC-MS/MS) to support the identity and purity of ovalbumin from fermentation by *T. reesei* in samples from three non-consecutive production batches.

NOVA lots ID-7695, ID-6309, and ID-8711 showed comparable results, with 85.8%, 90.2%, and 90.9% of minimum ovalbumin content, respectively. The sequence coverage for ovalbumin ranged between 98% and 100% in all samples. The remaining components of the protein fraction (9.1% to 14.8%) correspond to host proteins.

When assessed individually, a total of 387, 261, and 308 peptides from *T. reesei* were identified in NOVA lots ID-7695, ID-6309, and ID-8711, respectively. Overall, a total of 541 peptides from the production organism were identified across all three batches. A summary of these findings is provided in Table 9.

Of the 541 *T. reesei* peptides identified, 17 were found in amounts greater than 1% in at least one batch, with Transaldolase (EC 2.2.1.2, JGI accession 123026) being the most abundant in all three lots (10.8%, 11.5%, and 8.1% in ID-7695, ID-6309, and ID-8711, respectively). The remaining 16 peptides above 1% were present in quantities ranging from 1% to 4.4%. Below 1%, 75 *T. reesei* peptides were detected at levels between 1% and 0.1%, while 198 were at levels of 0.1% to 0.01%, and 251 were present at concentrations below 0.01%.

	Above 1%	1% to 0.1%	0.1% to 0.01%	0.01% to 0.001%	below 0.001%
Batch ID-7695	15	61	140	146	25
Batch 2 ID-6309	9	40	87	106	19
Batch 3 ID-8711	6	44	110	124	24
All NOVA batches	17	75	198	251	

An extensive review of the scientific literature on the safety of *T. reesei* as a production organism for food enzymes - many of which are GRAS - found no reports of allergenicity despite its widespread use, as detailed in **Sections 6.3.1 and 6.3.2**. Nevertheless, to assess potential allergenic cross-reactivity, Onego Bio conducted sequence homology testing of *T. reesei* proteins identified in NOVA at levels above 0.01%, using the AllergenOnline database from the Food Allergy Research and Resource Program (FARRP) at the University of Nebraska. The findings are discussed in **Section 6.3.5**.

Notably, to date, the FDA has not established threshold levels for any food allergens ([FDA, 2024](#)), likely reflecting the understanding that all proteins in food ingredients, regardless of concentration, may pose allergenicity concerns. Nevertheless, Onego Bio has set a conservative threshold of 0.01% for sequence homology testing of *T. reesei* proteins against known allergens. This threshold represents a practical yet precautionary approach, addressing potential allergenicity risks at low concentrations while acknowledging the unlikelihood of identifying allergens of food safety concern based on the established safety profile of *T. reesei*.

2.6. Physical or technical effect

NOVA from fermentation by *T. reesei* may be incorporated into diverse food product categories as detailed in **Section 1.3**, as a source of dietary protein providing nutritional value and functionality due to its foaming, gelling, and binding properties.

3. 21 CFR §170.235 DIETARY EXPOSURE

3.1. Intended use

NOVA is intended as a general-purpose protein ingredient, at a maximum use level of 25 g of ovalbumin per 100 g, in a wide range of food categories. These categories, as defined in 21 CFR § 170.3, include baked goods and baking mixes, beverages and beverage bases (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections and frostings, dairy product analogs, egg products, fats and oils, frozen dairy desserts and mixes, fruit and water ices, gelatins, puddings, and fillings, grain products and pastas, gravies and sauces, milk products, plant protein products, snack foods (protein and snack bars), and soft candy.

The proposed use of NOVA is not intended for foods where standards of identity would preclude such use, infant formula, meat, poultry, egg products, catfish, or any products that would require additional regulatory review by USDA.

3.2. Estimated daily Intake

3.2.1. Methods

The estimated daily intake (EDI) of NOVA, assuming a maximum use of 25 g of ovalbumin per 100 g in the selected food categories, was calculated for the total U.S. population (ages 2 and older, “2+ year”) and four subpopulations: 2-5-year-old children (“2-5 year”), 6-12-year-old children (“6-12 year”), 13-18-year-old adolescents (“13-18 year”), and 19+ year-old adults (“19+ year”). Food consumption data from the 2017–2020 pre-pandemic National Health and Nutrition Examination Surveys (NHANES), hereafter referred to as NHANES 2017-2020 pp, were used in this calculation. The data, methods, and results of the assessment are summarized below.

The food codes reported in NHANES 2017-2020 pp data were reviewed. **Table 10** provides an overview of the food types where NOVA is proposed for use, along with descriptions of representative food categories included in the analysis and the maximum intended use levels of NOVA.

Table 10. Intended uses of NOVA, listed by food use, in representative foods selected for EDI analysis			
Food use (21 CFR § 170.3(n))	Representative examples	Description of selected food categories reported in NHANES 2017-2020 pp to estimate exposure	Maximum use level (g per 100 g food as consumed)
Baked goods and baking mixes	Breads and rolls	Bread (yeast breads, all types), garlic bread (all types), savory rolls, bagels and English muffins (all types)	3
	Batter mixes	Batter mixes	3
	Cakes	Cakes (all types, including cheesecake), cupcakes, tiramisu	6
	Cookies	Biscuits (all types)	3
	Pies, cobblers, fruit crisps, turnovers and other pastries (excluding products regulated by USDA/FSIS)	Doughnuts, sweet rolls, pastries (all types), cobblers (all types), pies (all types), fruit crisps/fritters (all types), strudel (all types), baklava, basbousa, turnover (all types), muffins, quick breads	5
	Quiches	Quiche (all types)	12
	Pancakes, waffles, French toast	Pancakes (all types), waffles (all types), French toast (all types)	5
	Protein breads	Bread (high protein)	25
	Souffles	Cheese, spinach and squash souffle	8
Beverages and beverage bases, nonalcoholic	Smoothies	Fruit smoothies (all types), vegetable smoothies	12
	Protein smoothies	Fruit and vegetable smoothies with added protein	25
	Powdered shake mixes	Nutritional drinks or shakes (all types), nutritional powder mixes (all types)	25
	Eggnog	Eggnog	4

Table 10. Intended uses of NOVA, listed by food use, in representative foods selected for EDI analysis			
Food use (21 CFR § 170.3(n))	Representative examples	Description of selected food categories reported in NHANES 2017-2020 pp to estimate exposure	Maximum use level (g per 100 g food as consumed)
Beverages, alcoholic	Egg white cocktails	Cocktails containing egg white, including whiskey sour, sloe gin fizz, White Russian, gin fizz and liqueur with cream	4
Breakfast cereals	Ready-to-eat cereals	Ready-to-eat cereal (both lower and higher sugar)	10
Cheeses	Breaded mozzarella sticks	Mozzarella sticks, breaded, baked, or fried	6
Confections and frostings	Marshmallows	Marshmallows (as candy and topping)	3
Dairy product analogs	Milk substitute beverages	Soy milk, almond milk, rice milk and coconut milk	10
Egg analogs	Egg analogs	Conventional eggs, including omelets and scrambled eggs and egg substitutes	12
Fats and oils	Salad dressings and vegetable oils	Dressings (all types)	10
	Mayonnaise	Mayonnaise (all types), sandwich spread, tartar	10
Frozen dairy desserts and mixes	Ice cream, frozen yogurt, sherbet	Frozen yogurt (all types), ice cream (all types), ice cream bars/cones (all types) gelato, sherbet, fudgesicle	8
Fruit and water ices	Frozen flavored and sweetened ice and pops	Frozen fruit juice bars, sorbet, Italian ice, popsicles, freezer pop, snow cones	8
Gelatins, puddings, and fillings	Gelatin, pudding, and custards	Pudding (all types), custards, flan, gelatin desserts, crème brulee, firni, barfi, meringues, trifle, haupia	5
Grain products and pastas	Plain pasta	Pasta, noodles, cooked grains	10
	Egg rolls and dumplings (excluding products regulated by USDA/FSIS)	Egg rolls (all types), Wonton, boa buns	6
Gravies and sauces	Sauces and dips (excluding products regulated by USDA/FSIS)	Chipotle/dill/onion/ranch/spinach/vegetabl e/cheese/artichoke dip, Hollandaise and vegetarian gravy	10
Milk products	Milk and milk-based drinks	Dairy-based milk (all types, including flavored milk) milk shakes (all types)	10
Plant protein products	Meat analogs	Meat analog/vegetarian/meatless products including frankfurter, bacon strip, breakfast link, chicken, luncheon slide, meatball, vegetarian burger/patty, sandwich spread (meat substitute type), meat substitute	10
Protein and snack bars	Cereal or granola bars	Cereal or granola bars (all types)	10
	Protein bars	Nutrition bars	25
Soft candy	Snack bars	Caramel, chocolate candy/bars/chips, fudge, peanut bar, pralines	3
The estimates for the intake of NOVA were generated using the maximum use level for the intended food use with food consumption data from NHANES 2017-2020 pp with DaDiet Software.			

Consumption data from individual dietary records, detailing food items ingested by each survey participant (What We Eat in America (WWEIA) dietary component of the NHANES conducted in 2017-2020 pp, were used to generate estimates for the intake of NOVA when used in the selected food categories, by the US population. Statistical analysis and data management were conducted in DaDiet Software (Version 17.04, Dazult Ltd). DaDiet Software is a web-based software tool that allows accurate estimate of exposure to nutrients and to substances added to foods, including contaminants, food additives and novel ingredients. The main input components are concentration (use level) data and food consumption data. Datasets are combined in the software to provide accurate and efficient exposure assessments.

Estimates for the daily intake of NOVA represent projected two-day averages for each individual from Day 1 and Day 2 of NHANES 2017-2020 pp; these average amounts comprised the distribution from which mean, and percentile intake estimates were determined. Mean and percentile estimates were generated incorporating survey weights to provide representative intakes for the entire US population. “*Per capita*” intake refers to the estimated intake of NOVA averaged over all individuals surveyed, regardless of whether they consumed food products in which NOVA is proposed for use and therefore includes individuals with “zero” intakes (i.e., those who reported no intake of food products containing NOVA during the 2 survey days). “*Consumer-only*” intake refers to the estimated intake of NOVA by those individuals who reported consuming food products in which the use of NOVA is currently under consideration. Individuals were considered “consumers” if they reported consumption of 1 or more food products in which NOVA is proposed for use on either Day 1 or Day 2 of the survey.

This approach and software to estimate daily intake has been reviewed by the FDA as part of different GRAS notices for which a “no questions letter” was issued (e.g.; [GRN 881](#), [GRN 943](#), [GRN 996](#), [GRN 997](#)).

3.2.2. Results

The estimated daily intake (EDI) for NOVA was calculated using DaDiet Software (Version 17.04, Dazult Ltd) and NHANES 2017-2020 pp consumption data and maximum proposed use levels in the selected food categories (**Table 10**). Intake estimates are provided on a *per capita* and *per user* basis at the mean and 90th percentile of intake and expressed in g per day (g/day) and g per kg body weight per day (g/kg body weight/day) for the total US population (2+ year) and 4 subpopulations (2-5 year, 6-12 year, 13-18 year, 19+ year). The total EDI of NOVA from the consumption of any of the proposed food uses are presented in Table 11 (in g/day) and Table 12 (in g/kg body weight/day).

Among the general US population (2+ year), 97.4% were identified as consumers of foods proposed to contain NOVA. The EDI of NOVA among the US population 2+year at the per user mean and 90th percentile of intake was 18.89 g NOVA/day (0.32 g NOVA/kg bw/day) and 39.17 g NOVA/day (0.74 g NOVA /kg bw/day), respectively. Per user mean intake of NOVA ranged from 16.63 g NOVA /day among children 2-5 year to 19.86 g NOVA /day among children 6-12 year. On body weight basis, the highest per user mean intake is among children 2-5 year at 0.97 g NOVA /kg bw/day. The per user 90th intake estimates of NOVA range from 34.98 g NOVA /day among children 2-5 year, to 41.99 g NOVA /day among children 6-12 year. The highest per user 90th percentile of intake on a bodyweight basis are among children 2-5 year at 2.26 g NOVA /kg bw/day.

Table 11. Two-day average estimated daily intake (EDI) for NOVA (g/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.						
Population	N*	% User	Per Capita		Per User	
			Mean	90 th percentile	Mean	90 th percentile
US population (2+ year)	9,855	97.4%	18.41	38.64	18.89	39.17
Children (2-5 year)	796	99.4%	16.54	34.98	16.63	34.98
Children (6-12 year)	1,375	99.6%	19.82	41.98	19.86	41.99
Adolescents (13-18 year)	1,101	97.0%	17.21	36.88	17.53	37.53
Adults (19+ year)	6,583	96.8%	18.49	38.64	19.07	39.17

*Unweighted number of users, % users, and *per capita* and *per user* estimates based on NHANES 2017-2020 pp were derived using the statistical weights provided by the National Center for Health Statistics (NCHS).

Table 12. Two-day average estimated daily intake (EDI) for NOVA (g/kg bw/day) from all proposed food uses among the total US population two years and older (2+ year) & subpopulations.						
Population	N*	% User	Per Capita		Per User	
			Mean	90 th percentile	Mean	90 th percentile
US population (2+ year)	9,756	97.4%	0.31	0.73	0.32	0.74
Children (2-5 year)	779	99.4%	0.97	2.26	0.97	2.26
Children (6-12 year)	1,367	99.6%	0.59	1.35	0.59	1.35
Adolescents (13-18 year)	1,093	97.0%	0.27	0.57	0.28	0.57
Adults (19+ year)	6,517	96.8%	0.24	0.52	0.24	0.52
*Unweighted number of users, % users, and <i>per capita</i> and <i>per user</i> estimates based on NHANES 2017-2020 pp were derived using the statistical weights provided by the National Center for Health Statistics (NCHS).						

Notably, these exposure values represent a worst-case scenario and are considered an overestimation, as they assume the product is consumed at maximum use levels across all proposed food categories. The actual exposure to NOVA is anticipated to be significantly lower, as Onego Bio may not use the maximum intended use level of NOVA in all products, and not all consumers may select products with NOVA at all eating occasions.

NOVA is intended to replace fresh or powdered whole eggs and egg white protein powder in the selected food categories and will be substitutional for animal-derived ovalbumin in the diet. Given the proposed uses for ovalbumin from fermentation by *T. reesei*, the cumulative exposure to ovalbumin (both from dietary exposure from eggs and egg-containing products and from the intended uses of NOVA) is not expected to increase in the general population.

3.3. Exposure to undesirable substances

The consumption of NOVA under the proposed uses is not expected to result in exposure to undesirable substances of food safety concern. As detailed in **Section 2.5**, levels of impurities, namely heavy metals, adventitious agents, and mycotoxins in NOVA are all below limits of safety concern. The product contains no residual recombinant DNA or viable cells from the production organism.

4. 21 CFR §170.240 SELF-LIMITING LEVELS OF USE

The use of ovalbumin from fermentation by *T. reesei* as a food ingredient is not self-limiting. The maximum use levels in food are as detailed in **Section 3**.

5. 21 CFR §170.245 COMMON USE IN FOOD BEFORE 1958

As a key protein component of hen egg white, ovalbumin has been a part of the human diet for centuries. However, though substantially equivalent, ovalbumin from fermentation by *T. reesei* was not introduced into the food chain before 1958. As such, this GRAS determination is based on scientific procedures rather than common use in food prior to 1958.

6. 21 CFR §170.250 NARRATIVE ON THE CONCLUSION OF GRAS STATUS

6.1. Introduction

The subject of this GRAS determination is recombinant ovalbumin produced by Onego Bio via precision fermentation using a genetically modified strain of filamentous fungus *T. reesei*. Onego Bio has determined the product to be GRAS based on scientific procedures, in accordance with 21 CFR § 170.30(a) and § 170.30(b). The GRAS status is supported by publicly available safety data on ovalbumin from fermentation and the production organism, *T. reesei*.

NOVA is a water-soluble powder with a protein content of minimum 75%, out of which at least 80% is ovalbumin. NOVA is substantially equivalent to its native counterpart and can be used as a general-purpose ingredient to replace eggs and white protein powder in a range of food products, as a source of dietary protein providing nutritional value and functionality due to its foaming, gelling, and binding properties. These uses are not expected to increase the overall protein consumption of the general population.

The production platform for NOVA, *T. reesei*, is widely recognized as a safe production host for enzymes, with a long history of use in the food industry. It has been used to produce numerous GRAS proteins and enzymes, many of which have received “no questions” letters from the FDA.

The fermentation process takes place in a liquid culture medium composed of standard ingredients used in the food/enzyme industry in facilities that adhere to current cGMPs with the implementation of a Food Safety Plan in compliance with 21 CFR Part 117. All raw materials are of high purity and quality, regulated and suitable for their intended use and are food-grade, in accordance with 21 CFR § 170.230(c).

The food safety aspects of Onego’s ovalbumin from fermentation by *T. reesei*, the production process, and the production platform are discussed in the following sections.

6.2. Safety of ovalbumin

6.2.1. History of consumption of ovalbumin

Egg albumin, or ovalbumin (OVA), is the predominant protein in the egg white of *Gallus gallus* (chicken), accounting for about 54% of the total albumen by weight (Guha, 2019).

As a major component of egg white, OVA has an extensive history of safe consumption worldwide. Eggs have long been a fundamental part of the human diet due to their rich nutrient content and culinary versatility, with ovalbumin significantly contributing to their nutritional value and functional properties. Ovalbumin is an excellent source of high-quality dietary protein, as it contains all essential amino acids and is highly digestible. Additionally, its intrinsic functional properties – gelling, foaming, binding, emulsifying – make it very valuable for the food industry (Razi, 2023)

OVA was one of the first proteins to be purified from egg whites and it has also been expressed in several microbial platforms, including *E. coli*, *P. pastoris*, and *T. reesei*, as reported in peer-reviewed scientific journals. The final product of recombinant production by precision fermentation is a protein powder that typically shows comparable functional properties to chicken ovalbumin and can be used as a replacement in food formulations (Järviö, 2021; Pereira, 2016; Ito & Matsudomi, 2005), offering a sustainable alternative to the conventional food industry to reduce its reliance on animal farming.

Recombinant ovalbumin produced by fermentation using *Komagataella phaffii* (*P. pastoris*) has also been determined GRAS in GRN 1104 (2022), which has received a “no questions” letter from the FDA. Other egg white proteins have also been determined GRAS (Table 13), such as ovomucoid from fermentation with *Komagataella phaffii* as described in GRN 967 (2020) and egg-white lysozyme isolated from hen eggs as described in GRN 64 (2000), both of which have also received “no questions” letters from the FDA. In a proposed rule from 1998 to include egg-white lysozyme in 21 CFR § 184, Direct Food Substances Affirmed as GRAS, the FDA states that that proteins derived from egg whites do not

pose toxicity concerns due to the long history of human consumption of egg whites and the lack of toxicity reports ([FDA, 1998](#)).

Table 13. Overview of GRAS notifications involving egg white proteins				
GRN No.	Year of closure	Substance	Use	FDA's letter
1104	2023	Egg-white protein (ovalbumin) produced by <i>Komagataella phaffii</i>	Intended for use as a substitute for egg-white protein in foods containing eggs; and as a source of protein in baked goods; prepared foods including plant-based meat alternatives and egg noodles and pasta; ready-to-eat cereals; scrambled eggs and egg patties; pancakes and waffles; protein and snack bars; salad dressings; eggnog; egg white cocktails, and powdered shake mixes at levels in accordance with current good manufacturing practices; and as a fining agent in wine.	FDA has no questions
967	2021	Soluble egg-white protein (ovomucoid) produced by <i>Komagataella phaffii</i>	Intended for use as a substitute for egg-white protein in foods containing eggs; and as a source of protein in nutritional powders and drinks; bars; and certain snack foods at levels in accordance with current good manufacturing practices (excluding infant formula, or in any products under the jurisdiction of the United States Department of Agriculture).	FDA has no questions
64	2001	Egg white lysozyme isolated from hen eggs	For use on casings for Frankfurters and on cooked meat and poultry products as an antimicrobial agent	FDA has no questions

6.2.2. Absorption, distribution, metabolism and excretion (ADME) of ovalbumin

Being a major component of egg-white protein, OVA is digested and metabolized in the same way as other dietary proteins from egg, milk, muscle, and plants.

In general, dietary proteins are subjected to complex changes during ingestion, digestion and absorption. First, they are exposed to the acidic environment of the stomach, where they undergo partial hydrolysis by gastric enzymes, primarily pepsin. After passing through the gastric lumen, they undergo further hydrolysis in the small intestine by pancreatic enzymes such as trypsin, chymotrypsin, and carboxypeptidases, as well as peptidases released by brush border cells in the intestinal mucosa. These enzymes, most active at neutral to alkaline pH, ultimately break down proteins into a pool of free amino acids and di- and tri-peptides. While some of these proteolysis products pass into the large intestine, most are absorbed by enterocytes via solute carriers (SLCs) in the brush border and basolateral membranes which, via active transport mechanisms, deliver amino acids and peptides into the bloodstream (Torres, 2023; Jahan-Mihan, 2011).

Ovalbumin, specifically, has shown resistance to gastric proteolysis in the presence of pepsin at pH 2, in an *in vitro* gastrointestinal digestion model with two steps, mimicking stomach and duodenum conditions at 37°C (Martos, 2010). It becomes markedly more susceptible to cleavage in the duodenal phase, in the presence of bile salts and pancreatic proteinases at pH 7.

Once absorbed into the bloodstream, free amino acids are transported to the liver via the portal vein. In the liver, these amino acids undergo various metabolic pathways: they can be utilized for the synthesis of proteins, including plasma proteins, or converted into other biologically active molecules, such as neurotransmitters, hormones, and nucleotides. Additionally, amino acids can be deaminated, allowing their carbon skeletons to enter the citric acid cycle as intermediates, contributing to gluconeogenesis or energy production. The liver also plays a crucial role in regulating amino acid levels in the blood, ensuring a balanced supply to peripheral tissues where they are used for local protein synthesis and other cellular functions (Torres, 2023; Chandel, 2021).

Overall, the bioavailability of amino acids from dietary proteins is conditioned by protein digestibility. Egg proteins, particularly ovalbumin, are well-established as a high-quality protein source, providing all essential amino acids with a true ileal digestibility of 90.9% in humans. The digestibility of egg white proteins is influenced by structural changes induced during processing, such as heat treatment.

Notably, native egg white proteins are less digestible compared to those subjected to moderate heat treatments (60–90 °C). Such treatments alter the secondary and tertiary structures, exposing previously buried hydrophobic and sulfhydryl groups, thereby enhancing the accessibility of enzymatic cleavage sites and improving overall digestibility (Farjami, 2021; Evenepoel, 1998). Heat treatment can also modulate the allergenicity of egg white proteins, especially ovalbumin by destroying IgE and IgG-reactive epitopes, which reduces allergenic potential as will be detailed in section 6.2.3.

Given the demonstrated substantial equivalence to native ovalbumin in terms of thermostability and digestibility (section 2.1.3), the ADME profile of ovalbumin from fermentation by *T. reesei* is expected to resemble that of dietary proteins, and comparable to OVA derived from chicken eggs.

6.2.3. Allergenicity of ovalbumin

Eggs are one of the nine major food allergens identified by the Food Allergen Labeling and Consumer Protection Act of 2004 (FALCPA) and the FASTER Act of 2021, along with milk, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybean, and sesame. Together, these food groups account for 90% of food allergies in the US ([FDA, 2024](#); [USDA, 2024](#); [FDA, 2022a](#)). After milk, eggs are the second most common allergen in childhood, typically appearing in the first two years of life. In 50 to 80% of children, egg allergies resolve spontaneously by school age. Egg allergies rarely appear in adulthood (ECARF, 2016; Jiaqi, 2023; Caubet, 2011).

The most common egg allergens are found in the egg white, where, despite OVA (Gal d 2) being the most abundant protein, ovomucoid (Gal d 1) is the primary allergen. Other significant allergens in egg white are ovotransferrin (Gal d 3), egg white lysozyme (Gal d 4), and ovomucin (ECARF, 2016; Mathew, 2023; Caubet, 2011). In contrast, egg yolk allergies are less common, with chicken serum albumin (Gal d 5) being the main allergen in the yolk (Tomei, 2023; Quirce, 2001).

Egg allergy is mediated by immunoglobulin E (IgE). While healthy individuals can produce antigen-specific IgE without adverse effects, those with egg allergies experience an exaggerated immune response, even to minimal egg protein exposure. This can lead to a range of symptoms from cutaneous reactions (most common) to severe, potentially life-threatening anaphylaxis. A genetic predisposition likely contributes to egg allergies, as children of atopic individuals are more prone to developing this allergy. Asthmatic patients are particularly at high risk of severe reactions (Mathew, 2023; Caubet, 2011).

The allergenicity of Gal d 2 has been validated in various animal models mainly aimed at developing valid sensitization models for the study of food protein allergenicity. For instance, Knippels et al. (1999) explored oral sensitization with ovalbumin in different rat strains, administering 1 mg of OVA daily via gavage for six weeks. While all strains developed OVA-specific IgG responses, only Brown Norway (BN) rats also developed OVA-specific IgE responses, indicating that genetic background influences the outcomes of oral sensitization. This pattern of IgG and IgE responses observed in orally sensitized BN rats was later found to be comparable to that seen in food-allergic pediatric patients. Local and systemic immune-mediated effects observed in rats were in accordance with observations from human clinical practice (Knippels, 2000).

Similarly, a study by Batista et al. (2014) demonstrated that mice sensitized to ovalbumin through subcutaneous injection (2 x 10 mg OVA) exhibited food allergy-like symptoms after consuming a 14% OVA diet for seven days. These symptoms, including increased anti-OVA IgE, intestinal inflammation, and significant metabolic alterations, notably improved after 7 additional days of OVA ingestion, suggesting that prolonged OVA ingestion may mitigate the metabolic effects of experimental egg allergy.

It is important to note that food protein allergenicity largely depends on their resistance to heat and digestive enzymes, as food allergens must remain stable during cooking and digestion to reach the intestinal mucosa, where absorption and sensitization can occur (Astwood, 1996). OVA (Gal d 2) is known to be heat-labile, undergoing structural changes during cooking that reduce its immunogenic potential. This likely explains why some individuals with egg allergies can tolerate cooked and baked egg products (Caubet, 2011) a finding extensively documented in interventional studies involving sensitized patients (Upton, 2023; Lemon-Mulé, 2008). In contrast, ovomucoid (Gal d 1) is resistant to

heat and digestive enzymes, and its allergenicity remains largely unaffected by cooking (Dhanapala, 2015).

Given that the primary and secondary structures of NOVA, along with its thermostability, have been shown to be equivalent to those of native OVA through experimental data (**Section 2.1.3**), it is expected that NOVA from fermentation by *T. reesei* will exhibit an allergenic profile similar to that of OVA and egg white protein, as described above.

Furthermore, given that most products containing NOVA are intended to be cooked, the use of OVA as an ingredient, rather than whole eggs or egg-white protein, could somewhat reduce allergenicity concerns associated with egg allergens. However, in compliance with FALCPA (Public Law 108-282) and the FD&C Act section 403(w), all food products containing NOVA must still be labeled as containing egg. Labels will clearly indicate the presence of egg allergen, specifying the source ([FDA, 2022b](#)), to assist consumers in avoiding allergens.

6.2.4. Toxicity of ovalbumin

Given the long history of egg consumption worldwide, research has primarily focused on the allergenicity of OVA, with fewer studies addressing its toxicity. Generally, OVA is considered non-toxic and safe for consumption as part of a regular diet, with the primary concerns related to allergenicity rather than inherent toxicity. Most of the studies conducted with OVA focus on the benefits of its consumption instead.

In a study by Nicklin and Miller (1987), researchers explored the impact of early dietary exposure to OVA on the development of OVA-specific antibody responses in Piebald Virol Glaxo (PVG) rats. Pregnant and lactating rats were fed a diet containing 21% OVA, and their offspring were later challenged with OVA either parenterally (10 µg OVA) or orally (200 mg OVA). The study found that early exposure during gestation and the post-partum period led to increased biliary anti-OVA antibody activity while maintaining systemic tolerance following parenteral OVA challenge. This suggests that early dietary antigen exposure plays a crucial role in shaping immune responses later in life. Although not designed as a toxicity study, the absence of reported deaths (both control and treatment groups) and the seemingly well-tolerated prolonged OVA exposure in both adult and neonatal rats indirectly indicates a lack of developmental toxicity.

In a human clinical context, other studies have explored the potential effects of OVA and its hydrolysates on human health. Lucey et al. (2019) assessed the cardiovascular effects of OVA-derived protein hydrolysates in a double-blind, placebo-controlled crossover trial involving adults aged 50–70 with elevated systolic blood pressure. Despite promising animal and *in vitro* data, the study found no evidence that the hydrolysate reduced blood pressure or modified cardiovascular risk factors. However, it is worth noting that the treatment did not lead to any adverse effects, reinforcing the general safety of egg-derived proteins in human diets.

Further supporting the safety of egg-derived proteins, Nijssen et al. (2023) assessed the effects of egg-derived protein hydrolysates on arterial stiffness and cardiometabolic markers in adults with metabolic syndrome. While the preparation did not significantly impact arterial stiffness, it did modestly reduce fasting pulse pressure and acutely lowered central augmentation index and diastolic blood pressure, demonstrating some cardiovascular benefits. As in Lucey et al. (2029), no adverse effects were observed, further reinforcing the safety of egg protein hydrolysates in adults with metabolic syndrome.

While these studies suggest positive health effects, egg white protein has been linked to a reduction in biotin absorption, although this is not attributed to OVA itself. The interaction between avidin in raw egg white and biotin, which renders the vitamin non-absorbable, is a well-known phenomenon, though biotin deficiency from egg consumption remains rare (Saleem & Soos, 2023, Wallig & Keenan, 2013). These concerns, however, are primarily relevant to the consumption of raw egg whites and do not extend to the typical dietary intake of cooked egg proteins.

These studies collectively suggest that OVA and egg white protein are safe under normal dietary conditions, with no significant toxicity reported outside of allergenic responses.

6.3. Safety of ovalbumin from fermentation by *T. reesei*

6.3.1. Safety of the parental strain

Trichoderma reesei, the anamorph of *Hypocrea jecorina*, is a key microorganism in industrial enzyme production, widely recognized for its ability to secrete large quantities of cellulolytic enzymes. Initially isolated in 1944, the original strain QM6a and its derivatives have undergone extensive research and development, particularly in the context of cellulase production. The fungal species has been employed industrially since the early 1970s and serves as a model organism for lignocellulose degradation and fungal protein secretion (Paloheimo, 2016).

With the growing demand for alternative fuels in the 1970s, *T. reesei*'s capacity to hydrolyze cellulose into fermentable sugars became increasingly valuable. Successive advancements through random mutagenesis and, later, genetic engineering have significantly enhanced its enzyme productivity. Since the 1990s, *T. reesei* has evolved into a major platform for producing both homologous and heterologous enzymes, widely used in various industries, including feed, textiles, biopharma, and food (Rantasalo, 2019; Landowski, 2016; Paloheimo, 2016).

T. reesei is now the most common host for producing lignocellulose-degrading enzymes globally, valued for its suitability in large-scale cultivation and its long history of safe use (Aro, 2023). This organism is extensively used in food applications, with commercial enzyme products including amylase, beta-glucanase, pectinase, and xylanase, which are used in the bakery, dairy, brewery, and starch industries (Paloheimo, 2016; Lopes Ferreira, 2014).

T. reesei's exceptional secretion capabilities have made it a key platform for producing numerous GRAS proteins and enzymes. As of December 06, 2024, the FDA's GRAS Notice Inventory lists 32 notices for enzymes and proteins derived from *T. reesei* (including genetically modified strains), the earliest dating back to 1999. Of these, 24 have received 'no questions' letters, three are pending, and five were discontinued at the notifier's request. This demonstrates the FDA's confidence in the safety of recombinant proteins produced using *T. reesei* for a wide range of food applications. An overview of all GRAS notices involving *T. reesei* as a production organism (up until December 2024) is provided in Table 14.

Table 14. Overview of GRAS Notices involving proteins from <i>T. reesei</i>				
GRN No.	Year of closure	Substance	Intended use	FDA's letter
32	2000	Pectin lyase derived from <i>T. reesei</i> carrying a gene encoding pectin lyase from <i>Aspergillus niger</i>	For use in the preparation of fruit and vegetable juices, purees, and concentrates as a processing aid at levels ranging from 5 to 100 parts per million	FDA has no questions
230	2008	Chymosin enzyme preparation from <i>Trichoderma reesei</i> expressing the bovine prochymosin B gene	As an enzyme for milk coagulation during cheese making	FDA has no questions
315	2010	Transglucosidase enzyme preparation from <i>Trichoderma reesei</i> expressing the gene encoding transglucosidase from <i>Aspergillus niger</i>	Use in the production of isomalto-oligosaccharide syrups from starch and potable ethanol from molasses	FDA has no questions
333	2010	Acid fungal protease enzyme preparation from <i>Trichoderma reesei</i> expressing the gene encoding acid fungal protease from <i>T. reesei</i>	Use as an enzyme in grain processing (corn steeping), alcoholic beverage (potable alcohol) manufacture, and degumming of membranes during orange juice manufacturing	FDA has no questions

Table 14. Overview of GRAS Notices involving proteins from *T. reesei*

GRN No.	Year of closure	Substance	Intended use	FDA's letter
372	2011	Glucoamylase enzyme preparation from <i>Trichoderma reesei</i> expressing the glucoamylase gene from <i>T. reesei</i> (glucoamylase enzyme preparation)	Use as an enzyme in grain processing, including the manufacture of corn sweeteners such as high fructose corn syrup, brewing, potable alcohol manufacture, and lactic acid production	FDA has no questions
490	2014	Phospholipase A2 enzyme preparation from <i>Trichoderma reesei</i>	As an enzyme to degum oils	At the notifier's request, FDA ceased to evaluate this notice
524	2014	Phospholipase A2 enzyme preparation from <i>Trichoderma reesei</i> carrying a phospholipase A2 gene from <i>Aspergillus nishimurae</i>	For use as a processing aid in edible oil refining.	FDA has no questions
557	2015	Polygalacturonase from <i>Aspergillus tubingensis</i> expressed in <i>Trichoderma reesei</i>	For use in fruit and vegetable processing, wine, coffee and flavoring production, and grain treatment.	FDA has no questions
558	2015	Pectin esterase from <i>Aspergillus tubingensis</i> expressed in <i>Trichoderma reesei</i>	For use in fruit and vegetable processing, wine, coffee and flavoring production, and grain treatment at GMP use levels.	FDA has no questions
566	2015	Mannanase from <i>Trichoderma reesei</i>	For use in fruit and vegetable processing, oil processing, and coffee production at levels up to 26 milligrams TOS per kilogram of raw material.	FDA has no questions
567	2015	Xylanase from <i>Trichoderma reesei</i>	For use as a processing aid in brewing, potable alcohol, cereal beverage manufacture and bakery applications.	FDA has no questions
628	2016	endo-1,4-beta-xylanase from <i>Trichoderma reesei</i>	For use as a processing aid in the manufacture of bakery products, and potable alcohol.	FDA has no questions
631	2016	Triacylglycerol lipase from <i>Fusarium oxysporum</i> produced in <i>Trichoderma reesei</i>	For use in the production of baked goods not to exceed 10 mg TOS/kg	FDA has no questions
653	2016	Lysophospholipase from <i>Aspergillus nishimurae</i> expressed in <i>Trichoderma reesei</i>	For use as a processing aid in starch processing.	FDA has no questions
675	2017	Xylanase enzyme preparation produced by <i>Trichoderma reesei</i> carrying a xylanase gene from <i>Talaromyces leycettanus</i>	Intended for use as an enzyme in brewing, cereal beverage processing, potable alcohol processing, bakery applications, and grain processing from cereal grains such as corn, wheat, barley, and oats, at levels up to 46.1 milligrams Total Organic Solids per kilogram raw material.	FDA has no questions

Table 14. Overview of GRAS Notices involving proteins from *T. reesei*

GRN No.	Year of closure	Substance	Intended use	FDA's letter
680	2017	Alpha-L-arabinofuranosidase from <i>Talaromyces pinophilus</i> produced by <i>Trichoderma reesei</i>	For use as a processing aid to increase the yield of starch and gluten meal during the corn wet milling process at minimum levels necessary to achieve the desired effect and according to good manufacturing practices.	FDA has no questions
707	2017	Glucose oxidase from <i>Penicillium</i> produced in <i>Trichoderma reesei</i>	For use in baking processes at the minimum amount necessary to achieve the intended technical effect.	FDA has no questions
727	2018	Trehalase from <i>Trichoderma reesei</i> produced in <i>Trichoderma reesei</i>	For use as a processing aid during the fermentation process to produce organic acids, monosodium glutamate, and potable alcohol at 100 mg product/kg starch (36.72 mg TOS/kg starch).	FDA has no questions
756	2018	Endo-1,4-beta-glucanase from <i>Trichoderma reesei</i>	For use in the production and processing of potable alcohol from grain at levels consistent with current good manufacturing practices.	FDA has no questions
808	2019	Lipase from <i>Aspergillus tubingensis</i> produced in <i>Trichoderma reesei</i>	For use as a processing aid in baking at 21.2 mg TOS/kg, in brewing processes and cereal manufacturing at 52.2 mg TOS/kg, in pasta production at 2.03 mg TOS/kg, and in potable alcohol production at 3.6 mg TOS/kg.	FDA has no questions
853	2020	Lysozyme produced by <i>Trichoderma reesei</i>	For use as an enzyme in the production of ingredients produced by microbial fermentation such as xanthan gum, gellan gum, and yeast extracts to assist in the removal of cellular debris.	FDA has no questions
863	2020	β -lactoglobulin produced by <i>Trichoderma reesei</i>	Intended for use as a substitute for dairy and plant derived proteins in foods that currently use protein from milk or plants and a source of dietary protein.	FDA has no questions
891	2020	Cellulase produced by <i>Trichoderma reesei</i>	Intended for use as an enzyme in fruit and vegetable processing, wine processing, starch and grain processing, brewing and other cereal based beverages and potable/fuel alcohol.	FDA has no questions
964	2021	Lysophospholipase enzyme preparation from <i>Aspergillus niger</i> produced by <i>Trichoderma reesei</i>	Intended for use as an enzyme in carbohydrate processing (e.g., high fructose corn syrup) at 24.16 mg Total Organic Solids/kg raw material.	FDA has no questions

Table 14. Overview of GRAS Notices involving proteins from *T. reesei*

GRN No.	Year of closure	Substance	Intended use	FDA's letter
981	2022	Sterol esterase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene encoding sterol esterase from <i>Melanocarpus albomyces</i>	Intended for use as an enzyme in the manufacture of baked goods at up to 10 mg Total Organic solids per kg flour.	FDA has no questions
1048	Pending	Aminopeptidase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene encoding an aminopeptidase from <i>Aspergillus clavatus</i>	Intended for use as an enzyme at up to 2.1g Total Organic Solids (TOS)/kg protein in protein processing, at up to 4.3 g TOS/kg yeast in yeast processing, and at up to 7 g TOS/kg protein in flavoring production.	Pending
1173	Pending	Invertase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene for invertase from <i>Aspergillus niger</i>	Intended for use as an enzyme at up to 7 mg Total Organic Solids (TOS)/kg sucrose in the production of short chain fructooligosaccharides and up to 14 mg TOS/kg raw material in fruit and vegetable processing.	Pending
1102	2024	Invertase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene for invertase from <i>Aspergillus niger</i>	Intended for use as an enzyme at up to 7 mg Total Organic Solids (TOS)/kg sucrose in the production of short chain fructooligosaccharides and up to 14 mg TOS/raw material in fruit and vegetable processing.	At the notifier's request, FDA ceased to evaluate this notice
1110	2024	Polygalacturonase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene encoding a polygalacturonase from <i>Aspergillus kawachii</i>	Intended for use as an enzyme at up to 5 mg total organic solids (TOS)/kg raw material in the processing of fruits, vegetables, and coffee, and in the production of flavorings and wine.	At the notifier's request, FDA ceased to evaluate this notice
1121	2024	Fructanase enzyme preparation produced by <i>Trichoderma reesei</i> expressing the gene encoding fructanase from <i>Lactobacillus sp.</i>	Intended for use as an enzyme at up to 12 mg total organic solids (TOS)/kg raw material in baking.	At the notifier's request, FDA ceased to evaluate this notice
1187	Pending	Polygalacturonase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene encoding polygalacturonase from <i>Aspergillus luchuensis mut kawachii</i>	Intended for use as an enzyme at up to 5 mg total organic solids (TOS)/kg raw material in the processing of fruit, vegetables, and coffee, and in the production of flavorings and wine.	Pending
1150	2024	Triacylglycerol lipase enzyme preparation produced by <i>Trichoderma reesei</i> expressing a gene for triacylglycerol lipase from <i>Thermomyces lanuginosis</i>	Intended for use as an enzyme at up to 4 mg Total Organic Solids/kg cereal in the production of baked goods and other cereal based products.	At the notifier's request, FDA ceased to evaluate this notice

All industrial *T. reesei* strains can be traced back to the QM6a isolate (Paloheimo, 2016), which is non-pathogenic and does not produce fungal toxins or antibiotics under traditional enzyme production

conditions (Nevalainen, 1994). *T. reesei* is classified as Biosafety Level 1 (BSL1) by the American Type Culture Collection (ATCC) based on a risk assessment guided by the *Biosafety in Microbiological and Biomedical Laboratories (BMBL)* guideline from the US Department of Human Health and Human Services (ATCC, nd). BSL1 microorganisms are not known to consistently cause disease in healthy adults (US CDC, nd).

In 2012, the US Environmental Protection Agency (EPA) proposed a rule to add *T. reesei* strain QM6a and its derivatives to the list of recipient microorganisms in 40 CFR 725.420 that can be exempted from full notification and reporting procedures under the Toxic Substances Control Act (TSCA) for new microorganisms that are being manufactured into commerce (EPA 2012). The conditions for this exemption are only for the use in submerged industrial fermentation for enzyme (protein) production. The final rule was issued and became effective in 2020 (EPA, 2020), arguing that *T. reesei* “will not present an unreasonable risk of injury to health or the environment when used as a recipient microorganism provided that certain criteria for the introduced genetic material and the physical containment conditions are met”. As a result, *T. reesei* QM6a and its derivatives are now listed in 40 CFR 725.420(k).

To reach this conclusion, EPA conducted a comprehensive risk assessment of *T. reesei* QM6a (and its derivatives) focused on several key areas relevant to its safety, including the identification and classification of the microorganism, evaluation of its relationship to other microorganisms with potential health or environmental risks, and a review of its history of safe commercial use. EPA closely examined potential human health hazards – pathogenicity, toxicity, and mycotoxin production – and concluded that “*T. reesei* QM6a is not known to possess any virulence factors associated with colonization or disease such as adherence factors, penetration factors, necrotic factors, toxins, or the ability to grow at human body temperature, 37 °C. There are no reports in the literature on infection in healthy humans by *T. reesei* QM6a. There are no reports of harmful effects associated with the use of or exposure to *T. reesei* QM6a strains given decades of commercial use for enzyme production. The body of evidence indicates that *T. reesei* QM6a does not pose concerns regarding human pathogenicity” (EPA, 2020).

Numerous other publications support the safety of *T. reesei*, particularly strains derived from QM6a. For instance, a comprehensive risk assessment conducted by Health Canada (Health Canada, 2018) evaluated *T. reesei* using both published and unpublished data on the environmental and human-health risks of the microorganism. The review, which involved extensive literature searches covering terms related to *T. reesei* and its synonym *Hypocrea jecorina*, found no evidence suggesting that *T. reesei* is a human pathogen. Although closely related species, such as *T. longibrachiatum*, have been associated with infections in severely immunocompromised individuals, *T. reesei* itself has not been shown to pose similar risks. Experimental studies in immunosuppressed animals (mice, guinea pigs, and rabbits) have shown adverse effects only after high intravenous doses of viable spores (1.0×10^5 spores or higher), which far exceed typical exposure levels. *T. reesei* strains derived from QM6a are unlikely to cause infections in healthy or debilitated humans. Furthermore, these strains are also sensitive to commonly used antifungal drugs, which would be effective in the unlikely event of an infection.

The only known health concern associated with *T. reesei* QM6a is its ability to produce a peptide antibiotic called paracelsin, which has been demonstrated toxic in numerous animal studies (Health Canada, 2018, EPA, 2020). However, paracelsin has not been detected in enzyme products from *T. reesei* QM6a cultured under standard industrial fermentation conditions. Additionally, extensive toxicity studies on these enzyme products have shown no toxicity to laboratory animals. As such, the production of paracelsin is not expected to pose significant risks when the *T. reesei* is subjected to standard fermentation methods like those employed by Onego Bio, as outlined in 40 CFR part 725.3 (EPA, 2020).

This conclusion is supported by a number of GRAS notices (e.g., GRN 964, GRN 727, GRN 707, GRN 567, GRN 372, GRN 333, GRN 315, GRN 230), in which enzyme preparations from genetically modified *T. reesei* strains (all derived from QM6a) produced via submerged fermentation have undergone a range of *in vitro* and *in vivo* studies. These studies, including the bacterial reverse mutation assay (Ames test), chromosomal aberration assay in mammalian cells, and 14- to 90-day animal toxicity studies, have found the recombinant enzymes to be non-toxic.

Considering its proven safety, *T. reesei* complies with requirements for use at the lowest containment level at large scale according to Good Industrial Large-Scale Practice (GILSP), as defined by the OECD (OECD, 1998). The strain used to produce NOVA, *T. reesei* TR112, also meets the criteria for a safe

production microorganism established by Pariza & Foster (1983) and Pariza & Johnson (2001), as detailed below.

6.3.2. Safety of the production strain - Pariza and Johnson decision tree

The publications by Pariza & Foster (1983), Pariza & Johnson (2001), and Pariza et al. (2015), are foundational references in the safety assessment of recombinant enzymes (and proteins) intended for food use, providing widely recognized guidelines essential for GRAS determination. Both emphasize the safety of the production organism as the primary consideration determining the safety of recombinant food proteins. Ideal strains should be non-toxigenic, non-pathogenic, and have a history of safe use in food enzyme manufacture. All these requirements are met by *T. reesei* as discussed above.

For food enzymes and proteins produced in microbial GM platforms, strain lineages require thorough characterization of the GM host organism, safety evaluation of the introduced DNA, and assurance that the genetic modification methods employed are appropriate for food use. Pariza & Johnson (2001) have devised a decision tree for the effective safety assessment of such food enzymes, which can be applied to Onego's ovalbumin from fermentation by *T. reesei*. Following this guidance, Onego Bio has evaluated the safety of NOVA and its production platform, *T. reesei* TR112, as detailed in Table 15.

According to the Pariza et al., decision trees (Pariza, 2015; Pariza and Johnson, 2001), the genetic modification methods employed by Onego Bio on *T. reesei* QM6a to construct strain TR112 are appropriate for food use and NOVA produced in this platform is safe for human consumption, provided that it is substantially equivalent to its native counterpart, and it is produced under cGMP (Pariza & Foster, 1983). The substantial equivalence of NOVA to native OVA has been determined as presented in **Section 2.1.3** and is discussed below in **Section 6.3.3**.

Based on these results, no toxicological studies are needed to further justify the safety of the production platform used to manufacture Onego's NOVA. As concluded during the detailed assessment by the EPA outlined above, the safety of *T. reesei* is well-supported by extensive toxicological evaluations. These studies, including those conducted on both healthy and immunocompromised animal models, consistently demonstrate that *T. reesei* is non-pathogenic and does not pose a risk to human health (EPA, 2020). To our knowledge, based on a literature search in Google Scholar, PubMed, and ScienceDirect in October 2024, that is still the case. Search terms included "*Trichoderma reesei*", "*T. reesei*", "*T. reesei* AND toxicological studies", "*T. reesei* AND animal studies", "*T. reesei* AND toxigenicity", "*T. reesei* AND pathogenicity", "*T. reesei* AND human safety", "*T. reesei* AND food enzyme production".

Table 15. Decision tree for evaluating the safety of ovalbumin from fermentation with <i>T. reesei</i> strain TR112. Adapted from Pariza & Johnson (2001)				
Question		Answer	Details	Action
1	Is the production strain genetically modified?	Yes	The production strain to manufacture NOVA is <i>T. reesei</i> strain TR112, a genetically modified strain derived from <i>T. reesei</i> QM6a. For details on host and production strains, see Section 2.2 .	Go to 2
2	Is the production strain modified using rDNA techniques?	Yes	<i>T. reesei</i> strain TR112 was constructed by means of well-known genetic engineering techniques (Section 2.2.2).	Go to 3
3	Issues relating to the introduced DNA are addressed in 3a-3e			
3a	Do the expressed enzyme [protein] product(s) which are encoded by the introduced DNA have a history of safe use in food?	Yes	OVA has a long history of safe use in food, as the major component of chicken egg white (Section 6.2). In this GRAS determination, NOVA has been determined substantially equivalent to native OVA by comparing data on amino acid sequence, molecular weight, glycosylation pattern, secondary structure, digestibility, and thermal stability (Section 2.1.3).	Go to 3c
3c	Is the test article free of transferable antibiotic resistance gene DNA and genetic elements encoding virulence factors and/or toxins associated with pathogenicity?	Yes	As described in Section 2.2.2 , the expression cassette used in the construction of <i>T. reesei</i> strain TR112 does not contain any AMR genes or mobile genetic elements, which was confirmed through a bioinformatics assessment in which the WGS was interrogated for AMR genes and virulence factors. No antimicrobial resistance, virulence genes or plasmids were found in the <i>T. reesei</i> TR112 genome assembly.	Go to 3e
3e	Is all other introduced DNA well characterized and free of attributes that would render it unsafe for constructing microorganisms to be used to produce food-grade products?	Yes	As detailed in Section 2.2.2 , the DNA inserted in the host strain QM6a is well characterized and free of unsafe attributes. The expression cassette contained the gene sequence for OVA from <i>Gallus gallus</i> , codon-optimized for efficient expression (preserving original amino acid sequence), and DNA homologous to the desired integration site. A bioinformatics assessment revealed that the <i>T. reesei</i> TR112 genome assembly contains only one copy of the OVA gene. The insertion site was also confirmed.	Go to 4
4	Is the introduced DNA randomly integrated into the chromosome?	Yes	WGS was used to confirm the correct insertion of the expression cassette. The insertion copy number was confirmed by qPCR.	Go to 5
5	Is the production strain sufficiently well characterized so that one may reasonably conclude that unintended pleiotropic effects which may result in the synthesis of toxins or other unsafe metabolites will not arise due to the genetic modification method that was employed?	Yes	The identity of the production strain was confirmed by WGS, and its genetic stability was demonstrated through consistent OVA expression levels in multi-generational studies (Section 2.2.3). A bioinformatics assessment was conducted on <i>T. reesei</i> TR112 to screen for genes and gene clusters related to secondary metabolite biosynthesis. Although the type and variety of the genes and gene clusters found do not directly indicate the production of toxic secondary metabolites, the results suggested that <i>T. reesei</i> TR112, like all fungal species, has the potential to produce mycotoxins under specific conditions. Analytical data on mycotoxin determination on 3 non-consecutive production batches of NOVA (Section 2.5) confirmed that no mycotoxins of food safety concern are produced by <i>T. reesei</i> TR112 under the manufacturing conditions employed by Onego Bio.	Go to 6
6	Is the production strain derived from a safe lineage, as previously demonstrated by repeated assessment via this evaluation procedure?	Yes	The safety of the host strain, <i>T. reesei</i> QM6, has been evaluated using the Pariza & Johnson decision tree in multiple GRAS notices that have received "no questions" letters from the FDA.	ACCEPT TEST ARTICLE

6.3.3. Safety of NOVA

6.3.3.1. Substantial equivalence to native ovalbumin

As outlined in **Section 2.1.3**, Onego Bio has conducted a comprehensive structural characterization of ovalbumin derived from fermentation by *T. reesei*, demonstrating its substantial equivalence to the native chicken ovalbumin, which has a well-established history of use in the human diet as the main component of chicken egg white.

The amino acid sequence of NOVA has been confirmed to be 100% identical to that of native chicken OVA (**Section 2.1.3.1**), with only minor differences in amino acid composition (Table 8, Section 2.5) which were expected and can be attributed to the different production organisms. Notably, Onego Bio has intentionally designed the expression cassette to match the amino acid sequence of mature chicken OVA, ensuring that the ovalbumin produced by *T. reesei* corresponds exactly to the OVA molecule commonly consumed as part of a regular diet from eggs and egg-containing products. Furthermore, experimental analysis confirmed that the molecular weight and secondary structure of NOVA are equivalent to native OVA, as demonstrated by SDS-PAGE and CD spectroscopy (**Sections 2.1.3.3 and 2.1.3.4**).

The main structural difference between NOVA and its native counterpart lies in the post-translational modification (PTM) pattern, which was expected due to differences in the biosynthetic machinery of *T. reesei* compared to that of chickens. Native ovalbumin is phosphorylated, with phosphate groups at Ser-68 and Ser-344 (Harvey, 2000), whereas NOVA lacks phosphorylation. Although *T. reesei* has been reported to perform this PTM (Ramos-Pedersoli, 2021; Rodriguez-Iglesias and Schmoll, 2019) its phosphorylation capabilities are limited compared to those of higher organisms.

Additionally, NOVA's glycosylation pattern was found to differ slightly from that of native OVA. LC-MS glycan analysis revealed the presence of mannose-based N-glycans with chains ranging from 5 to 11 mannose units, linked to the polypeptide chain by GlcNAc residues (**Section 2.1.3.2**). This glycosylation pattern is consistent with proteins produced in *T. reesei* (Kolaczowski et al., 2020; Beckham et al., 2012), and while it is not identical to that of native ovalbumin, it remains highly similar - both NOVA and native OVA are primarily glycosylated with mannose-based structures and share one predominate glycoform (Man)₅(GlcNAc)₂, which is the main N-glycan on NOVA. The overall difference in glycosylation is consistent with what is expected from fungal expression systems, which typically produce glycosylation patterns that are closer to those of higher eukaryotes than bacterial or yeast systems (Garvey, 2022; Su, 2012). However, as these differences could introduce food safety concerns, the stability of NOVA was further assessed.

Despite minor differences in PTMs, the stability, function, and safety of ovalbumin from fermentation by *T. reesei* were shown to be unaffected, as demonstrated by thermal stability and digestibility comparisons with its native counterpart (**Section 2.1.3.5**). The thermostability of NOVA was assessed by measuring its melting temperature (T_m), which indicates the protein's ability to maintain its three-dimensional structure under heat stress. NOVA's T_m was found to be equivalent to that of native OVA, suggesting that NOVA will perform similarly under food processing conditions, such as cooking or other thermal treatments, thereby ensuring its safety in food products.

In addition, protein digestibility data (PDCAAS) confirmed that NOVA is digested in a manner comparable to native OVA, further supporting their substantial equivalence. The similarity in both thermal stability and digestibility demonstrates that NOVA provides the same nutritional benefits, particularly in terms of essential amino acids, without altering its safety or bioactivity.

Based on the comprehensive structural characterization, thermal stability, and digestibility data, it can be concluded that NOVA produced by *T. reesei* is substantially equivalent to native chicken OVA. The minor differences in phosphorylation and glycosylation do not affect the stability, digestibility, or nutritional properties of NOVA, confirming its suitability for human consumption and ensuring that NOVA will provide the same nutritional benefits as native OVA without posing any additional safety concerns.

6.3.3.2. Product specifications and characterization

The batch analysis of NOVA produced through fermentation by *T. reesei* (**Section 2.5**), conducted in three non-consecutive production lots, confirms that the product is well-characterized and meets the quality specifications outlined in **Section 2.4**.

NOVA does not contain microbiological and chemical impurities above limits of food safety concern, including heavy metals and mycotoxins. The product is also confirmed free of residual viable cells and recombinant DNA from the production organism. NOVA contains a protein level of at least 75%, with a minimum of 80% of the total protein being Onego's recombinant ovalbumin, and a maximum of 15% originating from the host strain. The consistency of batch analysis results across all lots demonstrates the robustness of the manufacturing process, which further supports the safety of the final product.

6.3.3.3. Dietary exposure to protein and ovalbumin

As described in **Section 3**, the proposed uses of Onego's recombinant ovalbumin are substitutional for existing uses of egg and egg white and are not expected to increase the overall dietary protein consumption of the general population.

The EDI of NOVA for the general US population is 0.32 g/kg bw/day at the mean intake levels (0.74 g/kg bw/day at the 90th percentile). These values fall below the Acceptable Macronutrient Distribution Range (AMDR) for protein, which is 1.05 to 3.67 g/kg bw/day as recommended in the US by the Food and Nutrition Board (FNB) of the Institute of Medicine (Wolfe, 2017). In terms of grams per day, the intake NOVA at its intended conditions of use is also below the FDA's Daily Reference Value (DRV) of 50 g of protein per day for a 2,000-calorie diet (21 CFR § 101.9(c)(9)), with an EDI of 18.89 g/day at mean intake levels (39.17 g/day at the 90th percentile). Furthermore, exposure to protein from NOVA will be even lower, as the product contains at least 75% protein, of which at least 80% is ovalbumin as detailed in **Section 2.5**. It is also worth mentioning that these exposure values represent a worst-case scenario, assuming maximum use levels across all proposed food categories, and are therefore an overestimation.

NOVA is intended to replace fresh or powdered whole eggs and egg white protein powder in the selected food categories and will be substitutional for animal-derived ovalbumin in the diet. Given the proposed uses for ovalbumin from fermentation by *T. reesei*, the cumulative exposure to ovalbumin (both from dietary exposure from eggs and egg-containing products and from the intended uses of NOVA) is not expected to increase in the general population. Consumer exposure to dietary protein is not expected to increase either, as the majority of the population's protein intake comes from unprocessed foods, including meat, poultry, fish, and legumes, and will likely continue to do so.

Ovalbumin has a long history of safe consumption as a dietary protein, as detailed in **Section 6.2**. Given the substantial equivalence of NOVA to its native counterpart and consistent compliance to quality specifications of the final product, we anticipate that this established safety profile extends to the consumption of NOVA under its proposed conditions of use. The proposed uses and levels of use of NOVA are not expected to increase exposure or pose unintended safety risks to consumers.

6.3.4. Safety of the production process

The production strain and manufacturing process for Onego's NOVA from fermentation by *T. reesei* are not associated with the generation of any known toxicants.

Ovalbumin from precision fermentation with *T. reesei* is produced by submerged fermentation of a *T. reesei* strain that has been genetically modified. In line with the Pariza-and-Johnson decision tree (**Section 6.3.2**) for the safety evaluation of microbially-produced proteins, the genetic modification methods employed by Onego Bio are appropriate for food use. *T. reesei* has a long history of safe use in food enzyme manufacture and has been widely reported as non-toxigenic and non-pathogenic to humans, as discussed in **Section 6.3.1**.

NOVA is manufactured in facilities that meet US regulations for food production by adhering to cGMPs with the implementation of a Food Safety Plan in compliance with 21 CFR Part 117, and the product meets predetermined batch-release specifications (see **Section 2.4**).

All raw materials and processing aids used in the manufacturing process of ovalbumin from fermentation with *T. reesei* are of high purity and quality, regulated and suitable for their intended use and are food-grade, in accordance with 21 CFR § 170.230(c). All food contact materials used in compliance with the FD&C Act and any FDA implementation.

No antibiotics are used in the manufacturing process of ovalbumin from fermentation with *T. reesei*.

6.3.5. Allergenicity of NOVA

None of the raw materials or the production organism used to manufacture NOVA are (or are derived from) major food allergens as defined by the Food Allergen Labeling and Consumer Protection Act (FALCPA) of 2004 and the Food Allergy Safety, Treatment, Education and Research (FASTER) Act of 2021.

Besides ovalbumin, which is the main constituent of the protein fraction in NOVA (85.8% to 90.9%) and whose allergenicity is discussed in **Section 6.2.3**, no significant allergens are expected in Onego's finished NOVA preparation. To address allergenicity concerns, all food products containing NOVA will be clearly labeled to indicate the presence of egg, in accordance with the requirements of the Food Allergen Labeling and Consumer Protection Act (FALCPA) (Public Law 108-282) and Section 403 (w) of the FD&C Act.

Allergenicity concerns are not expected to arise from the production platform of NOVA. *T. reesei* has a well-established history of safe use in food enzyme production and is not reported in the scientific literature as a known producer of allergens. As detailed in **Section 6.3.1**, a review of the FDA GRAS Notice Inventory identified 32 GRAS notices for proteins produced using *T. reesei*. None of the 29 publicly available notices, including those withdrawn by the applicant, report allergenicity concerns related to the production organism. Furthermore, no scientific publications or reviews have documented widespread allergenic reactions resulting from the ingestion of proteins derived from fermentation by *T. reesei*.

As further discussed in **Section 6.3.1**, *T. reesei* is widely regarded as a safe organism for food enzyme production. While repeated occupational exposure to commercial enzyme preparations (e.g., cellulases) derived from *T. reesei* or other *Trichoderma* species has occasionally been associated with allergic responses such as dermatitis, rhinitis, and asthma due to handling or dust inhalation, such cases are rare (Health Canada, 2018). Importantly, the likelihood of these occupational aeroallergen or contact allergen sensitivities leading to food-related allergic reactions is considered extremely low.

Nonetheless, Onego Bio evaluated the allergenic potential of residual *T. reesei* proteins in the final ovalbumin preparation by analyzing the peptides and proteins from the production organism present in NOVA. To this end, the protein fraction in three non-consecutive production batches (ID-7695, ID-6309, and ID-8711) was characterized through quantitative proteomic analysis (LC-MS/MS), as described in **Section 2.5.2**. The analysis confirmed that ovalbumin was the predominant protein in all batches. In addition to ovalbumin, 541 peptides derived from *T. reesei* were identified in NOVA.

All *T. reesei* amino acid sequences detected at levels exceeding 0.01% (290 sequences) were further assessed for sequence homology using the AllergenOnline database (version 22; <http://www.allergenonline.org>), maintained by the Food Allergy Research and Resource Program (FARRP) at the University of Nebraska. This analysis aimed to identify potential homology between the identified *T. reesei* peptides and known allergens, to assess the risk of allergenic cross-reactivity that could potentially trigger reactions in sensitized individuals consuming NOVA.

Sequence homology testing took place in two steps. In the first step, we evaluated potential cross reactivity with documented allergenic proteins using an 80-amino-acid sliding window search with a 35% homology threshold, as recommended by the Codex Alimentarius Commission (FAO, 2001). It is worth noting, however, that sequences with > 35% identity over an 80-amino-acid window are common among highly conserved proteins and this threshold alone is not predictive of allergenicity, often resulting in a high number of false positives (Abdelmoteleb, 2021; EFSA, 2022). The scientific literature indicates that greater sequence or structural conservation (typically > 50% identity) is generally required for allergenic cross-reactivity, while sequences with at least 25% identity over 200 or more amino acids

may only indicate probable homology (Pearson, 1996). Matches with < 50% identity across the full-length sequence are unlikely to result in cross-reactivity, whereas clinically significant IgE cross-reactivity is usually observed for proteins with > 70% identity (Aalberse, 2000; Goodman et al., 2008; Abdelmoteleb et al., 2021; Scaife et al., 2024).

Thus, in the second step, matches showing > 35% identity in the 80-amino acid sliding window alignment were further evaluated using to apply a more realistic indicator of potential cross-reactivity, i.e., identity match across the full sequence combined with small *E*-scores (expectation scores), which indicate likely functional and biologically relevant similarities that could suggest allergy or cross-reactivity (Aalberse, 2000; Goodman et al., 2008; Abdelmoteleb et al., 2021). Only matches with > 50% identity over the full sequence and an *E*-value < 1x10⁻⁷ were included in the present allergenicity assessment.

Using sliding 80-amino acid window searches, 91 of the 290 proteins from *T. reesei* detected at levels above 0.01% met the > 35% identity threshold over at least one window. These 91 proteins were then analyzed for matches with full-length allergen sequences in AllergenOnline, applying a > 50% identity threshold. Among these, 23 proteins were identified as sharing at least 50% identity to an allergen over their full length (Table 16). A total of 138 potential allergens were retrieved (Table 17) and individually assessed for food safety concerns based on information from the [WHO/IUIS Allergen Nomenclature Subcommittee](#) and literature reviews.

Table 16. <i>T. reesei</i> proteins above 0.01% in 3 non-consecutive lots of NOVA, sharing > 50% identity with allergens from the AllergenOnline database					
	JGI protein ID	Protein Description	Abundance in NOVA batches (%)		
			Batch ID-7695	Batch 2 ID-6309	Batch 3 ID-8711
0	---	Ovalbumin	85.80	90.20	90.90
1	123026	Transaldolase (EC 2.2.1.2)	10.81	11.53	8.14
2	124005	Uncharacterized	1.62	2.35	0.85
3	121789	Triosephosphate isomerase (EC 5.3.1.1)	1.11	0.76	0.44
4	52809	Uncharacterized	0.48	0.17	0.15
5	78992	Uncharacterized	0.24	0.05	0.10
6	122920	Endoplasmic reticulum chaperone BiP (EC 3.6.4.10)	0.23	0.12	0.14
7	121820	5-methyltetrahydropteroyltriglutamate-homocysteine S-methyltransferase (EC 2.1.1.14)	0.23	0.02	0.04
8	45098	Uncharacterized	0.16	0.07	-
9	72199	Uncharacterized	-	-	0.11
10	121127	Beta-xylosidase (EC 3.2.1.37)	2.15	0.02	0.05
11	121915	TCTP (Translationally controlled tumor protein)	0.10	-	-
12	121977	Cytoplasmic cyclophilin	0.04	0.002	0.01
13	123859	Uncharacterized	0.03	0.01	0.01
14	78401	Uncharacterized	0.03	0.02	0.03
15	123922	Peptidyl-prolyl isomerase (cyclophilin)	0.03	0.002	-
16	123088	Uncharacterized	0.02	0.02	0.01
17	120568	Enolase	0.02	0.01	-
18	124265	Uncharacterized	0.01	-	-
19	122257	Uncharacterized	0.01	-	0.001
20	119722	Uncharacterized	0.008	0.004	0.073
21	103472	Uncharacterized	0.007	0.005	0.016
22	123029	Copper/zinc superoxide dismutase	0.004	0.0004	0.012
23	61956	Uncharacterized	-	0.012	0.004

Table 17. Allergen matches for *T. reesei* proteins in NOVA with > 50% sequence identity to AllergenOnline entries

	JGI protein ID	Protein length (Nr. of AA)	Matched allergen						
			Matched allergen	Source organism	Length (Nr. of AA)	E-value	Best Percent ID Over 80 AA	Percent ID Over Full Sequence	GenBank Accession
1	123026	324	Transaldolase	<i>Fusarium proliferatum</i>	323	1.1e-119	93.80%	88.20%	AHY02994
				<i>Penicillium chrysogenum</i>	324	2.1e-103	82.50%	75.90%	ADK27483
				<i>Cladosporium cladosporioides</i>	322	2.3e-102	82.50%	76.10%	ADK47394
2	124005	129	Thioredoxin-like protein	<i>Fusarium culmorum</i>	120	5.4e-038	77.52%	72.50%	AAL79931
3	121789	248	Triosephosphate isomerase	<i>Triticum aestivum</i>	246	4.5e-061	73.80%	58.50%	CAC14917
				<i>Procambarus clarkii</i>	242	2.4e-056	72.50%	58.30%	AEB54655
				<i>Scylla paramamosain</i>	242	5.3e-056	71.30%	57.00%	APP94292
				<i>Crangon crangon</i>	243	4.2e-057	68.80%	57.60%	ACR43476
			Der f 25 allergen	<i>Dermatophagoides farinae</i>	244	3.9e-058	62.50%	55.30%	AIO08860
			Triosephosphate isomerase	<i>Pangasianodon hypophthalmus</i>	242	2.8e-056	62.50%	54.50%	XP_026795867
				<i>Dermatophagoides farinae</i>	244	3.6e-057	61.30%	56.60%	AGC56216
				<i>Salmo salar</i>	244	3.9e-058	61.30%	55.70%	ACM09737
4	52809	743	Putative heat shock protein	<i>Malassezia sympodialis</i>	756	5.2e-126	67.55%	50.40%	CAD20981
			Allergen	<i>Malassezia sympodialis</i> ATCC 42132	756	2.0e-126	67.55%	50.40%	CCU97864
5	78992	1049	Disulfide-isomerase	<i>Alternaria alternata</i>	379	2.0e-102	78.70%	59.60%	Q00002
6	122920	664	Heat shock protein	<i>Aedes aegypti</i>	654	1.8e-189	87.50%	67.70%	ABF18258
				<i>Penicillium citrinum</i>	488	9.8e-126	82.50%	62.10%	Q92260
				<i>Tyrophagus putrescentiae</i>	624	2.6e-168	82.50%	62.20%	AOD75395
				Allergen Der p 28	<i>Dermatophagoides pteronyssinus</i>	608	8.9e-166	80.20%	62.30%
			<i>Dermatophagoides farinae</i>		608	8.9e-165	80.20%	62.20%	AIO08848
			Heat shock protein		<i>Davidiella tassiana</i>	614	1.6e-164	80.20%	62.70%
				<i>Dermatophagoides farinae</i>	571	9.7e-136	78.70%	58.10%	AGC56218
7	121820	766	Sal k3 pollen allergen	<i>Salsola kali</i>	766	1.0e-155	73.80%	50.00%	ACO34814
8	45098	654	Heat shock protein	<i>Penicillium citrinum</i>	520	1.3e-130	98.80%	81.30%	Q92260
				<i>Davidiella tassiana</i>	650	1.9e-175	96.30%	83.80%	P40918
				<i>Dermatophagoides farinae</i>	581	3.8e-135	96.30%	76.10%	AGC56218
				<i>Tyrophagus putrescentiae</i>	660	1.4e-160	92.53%	74.70%	AOD75395
			Der f 28 allergen	<i>Dermatophagoides farinae</i>	649	9.1e-151	91.25%	70.70%	QAT18639
				<i>Dermatophagoides pteronyssinus</i>	653	1.7e-152	91.25%	71.50%	AIO08848
			Heat shock protein	<i>Aedes aegypti</i>	620	4.7e-131	85.00%	65.20%	ABF18258
				<i>Alternaria alternata</i>	156	8.1e-037	81.30%	80.10%	P78983
9	72199	676	Heat shock protein	<i>Penicillium citrinum</i>	542	1.5e-095	98.80%	78.00%	Q92260

Table 17. Allergen matches for *T. reesei* proteins in NOVA with > 50% sequence identity to AllergenOnline entries

	JGI protein ID	Protein length (Nr. of AA)	Matched allergen						
			Matched allergen	Source organism	Length (Nr. of AA)	E-value	Best Percent ID Over 80 AA	Percent ID Over Full Sequence	GenBank Accession
				<i>Davidiella tassiana</i>	672	3.2e-103	96.30%	81.10%	P40918
				<i>Dermatophagoides farinae</i>	603	3.1e-068	96.30%	73.30%	AGC56218
				<i>Tyrophagus putrescentiae</i>	682	4.5e-068	92.53%	72.30%	AOD75395
			Allergen Der p 28	<i>Dermatophagoides pteronyssinus</i>	675	3.3e-065	91.25%	69.20%	QAT18639
			Der f 28 allergen	<i>Dermatophagoides farinae</i>	671	1.1e-063	91.25%	68.40%	AIO08848
			Heat shock protein	<i>Aedes aegypti</i>	642	1.7e-072	85.00%	62.90%	ABF18258
				<i>Alternaria alternata</i>	156	1.8e-037	81.30%	80.10%	P78983
10	121127	797	Beta-xylosidase	<i>Aspergillus niger</i>	799	1.3e-125	81.30%	63.30%	AAD13106
			Xylosidase	<i>Aspergillus niger</i>	799	5.8e-126	81.30%	63.30%	CAB06417
11	121915	170	TCTP (Translationally controlled tumor protein)	<i>Alternaria alternata</i>	171	1.2e-065	73.80%	71.30%	ABI26088
				<i>Davidiella tassiana</i>	171	1.0e-054	61.30%	59.10%	A1KXP4
12	121977	227	Cyclophilin	<i>Aspergillus fumigatus</i>	154	2.0e-046	91.20%	74.00%	CAI78448
			Allergen	<i>Malassezia sympodialis</i>	150	6.6e-047	90.00%	76.70%	CAA09884
			Per a 18 allergen	<i>Periplaneta americana</i>	181	7.6e-043	87.50%	63.00%	UZC36341
			PPIase, partial	<i>Aspergillus fumigatus</i>	188	1.4e-046	86.30%	62.20%	CAB44442
			Cyclophilin	<i>Rhizopus oryzae</i>	176	6.3e-041	86.30%	62.50%	ALM24136
			Der f Mal f 6 allergen	<i>Dermatophagoides farinae</i>	166	5.9e-042	85.00%	65.70%	AAP35065
			Cyclophilin	<i>Arachis hypogaea</i>	189	1.1e-038	78.70%	56.10%	XP_025675300
				<i>Catharanthus roseus</i>	188	3.0e-039	78.70%	56.40%	CAA59468
				<i>Betula pendula</i>	188	1.8e-037	77.50%	55.90%	CAC84116
				<i>Daucus carota</i>	187	7.6e-039	77.50%	57.20%	AEY79726
			<i>Olea europaea</i>	190	1.2e-039	77.50%	58.40%	AVV30163	
13	123859	372	Allergen	<i>Malassezia sympodialis</i>	165	1.8e-037	82.50%	66.10%	CAA09884
			Cyclophilin	<i>Daucus carota</i>	168	3.8e-044	82.50%	69.60%	AEY79726
				<i>Olea europaea</i>	170	4.0e-045	81.30%	68.20%	AVV30163
				<i>Catharanthus roseus</i>	170	2.2e-044	81.30%	68.20%	CAA59468
				<i>Betula pendula</i>	170	2.2e-044	81.30%	69.40%	CAC84116
				<i>Arachis hypogaea</i>	170	1.3e-045	81.30%	70.00%	XP_025675300
			PPIase, partial	<i>Aspergillus fumigatus</i>	183	5.8e-030	77.50%	53.60%	CAB44442
			Cyclophilin	<i>Aspergillus fumigatus</i>	168	1.9e-038	77.50%	63.70%	CAI78448
			Der f Mal f 6 allergen	<i>Dermatophagoides farinae</i>	169	2.6e-038	72.50%	62.70%	AAP35065
			Cyclophilin	<i>Rhizopus oryzae</i>	169	1.4e-036	72.50%	60.90%	ALM24136
Per a 18 allergen	<i>Periplaneta americana</i>	169	1.2e-037	72.50%	60.90%	UZC36341			
14	78401	1103	Per a 20 allergen	<i>Periplaneta americana</i>	222	1.8e-053	65.00%	50.50%	UZC36343
15	123922	207	Allergen	<i>Malassezia sympodialis</i>	158	4.9e-048	85.00%	67.70%	CAA09884

Table 17. Allergen matches for *T. reesei* proteins in NOVA with > 50% sequence identity to AllergenOnline entries

	JGI protein ID	Protein length (Nr. of AA)	Matched allergen						
			Matched allergen	Source organism	Length (Nr. of AA)	E-value	Best Percent ID Over 80 AA	Percent ID Over Full Sequence	GenBank Accession
			Cyclophilin	<i>Aspergillus fumigatus</i>	161	5.3e-049	83.80%	67.10%	CAI78448
			PPase, partial	<i>Aspergillus fumigatus</i>	174	1.6e-043	81.30%	58.60%	CAB44442
			Per a 18 allergen	<i>Periplaneta americana</i>	161	1.4e-046	81.30%	63.40%	UZC36341
			Cyclophilin	<i>Rhizopus oryzae</i>	161	2.0e-047	80.00%	64.60%	ALM24136
			Mal f 6 allergen	<i>Dermatophagoides farinae</i>	161	1.8e-048	78.70%	65.80%	AAP35065
			Cyclophilin	<i>Daucus carota</i>	167	6.0e-045	75.00%	62.90%	AEY79726
				<i>Catharanthus roseus</i>	168	6.2e-043	75.00%	60.10%	CAA59468
				<i>Betula pendula</i>	168	7.2e-045	75.00%	63.70%	CAC84116
				<i>Olea europaea</i>	168	8.5e-045	75.00%	61.30%	AVV30163
				<i>Arachis hypogaea</i>	168	3.4e-044	73.80%	60.70%	XP_025675300
16	123088	Manganese superoxide dismutase	<i>Alternaria alternata</i>	191	5.4e-062	79.00%	69.10%	AGS80276	
			<i>Aspergillus fumigatus</i>	199	1.3e-057	70.00%	63.30%	Q92450	
			<i>Aspergillus fumigatus</i>	212	2.8e-057	70.00%	60.80%	AAB60779	
			<i>Hevea brasiliensis</i>	219	1.1e-042	62.50%	48.40%	AAA16792	
			<i>Pistacia vera</i>	199	1.7e-043	62.50%	51.30%	ABR29644	
			<i>Hevea brasiliensis</i>	199	9.0e-043	62.50%	51.30%	CAB53458	
			<i>Hevea brasiliensis</i>	199	4.1e-042	62.50%	50.80%	CAC13961	
		Allergen	<i>Malassezia sympodialis</i> ATCC 42132	193	2.7e-043	61.30%	51.80%	CCV00099	
Manganese superoxide dismutase	<i>Malassezia sympodialis</i>	193	3.4e-043	61.30%	51.80%	CAD68071			
17	120568	438	Enolase	<i>Aspergillus fumigatus</i>	438	1.5e-166	95.00%	87.70%	Q96X30
				<i>Aspergillus fumigatus</i>	438	7.1e-166	95.00%	87.20%	AAK49451
				<i>Penicillium citrinum</i>	438	2.0e-166	93.80%	87.70%	AAK51201
				<i>Alternaria alternata</i>	438	8.0e-162	91.20%	85.40%	Q9HDT3
				<i>Davidiella tassiana</i>	440	6.1e-159	90.00%	83.60%	CAA55070
				<i>Davidiella tassiana</i>	440	2.6e-160	90.00%	84.30%	P42040
				<i>Candida albicans</i>	438	4.4e-138	90.00%	74.70%	P30575
				<i>Rhodotorula mucilaginosa</i>	441	5.8e-141	90.00%	76.00%	AAP30720
				<i>Cochliobolus lunatus</i>	440	1.9e-148	88.70%	79.30%	AAK67491
				<i>Hevea brasiliensis</i>	447	5.4e-107	76.20%	61.30%	Q9LEI9
				<i>Hevea brasiliensis</i>	447	1.3e-107	76.20%	61.50%	CAC00532
				<i>Platanus x acerifolia</i>	446	2.0e-108	76.20%	61.70%	UYQ90943
			Per a 14 allergen	<i>Periplaneta americana</i>	436	3.4e-118	75.00%	64.00%	UZC36337
			Enolase	<i>Salmo salar</i>	436	3.7e-117	75.00%	63.10%	CBL79146
				<i>Pangasianodon hypophthalmus</i>	338	3.3e-091	73.80%	64.80%	XP_026777250

Table 17. Allergen matches for *T. reesei* proteins in NOVA with > 50% sequence identity to AllergenOnline entries

Table 17. Allergen matches for <i>T. reesei</i> proteins in NOVA with > 50% sequence identity to AllergenOnline entries									
	JGI protein ID	Protein length (Nr. of AA)	Matched allergen						
			Matched allergen	Source organism	Length (Nr. of AA)	E-value	Best Percent ID Over 80 AA	Percent ID Over Full Sequence	GenBank Accession
				<i>Gallus gallus</i>	434	1.6e-111	73.80%	62.20%	NP_990450
				<i>Salmo salar</i>	434	2.5e-116	72.50%	63.80%	ACH70931
				<i>Thunnus albacares</i>	436	2.2e-115	71.30%	61.90%	I0J1J1
				<i>Fenneropenaeus merguensis</i>	118	6.7e-023	58.70%	53.40%	AEM89226
18	124265	229	Elongation factor 1 beta-like	<i>Penicillium citrinum</i>	230	4.1e-076	80.20%	64.80%	AAR17475
19	122257	496	Aldehyde dehydrogenase (NAD+)	<i>Davidiella tassiana</i>	493	5.4e-165	82.50%	72.80%	CAA55072
				<i>Alternaria alternata</i>	494	1.6e-162	81.30%	72.10%	CAA55071
				<i>Tyrophagus putrescentiae</i>	475	5.4e-106	70.01%	52.80%	AOD75396
20	119722	664	glycoside hydrolase family 15 and carbohydrate-binding module family 20	<i>Schizophyllum commune</i> H4-8	611	1.6e-113	70.70%	50.20%	XP_003030591
21	103472	166	Peroxisomal membrane protein	<i>Penicillium citrinum</i>	165	3.8e-051	69.10%	56.40%	AAD42074
				<i>Aspergillus fumigatus</i>	168	8.6e-049	61.70%	53.60%	AAB95638
22	123029	154	superoxide dismutase	<i>Ambrosia trifida</i>	152	4.7e-043	66.26%	61.80%	QVG59417
			Olive pollen allergen	<i>Olea europaea</i>	152	9.3e-039	60.04%	55.30%	ABX54869
				<i>Olea europaea</i>	152	3.8e-039	60.04%	55.90%	ABX26140
				<i>Olea europaea</i>	152	2.3e-039	60.04%	55.90%	ABX54844
				<i>Olea europaea</i>	152	2.3e-039	60.04%	55.90%	ABX54859
				<i>Olea europaea</i>	152	3.8e-039	60.04%	55.30%	ABX54877
				<i>Olea europaea</i>	152	5.4e-039	60.04%	55.90%	ABX26141
				<i>Olea europaea</i>	152	2.3e-039	60.04%	55.90%	ABX26143
			Zn superoxide dismutase	<i>Olea europaea</i>	152	1.9e-039	60.04%	55.90%	CAD21706
			Olive pollen allergen	<i>Olea europaea</i>	152	3.2e-039	60.04%	55.90%	ABX54849
				<i>Olea europaea</i>	152	7.8e-040	60.04%	55.90%	ABX26134
				<i>Olea europaea</i>	152	9.3e-039	60.04%	55.30%	ABX26147
				<i>Olea europaea</i>	152	1.1e-038	60.04%	55.30%	ABX26145
				<i>Olea europaea</i>	152	1.6e-039	60.04%	56.60%	ABX54842
				<i>Olea europaea</i>	152	2.3e-039	60.04%	55.90%	ABX26131
				<i>Olea europaea</i>	152	7.7e-038	60.04%	53.90%	ABX26139
				<i>Olea europaea</i>	152	5.4e-039	60.00%	55.30%	ABX54855
				<i>Olea europaea</i>	152	3.8e-039	58.79%	55.30%	ABX54862
				<i>Olea europaea</i>	152	1.1e-038	58.76%	55.90%	ABX54876
				<i>Olea europaea</i>	152	4.6e-039	58.76%	55.30%	ABP58627

Table 17. Allergen matches for <i>T. reesei</i> proteins in NOVA with > 50% sequence identity to AllergenOnline entries								
	JGI protein ID	Protein length (Nr. of AA)	Matched allergen					
			Matched allergen	Source organism	Length (Nr. of AA)	E-value	Best Percent ID Over 80 AA	Percent ID Over Full Sequence
23	61956	187		<i>Olea europaea</i>	152	4.9e-036	57.50%	53.90%
				<i>Olea europaea</i>	152	1.3e-037	57.50%	53.90%
				<i>Olea europaea</i>	152	7.7e-038	57.50%	54.60%
				<i>Olea europaea</i>	152	4.1e-035	57.50%	53.90%
			TCTP (Translationally controlled tumor protein)	<i>Alternaria alternata</i>	188	2.8e-055	73.80%	63.30%
				<i>Davidiella tassiana</i>	187	1.1e-044	58.76%	51.90%

As detailed in **Table 17**, sequence matches included various protein groups such as transaldolases, heat shock proteins, xylosidases, peptidyl-prolyl cis-trans isomerases (cyclophilins), superoxide dismutases, enolases, elongation factors, aldehyde dehydrogenases, peroxisomal membrane proteins, translationally-controlled tumor proteins, triosephosphate isomerases, and a disulfide-isomerase. Many of these proteins are evolutionarily conserved across species and genera, and their biological importance in a wide range of organisms results in significant sequence homology across diverse species (Dogan, 2013; Sharan, 2005).

The majority of identified matches were aeroallergens and contact allergens derived from mites (*Dermatophagoides* spp. and *Tyrophagus putrescentiae*), molds and yeasts (e.g., *Aspergillus* spp., *Penicillium* spp., *Alternaria alternata*, *Cladosporium cladosporioides*, *Malassezia sympodialis*, and others), and insects (*Periplaneta americana* and *Aedes aegypti*). Matches were also found with pollen allergens from edible and non-edible plants (e.g., *Olea europaea*, *Salsola kali*, *Catharanthus roseus*, and others) as well as contact allergens from non-edible plants (e.g., *Hevea brasiliensis*). It is important to note that the AllergenOnline database includes proteins from a variety of sources, not only food allergens, but also airway, contact, venom, and salivary allergens with IgE binding data. In Onego's case, the identified matches to aeroallergens and contact allergens are primarily associated with respiratory allergies and contact dermatitis, not food-related allergic reactions. A review of the scientific literature found no evidence linking these allergens to allergic reactions upon ingestion, further supporting the conclusion that they are unlikely to pose a food safety concern. Therefore, these allergens are not considered relevant to the allergenicity assessment of *T. reesei* as the production organism for NOVA.

Notably, a few matches with minor allergens from major allergenic food sources were identified for 5 of the 23 potentially allergenic *T. reesei* proteins in NOVA (proteins 3, 12, 13, 15, and 17 as listed in **Table 17**). These included matches to cyclophilins, enolases, and triosephosphate isomerases from fish (*Salmo salar*, *Thunnus albacares*, and *Pangasianodon hypophthalmus*), crustaceans (*Procambarus clarkii*, *Scylla paramamosain*, *Crangon crangon* and *Fenneropenaeus merguensis*), peanut (*Arachis hypogaea*), and wheat (*Triticum aestivum*).

Three of these five proteins were identified to share significant sequence homology with a cyclophilin from peanut, *Arachis hypogaea* (GenBank accession XP_025675300, WHO/IUIS allergen Ara h 18). Ara h 18 is not part of the pool of major peanut allergens (Ara h 1, Ara h 2, Ara h 3, and Ara h 6) that are associated with severe allergic symptoms and anaphylaxis. Instead, Ara h 18 is classified as a minor allergen and is considered a cross-reactive determinant, with IgE-binding properties attributed to sensitization to other allergen sources, such as pollen, rather than direct peanut exposure (Mattsson, 2021). Furthermore, Ara h 18 has not been associated with severe allergic symptoms upon exposure and is unlikely to pose a food safety concern (Mattsson, 2021). Cyclophilins are essential enzymes for most organisms, widely distributed in nature, and highly conserved across prokaryotes and eukaryotes; they have been identified in plants, insects, fungi, bacteria, and mammals, including humans (Singh, 2023; Wang and Heitman, 2005). Due to their conservation, cyclophilins commonly appear in *in silico* sequence homology tests for allergens. However, major allergens are generally not highly conserved across species, which explains why individuals sensitized to a specific food allergen typically do not exhibit allergic reactions to a wide range of unrelated food sources (Abdelmoteleb, 2021). Notable exceptions include tropomyosins, which are conserved among shellfish and certain insects (Smeekens, 2024; Kamath, 2023). Importantly, no clinical evidence suggests widespread cross-reactivity involving cyclophilins. Based on these considerations, the observed matches between *T. reesei* proteins and Ara h 18 do not present a significant allergenic risk to NOVA consumers.

One more protein from *T. reesei* showed sequence homology greater than 50% with a set of triosephosphate isomerases (TPI) from wheat (*Triticum aestivum*), crawfish (*Procambarus clarkii*), mud crab (*Scylla paramamosain*), brown shrimp (*Crangon crangon*), iridescent shark catfish (*Pangasianodon hypophthalmus*), and Atlantic salmon (*Salmo salar*). The TPI from wheat (GenBank accession CAC14917, WHO/IUIS allergen Tri a 31) has been reported only as an aeroallergen associated with occupational baker's asthma or rhinitis due to inhalation exposure and is not a primary wheat allergen (Sander, 2011). Therefore, Tri a 31 is not relevant to this allergenicity assessment. Regarding the TPis from crawfish, mud crab, shrimp, catfish, and salmon (WHO/IUIS allergens Pro c 8, Scy p 8, Cra c 8, Pan h 8, and Sal s 8, respectively), none correspond to the major allergens of these food sources (Jasim, 2021; Ruethers, 2021; Yang, 2017; Bauermeister, 2011). TPI is a highly conserved enzyme involved in glycolysis, found in a wide range of organisms, from bacteria to humans,

where it plays a critical role in energy production (Myers and Palladino, 2023). Given that TPI is highly conserved across species and is essential for glycolysis in virtually all organisms, it is reasonable to expect that humans would generally be tolerant to homologous TPI proteins. This is further supported by the fact that there is no clinical evidence of widespread cross-reactivity associated with TPI. In Onego's case, none of the TPI matches identified for *T. reesei* proteins showed an identity greater than 70%, making clinically significant IgE cross-reactivity highly unlikely (Aalberse, 2000; Goodman et al., 2008; Abdelmoteleb et al., 2021; Sharma and Vitte, 2024). Therefore, the allergenicity risk posed by these TPI matches from food sources is not considered a significant concern for NOVA consumers.

The fifth *T. reesei* protein that matched minor allergens from major allergenic foods showed sequence homology with a series of enolases derived from Atlantic salmon (*Salmo salar*), iridescent shark catfish (*Pangasianodon hypophthalmus*), yellowfin tuna (*Thunnus albacares*), and banana prawn (*Fenneropenaeus merguensis*). While enolases from fish, shrimp, and chicken have been identified as IgE-binding proteins and, in some cases, linked to mediator release (Khanaruksombat, 2014; Kuehn, 2013; Andreassen, 2009), they do not correspond to the major allergens in fish or prawn (Kamath, 2023). Similarly, the enolases identified in this analysis—Sal s 2, Pan h 2, and Thu a 2—are not classified as major allergens (Ruethers, 2021; Faisal, 2019; Kuehn, 2014). Enolases are ubiquitous and among the most abundant cytosolic glycolytic enzymes, highly conserved across taxonomically diverse organisms, including humans (Morales-Amparano, 2021). Despite their high sequence conservation across taxonomical groups, widespread clinical cross-reactivity between enolases from all organisms producing homologs has not been reported. Cross-reactivity has primarily been observed between enolases acting as aeroallergens from fungi and yeasts, as well as enolases acting as food allergens within specific fish species (Morales-Amparano, 2021). As such, the matches to enolases derived from food sources are not considered to present a significant risk of allergenicity in the context of NOVA. Moreover, none of the enolase matches identified for *T. reesei* proteins exceeded 70% sequence identity, further reducing the likelihood of clinically relevant IgE cross-reactivity.

In conclusion, the sequence homology analysis of *T. reesei* proteins present in NOVA at levels exceeding 0.01% predominantly identified matches with proteins classified as aeroallergens and contact allergens, with no major food allergens detected. While some matches were observed with allergenic proteins from major allergenic foods, these were limited to minor allergens not associated with severe allergic reactions or anaphylaxis. These minor allergens are primarily conserved across diverse taxa due to their essential biological functions, reflecting their evolutionary importance rather than a heightened potential for cross-reactivity. Although IgE binding has been reported for some of these proteins, there is no evidence of widespread clinical cross-reactivity. Furthermore, none of the matched proteins demonstrated greater than 70% sequence identity across their full length with the *T. reesei* proteins in NOVA, a threshold generally associated with significant IgE cross-reactivity. This indicates a minimal likelihood of clinically relevant cross-reactivity.

It is also important to note that allergenicity risk is dose dependent. Minor allergens are unlikely to pose significant food safety concerns unless present in high concentrations (Abdelmoteleb, 2021). In the case of NOVA, the *T. reesei* proteins in final products are expected to be present at concentrations unlikely to elicit significant allergenic responses. Moreover, *T. reesei* has a well-established history of safe use in food enzyme production, with no documented reports of food allergies linked to *T. reesei*-derived proteins. This history, combined with the outcomes of the sequence homology analysis, strongly supports the conclusion that proteins from *T. reesei* in NOVA, other than ovalbumin, do not present a significant allergenic risk. The allergenicity concern posed by ovalbumin, as detailed throughout this document, will be addressed by clearly labeling all food products containing NOVA to indicate the presence of egg, in accordance with the requirements of the Food Allergen Labeling and Consumer Protection Act (FALCPA) (Public Law 108-282) and Section 403 (w) of the FD&C Act.

6.4. Summary of basis for GRAS determination

Onego Bio, Inc. has determined that NOVA from fermentation by *T. reesei* is GRAS for its intended use in food, based on the following:

- The substantial equivalence of Onego's NOVA to native OVA from chicken egg white.
- The long history of safe consumption of native ovalbumin as part of chicken egg white.

- The intended uses and the estimated consumption of NOVA, which will replace fresh or powdered whole eggs and egg white protein, without increasing the overall protein consumption of the general population.
- The ability to effectively mitigate the risk of allergenicity from ovalbumin consumption in sensitized individuals through clear labeling of products containing ovalbumin from *T. reesei*, indicating the presence of egg allergens, in full compliance with all food allergen labeling requirements in FALCPA, FASTER Act, and section 403(w) of the FD&C Act, and any FDA implementing regulations.
- The manufacturing conditions of Onego's recombinant ovalbumin, produced in compliance with cGMPs and a Food Safety Plan, in accordance with 21 CFR Part 117 and meeting appropriate food-grade specifications.
- The absence of potential contaminants such as heavy metals, mycotoxins, and pathogenic microorganisms, or their presence below toxicological and regulatory limits.
- The long history of safe use of *T. reesei* in the industrial production of recombinant GRAS enzymes, with the production strain fully characterized and meeting the criteria for a safe production microorganism as outlined by Pariza and Johnson (2001).
- The existence of supportive evidence from the successful GRAS notice for ovalbumin from fermentation by *Komagataella phaffii* (GRN 1104).

6.5. GRAS conclusion

Based on scientific procedures in accordance with 21 CFR 170.30 (a) and (b), Onego Bio has determined that ovalbumin from fermentation by *T. reesei* is GRAS for use in food when manufactured according to cGMPs and a Food Safety Plan, in compliance with 21 CFR Part 117 and meeting the specifications described herein. As such, the product is exempt from pre-market approval requirements of the FC&C Act.

7. 21 CFR §170.255 LIST OF SUPPORTING DATA

7.1. List of references

- Aalberse, R.C. (2000). Structural biology of allergens. *J Allergy Clin Immunol*, 106(2), 228-38.
- Abdelmoteleb, M., Zhang, C., Furey, B., Kozubal, M., Griffiths, H., Champeaud, M., Goodman, R.E. (2021). Evaluating potential risks of food allergy of novel food sources based on comparison of proteins predicted from genomes and compared to www.AllergenOnline.org. *Food Chem Toxicol*, 147, 111888.
- Andreassen, R., Lunner, S., Høyheim, B. (2009). Characterization of full-length sequenced cDNA inserts (FLICs) from Atlantic salmon (*Salmo salar*). *BMC Genomics*, 10, 502.
- Aro, N., Ercili-Cura, D., Andberg, M., Silventoinen, P., Lille, M., Hosia, W., Nordlund, E., Landowski, C.P. (2023). Production of bovine beta-lactoglobulin and hen egg ovalbumin by *Trichoderma reesei* using precision fermentation technology and testing of their techno-functional properties. *Food Research International*, 163, 112131.
- Astwood, J.D., Leach, J.N., Fuchs, R.L. (1996). Stability of food allergens to digestion *in vitro*. *Nat Biotechnol*, 14(10), 1269-1273.
- Batista, N.V., Pereira, R.V., Noviello, M.L., Dourado, L.P., Perez, D.A., Foureaux, G., Ferreira, A.J., Ferreira, A.V., Cara, D.C. (2014). Prolonged ingestion of ovalbumin diet by sensitized mice improves the metabolic consequences induced by experimental food allergy. *Clin Exp Immunol*, 178(3), 416-427.
- Bauermeister, K., Wangorsch, A., Garoffo, L.P., Reuter, A., Conti, A., Taylor, S.L., Lidholm, J., Dewitt, A.M., Enrique, E., Vieths, S., Holzhauser, T., Ballmer-Weber, B., Reese, G. (2011). Generation of a comprehensive panel of crustacean allergens from the North Sea Shrimp *Crangon crangon*. *Mol Immunol*, 48(15-16), 1983-1992.
- Beckham, G.T., Dai, Z., Matthews, J.F., Momany, M., Payne, C.M., Adney, W.S., Baker, S.E., Himmel, M.E. (2012). Harnessing glycosylation to improve cellulase activity. *Current Opinion in Biotechnology*, 23(3), 338-345.
- Caubet, J.C., Wang, J. (2011). Current understanding of egg allergy. *Pediatr Clin North Am*, 58(2), 427-43.
- Chandel, N.S. (2021). Amino acid metabolism. *Cold Spring Harb Perspect Biol*, 13(4), a040584.
- Dhanapala, P., De Silva, C., Doran, T., Suphioglu, C. (2015). Cracking the egg: An insight into egg hypersensitivity. *Molecular Immunology*, 66(2), 375-383.
- Doğan, T., Karaçalı, B. (2013). Automatic identification of highly conserved family regions and relationships in genome wide datasets including remote protein sequences. *PLoS One*, 12(8(9), e75458.
- ECARF. (2016). *Chicken egg allergy*. Retrieved December 6, 2024, from <https://www.ecarf.org/en/information-portal/allergies-overview/chicken-egg-allergy/>.
- EFSA. (2022). Scientific Opinion on development needs for the allergenicity and protein safety assessment of food and feed products derived from biotechnology. Retrieved December 6, 2024 from <https://efsa.onlinelibrary.wiley.com/doi/10.2903/j.efsa.2022.7044>
- EPA. (2012). Microorganisms; General exemptions from reporting requirements; Revisions to recipient organisms eligible for tier I and tier II exemptions, 77 Fed. Reg. 54499. Retrieved December 6, 2024 from <https://www.regulations.gov/document/EPA-HQ-OPPT-2011-0740-0001>.
- EPA. (2020). Microorganisms; General exemptions from reporting requirements; Revisions to recipient organisms eligible for tier I and tier II exemptions, 85 Fed. Reg. 13760. Retrieved December 6, 2024 from <https://www.federalregister.gov/documents/2020/03/10/2020-04746/microorganisms-general-exemptions-from-reporting-requirements-revisions-to-recipient-organisms>.
- Evenepoel, P., Geypens, B., Luybaerts, A., Hiele, M., Ghoo, Y., Rutgeerts, P. (1998). Digestibility of cooked and raw egg protein in humans as assessed by stable isotope techniques. *The Journal of Nutrition*, 128(10), 1716-1722.
- Faisal, M., Buckow, R., Vasiljevic, T., Donkor, O.N. (2019). Effect of simulated digestion on antigenicity of banana prawn (*Fenneropenaeus merguensis*) after high pressure processing at different temperatures. *Food Control*, 104, 187-192.
- FAO/WHO. (2001). Evaluation of Allergenicity of Genetically Modified Foods, Report of a Joint FAO/WHO Expert Consultation on Allergenicity of Foods Derived from Biotechnology, 22-25, <http://www.who.int/foodsafety/publications/gmo-allergenicity/en>.
- Farjami, T., Babaei, J., Nau, F., Dupont, D., Madadlou, A. (2021). Effects of thermal, non-thermal and emulsification processes on the gastrointestinal digestibility of egg white proteins. *Trends in Food Science & Technology*, 107, 45-56.

- FDA. (1998). Direct food substances affirmed as generally recognized as safe; Egg white lysozyme, 63 Fed.Reg. 21 CFR Part 184. Federal Register/Vol. 63, No. 49, March 13.
- FDA. (2022a). *Food Allergen labelling and Consumer Protection Act of 2004 (FALCPA)*. Retrieved December 6, 2024, from <https://www.fda.gov/food/food-allergensgluten-free-guidance-documents-regulatory-information/food-allergen-labeling-and-consumer-protection-act-2004-falcpa>.
- FDA. (2022b). *Questions and answers regarding food allergens, including the food allergen labeling requirements of the federal food, drug, and cosmetic act (edition 5): Guidance for industry*. Retrieved December 6, 2024, from <https://www.fda.gov/media/117410/download>.
- FDA. (2024). *Food allergies: What you need to know*. Retrieved December 6, 2024, from <https://www.fda.gov/food/buy-store-serve-safe-food/food-allergies-what-you-need-know>.
- Garvey M. (2022). Non-mammalian eukaryotic expression systems yeast and fungi in the production of biologics. *J Fungi (Basel)* 8(11):1179.
- GRN No. 64. Egg white lysozyme. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 230. Chymosin enzyme preparation from *Trichoderma reesei* expressing the bovine prochymosin B gene. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 315. Transglucosidase enzyme preparation from *Trichoderma reesei* expressing the gene encoding transglucosidase from *Aspergillus niger*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 333. Acid fungal protease enzyme preparation from *Trichoderma reesei* expressing the gene encoding acid fungal protease from *T. reesei*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 372. Glucoamylase enzyme preparation from *Trichoderma reesei* expressing the glucoamylase gene from *T. reesei* (glucoamylase enzyme preparation). Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 567. Xylanase from *Trichoderma reesei*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 707. Glucose oxidase from *Penicillium* produced in *Trichoderma reesei*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 727. Trehalase from *Trichoderma reesei* produced in *Trichoderma reesei*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 964. Lysophospholipase enzyme preparation from *Aspergillus niger* produced by *Trichoderma reesei*. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 967. Soluble egg-white protein produced by *Komagataella phaffii* strain GSD-1209. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- GRN No. 1104. Egg-white protein produced by *Komagataella phaffii* ATCC GSD-1235. Available on FDA website at: [GRAS Notices \(fda.gov\)](https://www.fda.gov/GRAS/GRAS-Notices/GRAS-Notices-fda.gov).
- Goodman, R.E., Vieths, S., Sampson, H.A., Hill, D., Ebisawa, M., Taylor, S.L., van Ree, R. (2008). Allergenicity assessment of genetically modified crops--what makes sense? *Nat Biotechnol*, 26(1), 73-81.
- Guha, S., Majumder, K., Mine, Y. (2019). Egg proteins. In L. Melton, F. Shahidi & P. Varelis (Eds.), *Encyclopedia of Food Chemistry* (pp. 74-84). Academic Press.
- Harvey, D.J., Wing, D.R., Küster, B., Wilson, I.B.H. (2000). Composition of N-linked carbohydrates from ovalbumin and co-purified glycoproteins. *J Am Soc Mass Spectrom*, 11(6), 564-571.
- Health Canada. (2018). Screening assessment of *Trichoderma reesei* strain ATCC 74252. Environment and Climate Change Canada, Health Canada. Retrieved December 6, 2024, from https://www.canada.ca/content/dam/eccc/documents/pdf/pded/trichoderma/EN_trichoderma_2018-02-21.pdf.
- Hoffmann, J.R., Falvo, M.J. (2004). Protein - which is best?. *J Sports Sci Med*, 3, 118-130.
- Huntington, J.A., Stein, P.E. (2001). Structure and properties of ovalbumin. *J Chromatogr B Biomed Appl*, 756(1-2), 189-198.
- Huynh, K., Partch, C.L. (2015). Analysis of protein stability and ligand interactions by thermal shift assay. *Curr Protoc Protein Sci*, 79, 28.9.1-28.9.14.
- Ito, K., Matsudomi, N. (2005). Structural characteristics of hen egg ovalbumin expressed in yeast *Pichia pastoris*. *Biosci Biotechnol Biochem*, 69(4), 755-761.
- Jahan-Mihan, A., Luhovyy, B.L., El Khoury, D., Anderson, G.H. (2011). Dietary proteins as determinants of metabolic and physiologic functions of the gastrointestinal tract. *Nutrients*, 3(5), 574-603.
- Järviö, N., Parviainen, T., Maljanen, N.L., Kobayashi, Y., Kujanpää, L., Ercili-Cura, D., Landowski, C.P., Ryyänänen, T., Nordlund, E., Tuomisto, H. (2021). Ovalbumin production using *Trichoderma*

- reesei* culture and low-carbon energy could mitigate the environmental impacts of chicken-egg-derived ovalbumin. *Nat Food*, 2, 1005–1013.
- Jasim, H.A., Misnan, R., Yadzir, Z.H.M., Abdullah, N., Bakhtiar, F., Arip, M., Ateshan, H.M., Keong, P.B. (2121). Identification of common and novel major crab allergens in *Scylla tranquebarica* and the allergen stability in untreated and vinegar-treated crab. *IJAAI*, 20(1), 76-87.
- Jiaqi, W., Yanjun, C. (2023). Research progress on the allergic mechanism, molecular properties, and immune cross-reactivity of the egg allergen Gal d 5. *Front Nutr*, 10, 1205671.
- Kamath, S.D., Bublin, M., Kitamura, K., Matsui, T., Lopata, A.L. (2023). Cross-reactive epitopes and their role in food allergy. *JACI*, 151(5), 1178 – 1190
- Khanaruksombat, S., Srisomsap, C., Chokchaichamnankit, D., Punyarit, P., Phiriyangkul, P. (2014) Identification of a novel allergen from muscle and various organs in banana shrimp (*Fenneropenaeus merguensis*). *Ann Allergy Asthma Immunol*, 113(3), 301-306.
- Kitabatake, N., Ishida, A., Doi, E. (1988). Physicochemical and functional properties of hen ovalbumin dephosphorylated by acid phosphatase. *Agric Biol Chem*, 52(4), 967-973.
- Knippels, L.M., Penninks, A.H., van Meeteren, M., Houben, G.F. (1999). Humoral and cellular immune responses in different rat strains on oral exposure to ovalbumin. *Food Chem Toxicol*, 37(8), 881-888.
- Knippels, L.M., van der Kleij, H.P., Koppelman, S.J., Houben, G.F., Penninks, A.H. (2000). Comparison of antibody responses to hen's egg and cow's milk proteins in orally sensitized rats and food-allergic patients. *Allergy*, 55(3), 251-258.
- Kołaczkowski, B.B., Schaller, K.S., Sørensen, T.H., Peters, G.H.J., Jensen, K., Kristian B. R. M. Krogh, K.B.R.M., Westh, P. (2020). Removal of N-linked glycans in cellobiohydrolase Cel7A from *Trichoderma reesei* reveals higher activity and binding affinity on crystalline cellulose. *Biotech for Biofuels & Bioprod*, 13, 136.
- Kuehn, A., Hilger, C., Lehnert-Weber, C., Codreanu-Morel, F., Morisset, M., Metz-Favre, C., Pauli, G., de Blay, F., Revets, D., Muller, C.P., Vogel, L., Vieths, S., Hentges, F. (2013). Identification of enolases and aldolases as important fish allergens in cod, salmon and tuna: component resolved diagnosis using parvalbumin and the new allergens. *Clin Exp Allergy*, 43(7), 811-822.
- Kuehn, A., Swoboda, I., Arumugam, K., Hilger, C., Hentges, F. (2014). Fish allergens at a glance: variable allergenicity of parvalbumins, the major fish allergens. *Front Immunol*, 5, 179
- Landowski, C.P., Mustalahti, E., Wahl, R. et al. (2016). Enabling low cost biopharmaceuticals: high level interferon alpha-2b production in *Trichoderma reesei*. *Microb Cell Fact*, 15, 104.
- Lemon-Mulé, H., Sampson, H.A., Sicherer, S.H., Shreffler, W.G., Noone, S., Nowak-Wegrzyn, A. (2008). Immunologic changes in children with egg allergy ingesting extensively heated egg. *J Allergy Clin Immunol*, 122(5), 977-983.
- Lopes Ferreira, N., Margeot, N., Blanquet, S., Berrin J.G. (2014). Use of cellulases from *Trichoderma reesei* in the twenty-first century—part I: Current industrial uses and future applications in the production of second ethanol generation. In V.K. Gupta, M. Schmoll, A. Herrera-Estrella, R.S. Upadhyay, I. Druzhinina, M. G. Tuohy (Eds.), *Biotechnology and Biology of Trichoderma* (pp 245-261). Elsevier.
- Lucey, A.J., Heneghan, C., Manning, E., Kroon, P.A., Kiely, M.A. (2019). Effect of an egg ovalbumin-derived protein hydrolysate on blood pressure and cardiovascular risk in adults with a mildly elevated blood pressure: a randomized placebo-controlled crossover trial. *Eur J Nutr* 58, 2823–2833.
- Martos, G., Contreras, P., Molina, E., López-Fandiño, R. (2010). Egg white ovalbumin digestion mimicking physiological conditions. *J Agric Food Chem*, 58(9), 5640-5648.
- Mathew, P., Pflieger, J.L. (2023). Egg Allergy. In: *StatPearls* [Internet]. StatPearls Publishing. Available from <https://www.ncbi.nlm.nih.gov/books/NBK538192/>
- Matsuoka R., Kurihara H., Nishijima N., Oda Y., Handa A. (2019). Egg white hydrolysate retains the nutritional value of proteins and is quickly absorbed in rats. *ScientificWorldJournal*, 27: 5475302.
- Mattsson, L., Valcour, A., Holmqvist, M., Larsson, H., Lidholm, J. (2021). Cyclophilin - A novel cross-reactive determinant in peanut. *Clin Exp Allergy*, 51(4), 620-622.
- Neagu, A.N., Jayathirtha, M., Baxter, E., Donnelly, M., Petre, B.A., Darie, C.C. (2022). Applications of tandem mass spectrometry (MS/MS) in protein analysis for biomedical research. *Molecules*, 27(8), 2411.
- Nevalainen, H., Suominen, P., Taimisto, K. (1994). On the safety of *Trichoderma reesei*. *Journal of Biotechnology*, 37(3), 193-200.
- Nevalainen, H., Peterson, R. (2014). Making recombinant proteins in filamentous fungi- are we expecting too much? *Front. Microbiol*, 5, 75.

- Nicklin S., Miller K. (1987). Naturally acquired tolerance to dietary antigen: effect of in utero and perinatal exposure on subsequent humoral immune competence in the rat. *J Reprod Immunol*, 10(2), 167-176.
- Nijssen, K.M.R., Joris, P.J., Mensink, R.P., Plat, J. (2023). Longer-term effects of the egg-protein hydrolysate NWT-03 on arterial stiffness and cardiometabolic risk markers in adults with metabolic syndrome: a randomized, double-blind, placebo-controlled, crossover trial. *Eur J Clin Nutr*. 77(10), 982-988.
- OECD. (1998). Biotechnology for clean industrial products and processes: Towards industrial sustainability. OECD Publishing.
- Paloheimo, M., Haarmann, T., Mäkinen, S., Vehmaanperä, J. (2016). Production of industrial enzymes in *Trichoderma reesei*. In M. Schmoll & C. Dattenböck (Eds.), *Gene Expression Systems in Fungi: Advancements and Applications* (pp. 23-57). Springer.
- Pariza, M.W., Foster, E.M. (1983). Determining the safety of enzymes used in food processing. *J Food Prot*, 46(5), 453-468.
- Pariza, M.W., Johnson, E.A. (2001). Evaluating the safety of microbial enzyme preparations used in food processing: update for a new century. *Regul Toxicol Pharmacol*, 33(2), 173-186.
- Pariza, M.W., Gillies, K.O., Kraak-Ripple, S.F., Leyer, G., Smith, A.B. (2015). Determining the safety of microbial cultures for consumption by humans and animals. *Regul Toxicol Pharmacol*, 73(1), 164-71.
- Pearson, W.R. (1996). Effective protein sequence comparison. *Methods Enzymol*, 266, 227-58.
- Pereira, M.M., Cruz, R.A., Almeida, M.R., Lima, Á.S., Coutinho, J.A., Freire, M.G.. (2016). Single-step purification of ovalbumin from egg white using aqueous biphasic systems. *Process Biochem.*, 51(6), 781-791.
- Quirce, S., Marañón, F., Umpiérrez, A., de las Heras, M., Fernández-Caldas, E., Sastre, J. (2001). Chicken serum albumin (Gal d 5) is a partially heat-labile inhalant and food allergen implicated in the bird-egg syndrome. *Allergy*, 56(8), 754-62.
- Ramos-Pedersoli, W., De Paula, R.G., Campos-Antoniêto, A.C., Batista-Carraro, C., Taveira, I.C., Batista-Maués, D., Pompeu-Martins, M., Costa-Ribeiro, L.F., De Lima-Damasio, A.R., Silva-Rocha, R., Rossi-Filho, A., Silva, R.N. (2021). Analysis of the phosphorylome of *Trichoderma reesei* cultivated on sugarcane bagasse suggests post-translational regulation of the secreted glycosyl hydrolase Cel7A. *Biotech Rep* 31, e00652.
- Rantasalo, A., Vitikainen, M., Paasikallio, T. et al. (2019). Novel genetic tools that enable highly pure protein production in *Trichoderma reesei*. *Sci Rep*, 9, 5032.
- Razi, S.M., Fahim, H., Amirabadi, S., Rashidinejad, A. (2023). An overview of the functional properties of egg white proteins and their application in the food industry. *Food Hydrocolloids*, 135, 108183.
- Rodriguez-Iglesias, A., Schmoll, M. (2019). Protein phosphatases regulate growth, development, cellulases and secondary metabolism in *Trichoderma reesei*. *Sci Rep* 9, 10995.
- Ruethers, T., Taki, A.C., Karnaneedi, S., Nie, S., Kalic, T., Dai, D., Daduang, S., Leeming, M., Williamson, N.A., Breiteneder, H., Mehr, S.S., Kamath, S.D., Campbell, D.E., Lopata, A.L. (2021). Expanding the allergen repertoire of salmon and catfish. *Allergy*, 76(5), 1443-1453.
- Sander, I., Rozynek, P., Rihs, H.P., Van Kampen, V., Chew, F.T., Lee, W.S., Kotschy-Lang, N., Merget, R., Brüning, T., Raulf-Heimsoth, M. (2011). Multiple wheat flour allergens and cross-reactive carbohydrate determinants bind IgE in baker's asthma. *Allergy*, 66(9), 1208-15.
- Scaife, K., Taylor, S.L., Pařenicová, L., Goodman, R.E., Vo, T.D., Leune, E., Abdelmoteleb, M., Dommels, Y. (2024). *In silico* evaluation of the potential allergenicity of a fungal biomass from *Rhizomucor pusillus* for use as a novel food ingredient. *Regul Toxicol Pharmacol* 150, 105629.
- Saleem, F., Soos, M.P. (2023). Biotin Deficiency. In: StatPearls [Internet]. StatPearls Publishing. Available from: <https://www.ncbi.nlm.nih.gov/books/NBK547751/>
- Sharma, E., Vitte, J. (2024). A systematic review of allergen cross-reactivity: Translating basic concepts into clinical relevance. *J Allergy Clin Immunol Glob*, 3(2), 100230.
- Sharan, R., Suthram, S., Kelley, R.M., Kuhn, T., McCuine, S., Uetz, P., Sittler, T., Karp, R.M., Ideker, T. (2005). Conserved patterns of protein interaction in multiple species. *Proc. Natl. Acad. Sci. U.S.A.*, 102(6), 1974-1979.
- Singh, M., Singh, H., Kaur, K., Shubhankar, S., Sing, S., Kaur, A., Singh, P. (2023). Characterization and regulation of salt upregulated cyclophilin from a halotolerant strain of *Penicillium oxalicum*. *Sci Rep* 13, 17433.
- Smeeckens, J.M., Kesselring, J.R., Bagley, K., Kulis, M.D. (2024). A mouse model of shrimp allergy with cross-reactivity to crab and lobster. *Methods Mol Biol*, 2717, 311-319.

- Su, X., Schmitz, G., Zhang, M., Mackie, R.I., Cann, I.K.O. (2012). Chapter One - Heterologous Gene Expression in Filamentous Fungi. In: Advances in Applied Microbiology. Editors: Geoffrey M. Gadd, Sima Sariaslani. Academic Press. Pages 81, 1-61.
- Tomei, L., Muraro, A., Giovannini, M., Barni, S., Liccioli, G., Paladini, E., Sarti, L., Pessina, B., Skypala, I., Novembre, E., Mori, F.. (2023). Hidden and rare food allergens in pediatric age. *Nutrients*, 15(6), 1386.
- Torres, N., Tobón-Cornejo, S., Velazquez-Villegas, L.A., Noriega, L.G., Alemán-Escondrillas, G., Tovar, A.R. (2023). Amino acid catabolism: an overlooked area of metabolism. *Nutrients*, 15(15), 3378.
- Upton, J., Wong, D., Nowak-Węgrzyn, A. (2024). Baked milk and egg diets revisited. *Annals of Allergy, Asthma & Immunol*, 132(3), 328-226.
- USDA. (2024). *Food allergies: The “Big-9”*. Retrieved December 6, 2024, from <https://www.fsis.usda.gov/food-safety/safe-food-handling-and-preparation/food-safety-basics/food-allergies-big-9>.
- Wang, P., Heitman, J. (2005). The cyclophilins. *Genome Biol*, 6(7), 226,
- Van Eerde, A., Várnai, A., Jameson, J.K., Paruch, L., Moen, A., Anonsen, J.H., Chylenski, P., Steen, H.S., Heldal, I., Bock, R., Eijsink, V.G.H., Liu-Clarke, J. (2020). In-depth characterization of *Trichoderma reesei* cellobiohydrolase TrCel7A produced in *Nicotiana benthamiana* reveals limitations of cellulase production in plants by host-specific post-translational modifications. *Plant Biotechnol J*, 18(3), 631-643.
- Wallig, M.A., Keenan, K.P. (2013). Chapter 36 - Nutritional Toxicologic Pathology. In: Haschek and Rousseaux's Handbook of Toxicologic Pathology (Third Edition). Editors: Wanda M. Haschek, Colin G. Rousseaux, Matthew A. Wallig. Academic Press. Pages 1077-1121.
- Wolfe, R.R., Cifelli, A.M., Kostas, G., Kim, I.Y. (2017). Optimizing protein intake in adults: interpretation and application of the recommended dietary allowance compared with the acceptable macronutrient distribution range. *Adv Nutr*, 8(2), 266-275.
- Yang, Y., Zhang, Y.X., Liu, M., Maleki, S.J., Zhang, M.L., Liu, Q.M., Cao, M.J., Su, W.J., Liu, G.M. (2017). Triosephosphate isomerase and filamin c share common epitopes as novel allergens of *Procambarus clarkii*. *J Agric Food Chem*, 65(4), 950-963.

APPENDIX I - Certificates of analysis (CoAs) of the presented production batches

Below, Onego Bio presents the CoAs of the three non-consecutive NOVA batches discussed in **Section 2.5** of this GRAS Determination. Personal information, including analyst and contact names, signatures, email addresses, and phone numbers, has been redacted. This does not qualify as confidential business information.

It should be noted that the term *Ei-eläinperäinen ovalbumiinjauhe* in the Batch Comments section of the CoAs refers to non-animal ovalbumin powder.

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Sample code Nr. 493-2024-00069527
Analytical Report Nr. AR-24-FL-033959-02

Date 14.10.2024 Page 1 / 1



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Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]
[REDACTED]

Client Reference:	OnegoID-7695		
Sample described as:	OnegoID-7695		
Sample reception date:	03.10.2024	Analysis Starting Date:	03.10.2024
Batch comment:	Ei-eläinperäinen ovalbumiinijauhe		

Results (MU)

FL077	FL	pH from spreads/margarines and foods	Method: AOAC 981.12 (1982)	Potentiometry
	pH		6,4 (± 0,1)	

SIGNATURE

[REDACTED] Analytical Service Manager Food Chemistry
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EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069541
Analytical Report Nr. AR-24-FL-033961-02

Date 14.10.2024 Page 1 / 1



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Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]
[REDACTED]

Client Reference:	onegolD-6309		
Sample described as:	OnegolD-6309		
Sample reception date:	03.10.2024	Analysis Starting Date:	03.10.2024
Batch comment:	Ei-eläinperäinen ovalbumiinjauhe		

Results (MU)

FL077	FL pH	pH from spreads/margarines and foods	Method: AOAC 981.12 (1982)	Potentiometry
			6,8 (± 0,1)	

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069555
Analytical Report Nr. AR-24-FL-033963-02

Date 14.10.2024 Page 1 / 1



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Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: onegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results (MU)

FL077 FL pH from spreads/margarines and foods Method: AOAC 981.12 (1982) Potentiometry
pH 6,7 ($\pm 0,1$)

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Analytical Service Manager Food Chemistry

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069528
Analytical Report Nr. AR-24-FL-034365-02

Date 25.10.2024 Page 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-034365-01/493-2024-00069528 dated 14/10/2024 which must be destroyed)

Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients			Results (MU)	
FL15N	FL	Fat content (foods), Gerhard	Method: NMKL 160 mod.	Gravimetry
(a)	Fat			<0,2 (± 0,0) g/100 g
FL036	FL	Protein (Kjeldahl)	Method: NMKL 6:2003, mod.	Kjeldahl (titrimetry)
(a)	Protein (Nx6,25)			85,1 (± 6,0) g/100 g
FL037	FL	Ash (foods, feeds)	Method: NMKL 173:2005, mod.	Gravimetry
(a)	Ash			3,0 (± 0,4) g/100 g
FL052	FL	Carbohydrates (calculated)	Calculation	
	Carbohydrates (calc.)			3,9 g/100 g
FL053	FL	Energy content	Method: EU dir. 90/496/EY	Calculation
	Energy value (kcal)			356 kcal/100 g
	Energy value (kJ)			1513 kJ/100 g
Analyses on: reanalysis				
FL033	FL	Moisture (foods, feeds)	Method: NMKL 23:1991, mod.	Gravimetry
(a)	Moisture			8,0 (± 1,0) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

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Sample code Nr. 493-2024-00069542
Analytical Report Nr. AR-24-FL-034367-02

Date 25.10.2024 Page 1 / 1



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Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients

Results (MU)

FL15N FL Fat content (foods), Gerhardt Method: NMKL 160 mod. Gravimetry
(a) Fat <0,2 ($\pm$ -0,0) g/100 g
FL036 FL Protein (Kjeldahl) Method: NMKL 6:2003, mod. Kjeldahl (titrimetry)
(a) Protein (Nx6,25) 85,8 ($\pm$ 6,0) g/100 g
FL037 FL Ash (foods, feeds) Method: NMKL 173:2005, mod. Gravimetry
(a) Ash 3,2 ($\pm$ 0,4) g/100 g
FL052 FL Carbohydrates (calculated) Calculation
Carbohydrates (calc.) 4,1 g/100 g
FL053 FL Energy content Method: EU dir. 90/496/EY Calculation
Energy value (kcal) 360 kcal/100 g
Energy value (kJ) 1528 kJ/100 g
Analyses on: Reanalysis
FL033 FL Moisture (foods, feeds) Method: NMKL 23:1991, mod. Gravimetry
(a) Moisture 6,9 ($\pm$ 0,8) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

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Analytical Service Manager Food Chemistry

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

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Sample code Nr. 493-2024-00069556
Analytical Report Nr. AR-24-FL-034369-03

Date 25.10.2024 Page 1 / 1



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Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients			Results (MU)	
FL15N	FL	Fat content (foods), Gerhardt	Method: NMKL 160 mod.	Gravimetry
(a)	Fat			<0,2 (± -0,0) g/100 g
FL036	FL	Protein (Kjeldahl)	Method: NMKL 6:2003, mod.	Kjeldahl (titrimetry)
(a)	Protein (Nx6,25)			84,6 (± 5,9) g/100 g
FL037	FL	Ash (foods, feeds)	Method: NMKL 173:2005, mod.	Gravimetry
(a)	Ash			4.4 (± 0.5) g/100 g
FL052	FL	Carbohydrates (calculated)	Calculation	
	Carbohydrates (calc.)			6,1 g/100 g
FL053	FL	Energy content	Method: EU dir. 90/496/EY	Calculation
	Energy value (kcal)			363 kcal/100 g
	Energy value (kJ)			1542 kJ/100 g
Analyses on: reanalysis				
FL033	FL	Moisture (foods, feeds)	Method: NMKL 23:1991, mod.	Gravimetry
(a)	Moisture			4,9 (± 0,6) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069529
Analytical Report Nr. AR-24-FL-034959-01

Date 18.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results (MU)

J8312 I6 Arsenic (ICP-MS, food) Method: DIN EN 15763:2010 (2010-04), mod. ICP-MS
(a) Arsenic (As) < 0,10 mg/kg
J8308 I6 Cadmium (Cd) Method: DIN EN 15763:2010 (2010-04), mod. ICP-MS
(a) Cadmium (Cd) 0,01 (± 0,01) mg/kg
J8306 I6 Lead (ICP-MS, food) Method: DIN EN 15763:2010 (2010-04), mod. ICP-MS
(a) Lead (Pb) 0,05 (± 0,04) mg/kg
JCHG2 I6 Mercury (Hg) Method: DIN EN 15763:2010 (2010-04), mod. ICP-MS
(a) Mercury (Hg) < 0,005 mg/kg

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[REDACTED]
Analytical Services Manager (ASM) CFI001

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00069543
Analytical Report Nr. AR-24-FL-035073-01

Date 18.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results (MU)

J8312	I6	Arsenic (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Arsenic (As)		< 0,10 mg/kg
J8308	I6	Cadmium (Cd)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Cadmium (Cd)		< 0,01 mg/kg
J8306	I6	Lead (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Lead (Pb)		0,06 (± 0,04) mg/kg
JCHG2	I6	Mercury (Hg)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Mercury (Hg)		< 0,005 mg/kg

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[REDACTED]
Analytical Service Manager

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00078458
Analytical Report Nr. AR-24-FL-038552-01

Date 14.11.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [REDACTED]

Sample described as: 8711
Sample reception date: 08.11.2024 Analysis Starting Date: 08.11.2024

Results (MU)

J8312	I6	Arsenic (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Arsenic (As)		< 0,10 mg/kg
J8308	I6	Cadmium (Cd)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Cadmium (Cd)		< 0,01 mg/kg
J8306	I6	Lead (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Lead (Pb)		0,05 (± 0,04) mg/kg
JCHG2	I6	Mercury (Hg)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Mercury (Hg)		< 0,005 mg/kg

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Analytical Services Manager (ASM) CFI001

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00069307
Analytical Report Nr. AR-24-FL-035412-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: [Redacted]

Sample described as: onegoID-7695
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)		Aerobic Plate Count 30°C	< 1000	cfu/g
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)		Coliforms 37°C	< 10	cfu/g
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)		Escherichia coli	< 10	cfu/g
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)		Listeria monocytogenes	Not Detected	/25 g
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)		Salmonella	Not Detected	/25 g
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)		Moulds	Est 20	cfu/g
(a)		Yeast	< 10	cfu/g

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[Redacted] Analytical Service Manager Food Microbiology

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069309
Analytical Report Nr. AR-24-FL-035414-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Sample described as: onegoID-6309
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)		Aerobic Plate Count 30°C	< 1000	cfu/g
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)		Coliforms 37°C	< 10	cfu/g
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)		Escherichia coli	< 10	cfu/g
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)		Listeria monocytogenes	Not Detected	/25 g
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)		Salmonella	Not Detected	/25 g
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)		Moulds	< 10	cfu/g
(a)		Yeast	< 10	cfu/g

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Analytical Service Manager Food Microbiology

EXPLANATORY NOTE

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(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069311
Analytical Report Nr. AR-24-FL-035416-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Sample described as: onegoID-8711
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)		Aerobic Plate Count 30°C	< 1000	cfu/g
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)		Coliforms 37°C	< 10	cfu/g
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)		Escherichia coli	< 10	cfu/g
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)		Listeria monocytogenes	Not Detected	/25 g
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)		Salmonella	Not Detected	/25 g
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)		Moulds	< 10	cfu/g
(a)		Yeast	< 10	cfu/g

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Analytical Service Manager Food Microbiology

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069526
Analytical Report Nr. AR-24-FL-035675-01

Date 23.10.2024 Page 1 / 2



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to: Erno.Laasonen@onego-bio.fi; Erna.Hanninen@onego-bio.fi

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score	1,17	
		Note	Protein digestibility provided by client	
		Protein Digestibility	1,00	
		Protein Digestibility Corrected Amino Acid Score	1,00	
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine	59,6	mg/g	
(a)	Arginine	50,1	mg/g	
(a)	Aspartic acid	85,7	mg/g	
(a)	Cystine	12,0	mg/g	
(a)	Glutamic Acid	131	mg/g	
(a)	Glycine	31,4	mg/g	
(a)	Histidine	19,6	mg/g	
(a)	Isoleucine	56,4	mg/g	
(a)	Leucine	80,7	mg/g	
(a)	Lysine	62,1	mg/g	
(a)	Methionine	37,8	mg/g	
(a)	Phenylalanine	61,5	mg/g	
(a)	Proline	34,3	mg/g	
(a)	Serine	66,4	mg/g	
(a)	Threonine	38,3	mg/g	
(a)	Tyrosine	37,6	mg/g	
(a)	Valine	60,9	mg/g	
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein	88,2	%	
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan	13,8	mg/g	

SIGNATURE

Analytical Services Manager (ASM) CFI001

Sample code Nr. 493-2024-00069526**Date** 23.10.2024 **Page 2 / 2****Analytical Report Nr.** AR-24-FL-035675-01**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FS - Eurofins Food Chemistry Testing US Madison, USA - ISO/IEC 17025:2017 A2LA 2918.01

Sample code Nr. 493-2024-00069540
Analytical Report Nr. AR-24-FL-035694-01

Date 23.10.2024 Page 1 / 2



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score		1,19
		Note	Protein digestibility provided by the client	
		Protein Digestibility		1,00
		Protein Digestibility Corrected Amino Acid Score		1,00
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine			57,2 mg/g
(a)	Arginine			50,7 mg/g
(a)	Aspartic acid			83,3 mg/g
(a)	Cystine			12,1 mg/g
(a)	Glutamic Acid			130 mg/g
(a)	Glycine			30,0 mg/g
(a)	Histidine			19,4 mg/g
(a)	Isoleucine			56,9 mg/g
(a)	Leucine			80,4 mg/g
(a)	Lysine			62,3 mg/g
(a)	Methionine			38,0 mg/g
(a)	Phenylalanine			61,0 mg/g
(a)	Proline			32,9 mg/g
(a)	Serine			65,7 mg/g
(a)	Threonine			37,4 mg/g
(a)	Tyrosine			36,2 mg/g
(a)	Valine			60,3 mg/g
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein			85,7 %
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan			12,9 mg/g

SIGNATURE

Analytical Services Manager (ASM) CFI001

Sample code Nr. 493-2024-00069540**Date** 23.10.2024 **Page 2 / 2****Analytical Report Nr.** AR-24-FL-035694-01**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

FS - Eurofins Food Chemistry Testing US Madison, USA - ISO/IEC 17025:2017 A2LA 2918.01

Sample code Nr. 493-2024-00069554
Analytical Report Nr. AR-24-FL-035703-02

Date 23.10.2024 Page 1 / 2



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score	1,20	
		Note	Protein digestibility provided by the client	
		Protein Digestibility	1,00	
		Protein Digestibility Corrected Amino Acid Score	1,00	
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine	58,6	mg/g	
(a)	Arginine	50,1	mg/g	
(a)	Aspartic acid	81,0	mg/g	
(a)	Cystine	12,4	mg/g	
(a)	Glutamic Acid	131	mg/g	
(a)	Glycine	29,0	mg/g	
(a)	Histidine	19,3	mg/g	
(a)	Isoleucine	56,7	mg/g	
(a)	Leucine	79,6	mg/g	
(a)	Lysine	61,2	mg/g	
(a)	Methionine	39,1	mg/g	
(a)	Phenylalanine	60,7	mg/g	
(a)	Proline	32,5	mg/g	
(a)	Serine	65,9	mg/g	
(a)	Threonine	35,2	mg/g	
(a)	Tyrosine	35,3	mg/g	
(a)	Valine	60,6	mg/g	
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein	84,6	%	
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan	12,7	mg/g	

SIGNATURE

Analytical Services Manager (ASM) CFI001

Sample code Nr. 493-2024-00069554**Date** 23.10.2024 **Page** 2 / 2**Analytical Report Nr.** AR-24-FL-035703-02**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

FS - Eurofins Food Chemistry Testing US Madison, USA - ISO/IEC 17025:2017 A2LA 2918.01

Sample code Nr. 493-2024-00069530
Analytical Report Nr. AR-24-FL-035160-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

JJ0EW	I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod.	IAC-LC-FLD
(a)	Aflatoxin B1		< 1,0 µg/kg	
(a)	Aflatoxin B2		< 1,0 µg/kg	
(a)	Aflatoxin G1		< 1,0 µg/kg	
(a)	Aflatoxin G2		< 1,0 µg/kg	
(a)	Sum of all positive Aflatoxins		not calculable µg/kg	
JJ613	I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding	Grinding
(a)	Sample preparation		done	
JJ0BG	I6	Fumonisin	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Fumonisin B1 (FB1)		< 20 µg/kg	
(a)	Fumonisin B2 (FB2)		< 20 µg/kg	
(a)	Fumonisin sum (B1+B2)		not calculable µg/kg	
JJ0FE	I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Deoxynivalenol (Vomitoxin)		< 20 µg/kg	
(a)	HT-2 Toxin		< 10 µg/kg	
(a)	sum T-2 HT-2 toxin		not calculable µg/kg	
(a)	T-2 Toxin		< 10 µg/kg	
(a)	Zearalenone (ZON)		< 10 µg/kg	
JJ0G5	I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod.	IAC-LC-FLD
(a)	Ochratoxin A		< 2,0 µg/kg	

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Analytical Service Manager

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAKKS D-PL-14602-01-00

Sample code Nr. 493-2024-00069544
Analytical Report Nr. AR-24-FL-035162-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

JJ0EW I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod. IAC-LC-FLD
(a) Aflatoxin B1	< 1,0 µg/kg	
(a) Aflatoxin B2	< 1,0 µg/kg	
(a) Aflatoxin G1	< 1,0 µg/kg	
(a) Aflatoxin G2	< 1,0 µg/kg	
(a) Sum of all positive Aflatoxins	not calculable µg/kg	
JJ613 I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding Grinding
(a) Sample preparation	done	
JJ0BG I6	Fumonisin	Method: Internal Method, LC-MS/MS LC-MS/MS
(a) Fumonisin B1 (FB1)	< 20 µg/kg	
(a) Fumonisin B2 (FB2)	< 20 µg/kg	
(a) Fumonisin sum (B1+B2)	not calculable µg/kg	
JJ0FE I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS LC-MS/MS
(a) Deoxynivalenol (Vomitoxin)	< 20 µg/kg	
(a) HT-2 Toxin	< 10 µg/kg	
(a) sum T-2 HT-2 toxin	not calculable µg/kg	
(a) T-2 Toxin	< 10 µg/kg	
(a) Zearalenone (ZON)	< 10 µg/kg	
JJ0G5 I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod. IAC-LC-FLD
(a) Ochratoxin A	< 2,0 µg/kg	

SIGNATURE

Analytical Service Manager

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00069558
Analytical Report Nr. AR-24-FL-035164-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd

Client Code:: FL0003394

Hämeentie 157
00560 HELSINKI
FINLAND

Copy to:

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

JJ0EW I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod. IAC-LC-FLD
(a) Aflatoxin B1	< 1,0 µg/kg	
(a) Aflatoxin B2	< 1,0 µg/kg	
(a) Aflatoxin G1	< 1,0 µg/kg	
(a) Aflatoxin G2	< 1,0 µg/kg	
(a) Sum of all positive Aflatoxins	not calculable µg/kg	
JJ613 I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding Grinding
(a) Sample preparation	done	
JJ0BG I6	Fumonisin	Method: Internal Method, LC-MS/MS LC-MS/MS
(a) Fumonisin B1 (FB1)	< 20 µg/kg	
(a) Fumonisin B2 (FB2)	< 20 µg/kg	
(a) Fumonisin sum (B1+B2)	not calculable µg/kg	
JJ0FE I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS LC-MS/MS
(a) Deoxynivalenol (Vomitoxin)	< 20 µg/kg	
(a) HT-2 Toxin	< 10 µg/kg	
(a) sum T-2 HT-2 toxin	not calculable µg/kg	
(a) T-2 Toxin	< 10 µg/kg	
(a) Zearalenone (ZON)	< 10 µg/kg	
JJ0G5 I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod. IAC-LC-FLD
(a) Ochratoxin A	< 2,0 µg/kg	

SIGNATURE

Analytical Service Manager

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

DEPARTMENT OF HEALTH AND HUMAN SERVICES Food and Drug Administration GENERALLY RECOGNIZED AS SAFE (GRAS) NOTICE (Subpart E of Part 170)	Form Approved: OMB No. 0910-0342; Expiration Date: 08/31/2025 (See last page for OMB Statement)	
	FDA USE ONLY	
	GRN NUMBER 001249	DATE OF RECEIPT Dec 19, 2024
	ESTIMATED DAILY INTAKE	INTENDED USE FOR INTERNET
	NAME FOR INTERNET	
KEYWORDS		

Transmit completed form and attachments electronically via the Electronic Submission Gateway (*see Instructions*); OR Transmit completed form and attachments in paper format or on physical media to: Office of Food Additive Safety (*HFS-200*), Center for Food Safety and Applied Nutrition, Food and Drug Administration, 5001 Campus Drive, College Park, MD 20740-3835.

SECTION A – INTRODUCTORY INFORMATION ABOUT THE SUBMISSION

1. Type of Submission (<i>Check one</i>)	
☒ New	☐ Amendment to GRN No. _____ ☐ Supplement to GRN No. _____
2. ☒ All electronic files included in this submission have been checked and found to be virus free. (<i>Check box to verify</i>)	
3. Most recent presubmission meeting (<i>if any</i>) with FDA on the subject substance (<i>yyyy/mm/dd</i>): 2024-10-30	
4. For Amendments or Supplements: Is your amendment or supplement submitted in response to a communication from FDA? (<i>Check one</i>)	
☐ Yes	If yes, enter the date of communication (<i>yyyy/mm/dd</i>): _____
☐ No	

SECTION B – INFORMATION ABOUT THE NOTIFIER

1a. Notifier	Name of Contact Person Maija Itkonen		Position or Title CEO	
	Organization (<i>if applicable</i>) Onego Bio, Inc.			
	Mailing Address (<i>number and street</i>) 9215 Brown Deer Road, Suite C			
City San Diego		State or Province California	Zip Code/Postal Code 92121	Country United States of America
Telephone Number +1 650 613 3712		Fax Number	E-Mail Address maija@onego.bio	
1b. Agent or Attorney (<i>if applicable</i>)	Name of Contact Person Diana Rueda		Position or Title Global Regulatory Affairs Manager	
	Organization (<i>if applicable</i>) Atova Regulatory Consulting SLU			
	Mailing Address (<i>number and street</i>) Passeig de Gracia, 50, 5			
City Barcelona		State or Province Barcelona	Zip Code/Postal Code 08019	Country Spain
Telephone Number +34 686 999 247		Fax Number	E-Mail Address diana@atovaconsulting.com	

SECTION C – GENERAL ADMINISTRATIVE INFORMATION

1. Name of notified substance, using an appropriately descriptive term

Ovalbumin from fermentation by *Trichoderma reesei*

2. Submission Format: *(Check appropriate box(es))*

☒ Electronic Submission Gateway

☐ Electronic files on physical media

☐ Paper

If applicable give number and type of physical media

3. For paper submissions only:

Number of volumes _____

Total number of pages _____

4. Does this submission incorporate any information in CFSAN's files? *(Check one)*

☐ Yes *(Proceed to Item 5)*

☒ No *(Proceed to Item 6)*

5. The submission incorporates information from a previous submission to FDA as indicated below *(Check all that apply)*

☐ a) GRAS Notice No. GRN _____

☐ b) GRAS Affirmation Petition No. GRP _____

☐ c) Food Additive Petition No. FAP _____

☐ d) Food Master File No. FMF _____

☐ e) Other or Additional *(describe or enter information as above)* _____

6. Statutory basis for conclusions of GRAS status *(Check one)*

☒ Scientific procedures *(21 CFR 170.30(a) and (b))*

☐ Experience based on common use in food *(21 CFR 170.30(a) and (c))*

7. Does the submission (including information that you are incorporating) contain information that you view as trade secret or as confidential commercial or financial information? (see 21 CFR 170.225(c)(8))

☐ Yes *(Proceed to Item 8)*

☒ No *(Proceed to Section D)*

8. Have you designated information in your submission that you view as trade secret or as confidential commercial or financial information *(Check all that apply)*

☐ Yes, information is designated at the place where it occurs in the submission

☐ No

9. Have you attached a redacted copy of some or all of the submission? *(Check one)*

☐ Yes, a redacted copy of the complete submission

☐ Yes, a redacted copy of part(s) of the submission

☐ No

SECTION D – INTENDED USE

1. Describe the intended conditions of use of the notified substance, including the foods in which the substance will be used, the levels of use in such foods, and the purposes for which the substance will be used, including, when appropriate, a description of a subpopulation expected to consume the notified substance.

Ovalbumin from fermentation by *T. reesei* is intended to be used as a general-purpose protein ingredient, at a maximum incorporation rate of 25%, in a wide range of food categories that include baked goods and baking mixes, beverages and beverage bases (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections and frostings, dairy product analogs, egg products, fats and oils, frozen dairy desserts and mixes, fruit and water ices, gelatins, puddings, and fillings, grain products and pastas, gravies and sauces, milk products, plant protein products, snack foods (protein and snack bars), and soft candy.

2. Does the intended use of the notified substance include any use in product(s) subject to regulation by the Food Safety and Inspection Service (FSIS) of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☒ No

3. If your submission contains trade secrets, do you authorize FDA to provide this information to the Food Safety and Inspection Service of the U.S. Department of Agriculture?

(Check one)

☐ Yes

☐ No, you ask us to exclude trade secrets from the information FDA will send to FSIS.

SECTION E – PARTS 2 -7 OF YOUR GRAS NOTICE

(check list to help ensure your submission is complete – PART 1 is addressed in other sections of this form)

- ☒ PART 2 of a GRAS notice: Identity, method of manufacture, specifications, and physical or technical effect (170.230).
- ☒ PART 3 of a GRAS notice: Dietary exposure (170.235).
- ☒ PART 4 of a GRAS notice: Self-limiting levels of use (170.240).
- ☒ PART 5 of a GRAS notice: Experience based on common use in foods before 1958 (170.245).
- ☒ PART 6 of a GRAS notice: Narrative (170.250).
- ☒ PART 7 of a GRAS notice: List of supporting data and information in your GRAS notice (170.255)

Other Information

Did you include any other information that you want FDA to consider in evaluating your GRAS notice?

☐ Yes ☒ No

Did you include this other information in the list of attachments?

☐ Yes ☐ No

SECTION F – SIGNATURE AND CERTIFICATION STATEMENTS

1. The undersigned is informing FDA that Onego Bio, Inc.

(name of notifier)

has concluded that the intended use(s) of Ovalbumin from fermentation by Trichoderma reesei

(name of notified substance)

described on this form, as discussed in the attached notice, is (are) not subject to the premarket approval requirements of the Federal Food, Drug, and Cosmetic Act based on your conclusion that the substance is generally recognized as safe recognized as safe under the conditions of its intended use in accordance with § 170.30.

2. Onego Bio, Inc. agrees to make the data and information that are the basis for the
(name of notifier) conclusion of GRAS status available to FDA if FDA asks to see them;
agrees to allow FDA to review and copy these data and information during customary business hours at the following location if FDA asks to do so; agrees to send these data and information to FDA if FDA asks to do so.

9215 Brown Deer Road, Suite C. San Diego, CA 92121, USA

(address of notifier or other location)

The notifying party certifies that this GRAS notice is a complete, representative, and balanced submission that includes unfavorable, as well as favorable information, pertinent to the evaluation of the safety and GRAS status of the use of the substance. The notifying party certifies that the information provided herein is accurate and complete to the best of his/her knowledge. Any knowing and willful misinterpretation is subject to criminal penalty pursuant to 18 U.S.C. 1001.

3. Signature of Responsible Official,
Agent, or Attorney

Diana Rueda

Digitally signed by Diana Rueda
Date: 2024.12.19 19:06:18 +08'00'

Printed Name and Title

Diana Rueda, Global Regulatory Affairs Manager

Date (mm/dd/yyyy)

12/18/2024

SECTION G – LIST OF ATTACHMENTS

List your attached files or documents containing your submission, forms, amendments or supplements, and other pertinent information. Clearly identify the attachment with appropriate descriptive file names (or titles for paper documents), preferably as suggested in the guidance associated with this form. Number your attachments consecutively. When submitting paper documents, enter the inclusive page numbers of each portion of the document below.

Attachment Number	Attachment Name	Folder Location (select from menu) (Page Number(s) for paper Copy Only)
	Form3667.pdf	Administrative
	GRAS_Cover_letter_Onego_Bio_Signed.pdf	Administrative
	GRAS_Cover_letter_Onego_Bio_Signed_38132.pdf	Administrative
	Confidential_GRAS_dossier_OnegoBio_Ovalbumin_T_reesei_Signed.pdf	Administrative

OMB Statement: Public reporting burden for this collection of information is estimated to average 170 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: Department of Health and Human Services, Food and Drug Administration, Office of Chief Information Officer, PRASStaff@fda.hhs.gov. (Please do NOT return the form to this address.). An agency may not conduct or sponsor, and a person is not required to respond to, a collection of information unless it displays a currently valid OMB control number.

From: [Diana Rueda](#)
To: [Kampmeyer, Christopher](#)
Subject: [EXTERNAL] Re: Clarification regarding your submission to the FDA GRAS Notification Program
Date: Wednesday, February 12, 2025 9:36:12 PM
Attachments: image001.png
image003.png
image004.png
GRAS_Appendix-1_OnegoBio_Ovalbumin_T_reesei_unredacted.pdf

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Mr. Kampmeyer,

Many thanks for your email. Below, please find the answers to your questions. I have highlighted them **in red**, for your reference.

- Q: In Part 1, the submission notes that information in the submission are not trade secret or confidential and therefore not exempt from the Freedom of Information Act. However, the submission file name contains the word, “Confidential” within. Could you please confirm the information in the file labeled “Confidential_GRAS_dossier_OnegoBio_Ovalbumin_T_reesei_Signed.pdf” does not actually contain confidential information?
A: I confirm that the file does not contain any confidential information. The use of the term “confidential” in the file title follows our standard protocol. However, the entire file submitted to the FDA can be considered non-confidential.
- Q: In addition, Appendix I contains certificates of analysis that have a variety of redactions, suggesting the information is confidential. Could you please provide a copy of an unredacted version of the CoAs in response to this email? Please note that we redact written signatures on GRNs posted to the GRN Inventory to be consistent with HHS policy; however, signatures would still be releasable under FOIA.
A: Please find attached the unredacted copy of the CoAs. It should be noted that the only information redacted pertains to personal details, such as names, email addresses, telephone numbers, and signatures. No data was redacted.

Please feel free to reach out if you have any further questions or require additional information during the review of this GRN. We will be happy to assist.

With kind regards,

Dr. Diana Rueda
Global Regulatory Affairs Manager

Based in Shanghai | Available until 12 PM CET

Emails received after this time will be answered starting at 2 AM CET the next day



Tel: [+86 150 2183 2447](tel:+8615021832447) | diana@atovaconsulting.com | www.atovaconsulting.com



The information contained in this message and/or attached file(s), sent by Atova Regulatory Consulting, SLU is confidential/privileged and is intended to be read solely by the person(s) to whom it is addressed. We hereby inform you that your data will be incorporated into the processing system owned by Atova Regulatory Consulting, SLU and that provided it complies with the requirements included in the law, you may exercise the rights of access, rectification, limitation of processing, deletion, portability and opposition by sending your request to the email address info@atovaconsulting.com. If you read this message and you are neither the intended recipient, employee or agent in charge of delivering the message to the recipient, or if you have received this message as the result of an error, we hereby inform you that distributing, sharing or reproducing any part of this message is totally forbidden and may be illegal. As a result, we request that you immediately inform us of the error and return this message to the address mentioned above. Thank you.

From: Kampmeyer, Christopher <Christopher.Kampmeyer@fda.hhs.gov>

Date: Thursday, 13 February 2025 at 00:02

To: Diana Rueda <diana@atovaconsulting.com>

Subject: Clarification regarding your submission to the FDA GRAS Notification Program

Dear Ms. Rueda,

This email is regarding your submission regarding uses of “ovalbumin from fermentation by *Trichoderma reesei*” in food. During our initial review, we noted the following points of clarification.

- In Part 1, the submission notes that information in the submission are not trade secret or confidential and therefore not exempt from the Freedom of Information Act. However, the submission file name contains the word, “Confidential” within. Could you please confirm the information in the file labeled “Confidential_GRAS_dossier_OnegoBio_Ovalbumin_T_reesei_Signed.pdf” does not actually contain confidential information?
- In addition, Appendix I contains certificates of analysis that have a variety of redactions, suggesting the information is confidential. Could you please provide a copy of an unredacted version of the CoAs in response to this email? Please note that we redact written signatures on GRNs posted to the GRN Inventory to be consistent with HHS policy; however, signatures would still be releasable under FOIA.

Best regards,

Chris

Chris Kampmeyer, M.S.

Regulatory Review Scientist

Regulatory Management Branch

Division of Food Ingredients

Office of Pre-Market Additive Safety

Office of Food Chemical Safety, Dietary Supplements, and Innovation

Human Foods Program

U.S. Food and Drug Administration

christopher.kampmeyer@fda.hhs.gov



Sample code Nr. 493-2024-00069527
Analytical Report Nr. AR-24-FL-033959-02

Date 14.10.2024 **Page** 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-033959-01/493-2024-00069527 dated 10/10/2024 which must be destroyed)

Onego Bio Ltd
Chris Landowski
 Hämeentie 157
 00560 HELSINKI
 FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference:	OnegoID-7695		
Sample described as:	OnegoID-7695		
Sample reception date:	03.10.2024	Analysis Starting Date:	03.10.2024
Batch comment:	Ei-eläinperäinen ovalbumiinijauhe		

Results (MU)

FL077	FL	pH from spreads/margarines and foods	Method: AOAC 981.12 (1982)	Potentiometry
	pH		6,4 (± 0,1)	

SIGNATURE

Iida Valta

Analytical Service Manager Food Chemistry
 +358444939285

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069541
Analytical Report Nr. AR-24-FL-033961-02

Date 14.10.2024 Page 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-033961-01/493-2024-00069541 dated 10/10/2024 which must be destroyed)

Onego Bio Ltd

Client Code:: FL0003394

Chris Landowski

Hämeentie 157

00560 HELSINKI

FINLAND

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference:	onegoID-6309		
Sample described as:	OnegoID-6309		
Sample reception date:	03.10.2024	Analysis Starting Date:	03.10.2024
Batch comment:	Ei-eläinperäinen ovalbumiinijauhe		

Results (MU)

FL077	FL	pH from spreads/margarines and foods	Method: AOAC 981.12 (1982)	Potentiometry
	pH		6,8 (± 0,1)	

SIGNATURE

Iida Valta

Analytical Service Manager Food Chemistry

+358444939285

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069555
Analytical Report Nr. AR-24-FL-033963-02

Date 14.10.2024 **Page** 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-033963-01/493-2024-00069555 dated 10/10/2024 which must be destroyed)

Onego Bio Ltd
Chris Landowski
 Hämeentie 157
 00560 HELSINKI
 FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference:	onegoID-8711		
Sample described as:	OnegoID-8711		
Sample reception date:	03.10.2024	Analysis Starting Date:	03.10.2024
Batch comment:	Ei-eläinperäinen ovalbumiinijauhe		

Results (MU)

FL077	FL	pH from spreads/margarines and foods	Method: AOAC 981.12 (1982)	Potentiometry
	pH		6,7 (± 0,1)	

SIGNATURE

iida Valta
 Analytical Service Manager Food Chemistry
 +358444939285

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND -

Sample code Nr. 493-2024-00069528
Analytical Report Nr. AR-24-FL-034365-02

Date 25.10.2024 Page 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-034365-01/493-2024-00069528 dated 14/10/2024 which must be destroyed)

Onego Bio Ltd
Chris Landowski
Hämeentie 157
00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients		Results (MU)	
FL15N	FL	Fat content (foods), Gerhardt Method: NMKL 160 mod.	Gravimetry
(a)	Fat		<0,2 (± 0,0) g/100 g
FL036	FL	Protein (Kjeldahl) Method: NMKL 6:2003, mod.	Kjeldahl (titrimetry)
(a)	Protein (Nx6,25)		85,1 (± 6,0) g/100 g
FL037	FL	Ash (foods, feeds) Method: NMKL 173:2005, mod.	Gravimetry
(a)	Ash		3,0 (± 0,4) g/100 g
FL052	FL	Carbohydrates (calculated) Calculation	
	Carbohydrates (calc.)		3,9 g/100 g
FL053	FL	Energy content Method: EU dir. 90/496/EY Calculation	
	Energy value (kcal)		356 kcal/100 g
	Energy value (kJ)		1513 kJ/100 g
Analyses on: reanalysis			
FL033	FL	Moisture (foods, feeds) Method: NMKL 23:1991, mod.	Gravimetry
(a)	Moisture		8,0 (± 1,0) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

SIGNATURE

Iida Valta
Analytical Service Manager Food Chemistry
+358444939285

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069542
Analytical Report Nr. AR-24-FL-034367-02

Date 25.10.2024 Page 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-034367-01/493-2024-00069542 dated 14/10/2024 which must be destroyed)

Onego Bio Ltd
Chris Landowski
Hämeentie 157
00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients			Results (MU)	
FL15N	FL	Fat content (foods), Gerhardt	Method: NMKL 160 mod.	Gravimetry
(a)	Fat			<0,2 (± -0,0) g/100 g
FL036	FL	Protein (Kjeldahl)	Method: NMKL 6:2003, mod.	Kjeldahl (titrimetry)
(a)	Protein (Nx6,25)			85,8 (± 6,0) g/100 g
FL037	FL	Ash (foods, feeds)	Method: NMKL 173:2005, mod.	Gravimetry
(a)	Ash			3,2 (± 0,4) g/100 g
FL052	FL	Carbohydrates (calculated)	Calculation	
	Carbohydrates (calc.)			4,1 g/100 g
FL053	FL	Energy content	Method: EU dir. 90/496/EY	Calculation
	Energy value (kcal)			360 kcal/100 g
	Energy value (kJ)			1528 kJ/100 g
Analyses on: Reanalysis				
FL033	FL	Moisture (foods, feeds)	Method: NMKL 23:1991, mod.	Gravimetry
(a)	Moisture			6,9 (± 0,8) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

SIGNATURE

Iida Valta

Analytical Service Manager Food Chemistry
+358444939285

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FL - Eurofins Scientific Finland Raisio (Food & Agro), FINLAND - SFS-EN ISO/IEC 17025:2017 FINAS T089

Sample code Nr. 493-2024-00069556
Analytical Report Nr. AR-24-FL-034369-03

Date 25.10.2024 Page 1 / 1



(*this report cancels and replaces the previous one, numbered AR-24-FL-034369-02/493-2024-00069556 dated 22/10/2024 which must be destroyed)

Onego Bio Ltd
Chris Landowski
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00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Food nutrients		Results (MU)	
FL15N	FL	Fat content (foods), Gerhardt Method: NMKL 160 mod.	Gravimetry
(a)	Fat		<0,2 (± 0,0) g/100 g
FL036	FL	Protein (Kjeldahl) Method: NMKL 6:2003, mod.	Kjeldahl (titrimetry)
(a)	Protein (Nx6,25)		84,6 (± 5,9) g/100 g
FL037	FL	Ash (foods, feeds) Method: NMKL 173:2005, mod.	Gravimetry
(a)	Ash		4,4 (± 0,5) g/100 g
FL052	FL	Carbohydrates (calculated) Calculation	
	Carbohydrates (calc.)		6,1 g/100 g
FL053	FL	Energy content Method: EU dir. 90/496/EY	Calculation
	Energy value (kcal)		363 kcal/100 g
	Energy value (kJ)		1542 kJ/100 g
Analyses on: reanalysis			
FL033	FL	Moisture (foods, feeds) Method: NMKL 23:1991, mod.	Gravimetry
(a)	Moisture		4,9 (± 0,6) g/100 g

New analytical report version. Result of FL033 (moisture) is substituted by result of reanalysis. Carbohydrate and energy value are calculated based on reanalysed result of moisture.

SIGNATURE

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EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

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Sample code Nr. 493-2024-00069529
Analytical Report Nr. AR-24-FL-034959-01

Date 18.10.2024 Page 1 / 1



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Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results (MU)

J8312	I6	Arsenic (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Arsenic (As)	< 0,10 mg/kg	
J8308	I6	Cadmium (Cd)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Cadmium (Cd)	0,01 (± 0,01) mg/kg	
J8306	I6	Lead (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Lead (Pb)	0,05 (± 0,04) mg/kg	
JCHG2	I6	Mercury (Hg)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Mercury (Hg)	< 0,005 mg/kg	

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EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00069543
Analytical Report Nr. AR-24-FL-035073-01

Date 18.10.2024 Page 1 / 1



Onego Bio Ltd
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Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results (MU)

J8312	I6	Arsenic (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)	Arsenic (As)			< 0,10 mg/kg
J8308	I6	Cadmium (Cd)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)	Cadmium (Cd)			< 0,01 mg/kg
J8306	I6	Lead (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)	Lead (Pb)			0,06 (± 0,04) mg/kg
JCHG2	I6	Mercury (Hg)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)	Mercury (Hg)			< 0,005 mg/kg

SIGNATURE

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EXPLANATORY NOTE

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I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00078458
Analytical Report Nr. AR-24-FL-038552-01

Date 14.11.2024 Page 1 / 1



Onego Bio Ltd
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FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio)

Sample described as: 8711
Sample reception date: 08.11.2024 Analysis Starting Date: 08.11.2024

Results (MU)

J8312	I6	Arsenic (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Arsenic (As)	< 0,10 mg/kg	
J8308	I6	Cadmium (Cd)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Cadmium (Cd)	< 0,01 mg/kg	
J8306	I6	Lead (ICP-MS, food)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Lead (Pb)	0,05 (± 0,04) mg/kg	
JCHG2	I6	Mercury (Hg)	Method: DIN EN 15763:2010 (2010-04), mod.	ICP-MS
(a)		Mercury (Hg)	< 0,005 mg/kg	

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(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAkkS D-PL-14602-01-00

Sample code Nr. 493-2024-00069307
Analytical Report Nr. AR-24-FL-035412-01

Date 21.10.2024 Page 1 / 1



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Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen (elviira.karkkainen@onego.bio)

Sample described as: onegoID-7695
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)		Aerobic Plate Count 30°C	< 1000	cfu/g
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)		Coliforms 37°C	< 10	cfu/g
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)		Escherichia coli	< 10	cfu/g
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)		Listeria monocytogenes	Not Detected	/25 g
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)		Salmonella	Not Detected	/25 g
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)		Moulds	Est 20	cfu/g
(a)		Yeast	< 10	cfu/g

SIGNATURE

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EXPLANATORY NOTE

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Sample code Nr. 493-2024-00069309
Analytical Report Nr. AR-24-FL-035414-01

Date 21.10.2024 Page 1 / 1



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Sample described as: onegoID-6309
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)	Aerobic Plate Count 30°C	< 1000 cfu/g		
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)	Coliforms 37°C	< 10 cfu/g		
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)	Escherichia coli	< 10 cfu/g		
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)	Listeria monocytogenes	Not Detected /25 g		
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)	Salmonella	Not Detected /25 g		
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)	Moulds	< 10 cfu/g		
(a)	Yeast	< 10 cfu/g		

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(MU) = Expanded measurement uncertainty (k=2)

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Sample code Nr. 493-2024-00069311
Analytical Report Nr. AR-24-FL-035416-01

Date 21.10.2024 Page 1 / 1



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Client Code:: FL0003394

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Sample described as: onegoID-8711
Sample reception date: 04.10.2024 Analysis Starting Date: 04.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe 23.9.2024

Results

UMH31	FL	Aerobic Plate Count 30°C <1000 >2500000 NMKL 86	Method: NMKL 86:2013	E-Cultural technique (non-chromogenic media)
(a)		Aerobic Plate Count 30°C	< 1000	cfu/g
UMZ1B	FL	Coliforms 37°C E <10 >10000 /g (1-2) VRB agar-P NMKL 44	Method: NMKL 44:2004	E-Cultural technique (non-chromogenic media)
(a)		Coliforms 37°C	< 10	cfu/g
ZM91M	FL	Escherichia coli E [FI Food] <10 >1 500 /g (1) TBX Agar-P ISO 16649-2	Method: ISO 16649-2:2001	E-Cultural technique (chromogenic media)
(a)		Escherichia coli	< 10	cfu/g
ZM920	FL	Listeria monocytogenes D Abs Pres /25 g AFNOR BRD 07/04-09/98	Method: BRD 07/04-09/98	D-Cultural technique (chromogenic media)
(a)		Listeria monocytogenes	Not Detected	/25 g
UM218	FL	Salmonella D Abs Pres /25 g BACGene Salmonella spp	Method: BACGene Salmonella spp	Real-time PCR
(a)		Salmonella	Not Detected	/25 g
UMNFZ	FL	Yeast-Moulds E <10 >4500 /g (1) DRBC Agar-S NMKL 98	Method: NMKL 98:2005 mod.	E-Cultural technique (non-chromogenic media)
(a)		Moulds	< 10	cfu/g
(a)		Yeast	< 10	cfu/g

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(MU) = Expanded measurement uncertainty (k=2)

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Sample code Nr. 493-2024-00069526
Analytical Report Nr. AR-24-FL-035675-01

Date 23.10.2024 Page 1 / 2



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(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score		1,17
		Note	Protein digestibility provided by client	
		Protein Digestibility		1,00
		Protein Digestibility Corrected Amino Acid Score		1,00
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine			59,6 mg/g
(a)	Arginine			50,1 mg/g
(a)	Aspartic acid			85,7 mg/g
(a)	Cystine			12,0 mg/g
(a)	Glutamic Acid			131 mg/g
(a)	Glycine			31,4 mg/g
(a)	Histidine			19,6 mg/g
(a)	Isoleucine			56,4 mg/g
(a)	Leucine			80,7 mg/g
(a)	Lysine			62,1 mg/g
(a)	Methionine			37,8 mg/g
(a)	Phenylalanine			61,5 mg/g
(a)	Proline			34,3 mg/g
(a)	Serine			66,4 mg/g
(a)	Threonine			38,3 mg/g
(a)	Tyrosine			37,6 mg/g
(a)	Valine			60,9 mg/g
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein			88,2 %
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan			13,8 mg/g

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Sample code Nr. 493-2024-00069526**Date** 23.10.2024**Page** 2 / 2**Analytical Report Nr.** AR-24-FL-035675-01**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

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Sample code Nr. 493-2024-00069540
Analytical Report Nr. AR-24-FL-035694-01

Date 23.10.2024 Page 1 / 2



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Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score		1,19
		Note	Protein digestibility provided by the client	
		Protein Digestibility		1,00
		Protein Digestibility Corrected Amino Acid Score		1,00
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine			57,2 mg/g
(a)	Arginine			50,7 mg/g
(a)	Aspartic acid			83,3 mg/g
(a)	Cystine			12,1 mg/g
(a)	Glutamic Acid			130 mg/g
(a)	Glycine			30,0 mg/g
(a)	Histidine			19,4 mg/g
(a)	Isoleucine			56,9 mg/g
(a)	Leucine			80,4 mg/g
(a)	Lysine			62,3 mg/g
(a)	Methionine			38,0 mg/g
(a)	Phenylalanine			61,0 mg/g
(a)	Proline			32,9 mg/g
(a)	Serine			65,7 mg/g
(a)	Threonine			37,4 mg/g
(a)	Tyrosine			36,2 mg/g
(a)	Valine			60,3 mg/g
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein			85,7 %
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan			12,9 mg/g

SIGNATURE



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Sample code Nr. 493-2024-00069540**Date** 23.10.2024**Page** 2 / 2**Analytical Report Nr.** AR-24-FL-035694-01**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty ($k=2$)

FS - Eurofins Food Chemistry Testing US Madison, USA - ISO/IEC 17025:2017 A2LA 2918.01

Sample code Nr. 493-2024-00069554
Analytical Report Nr. AR-24-FL-035703-02

Date 23.10.2024 Page 1 / 2



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Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results

FS0FU	FS	Protein Digestibility Corrected Amino Acid Score	Method: Federal Register, Vol 58, No 3, p2194	Calculation
		Amino Acid Score	1,20	
		Note	Protein digestibility provided by the client	
		Protein Digestibility	1,00	
		Protein Digestibility Corrected Amino Acid Score	1,00	
FS05F	FS	Total Amino Acids Profile	Method: Agilent Application Note 5990-4547, (2010); Analytical Biochemistry 177, 318-322; J of Chromatography, 431, pp 271-284	LC-FLD UV
(a)	Alanine	58,6	mg/g	
(a)	Arginine	50,1	mg/g	
(a)	Aspartic acid	81,0	mg/g	
(a)	Cystine	12,4	mg/g	
(a)	Glutamic Acid	131	mg/g	
(a)	Glycine	29,0	mg/g	
(a)	Histidine	19,3	mg/g	
(a)	Isoleucine	56,7	mg/g	
(a)	Leucine	79,6	mg/g	
(a)	Lysine	61,2	mg/g	
(a)	Methionine	39,1	mg/g	
(a)	Phenylalanine	60,7	mg/g	
(a)	Proline	32,5	mg/g	
(a)	Serine	65,9	mg/g	
(a)	Threonine	35,2	mg/g	
(a)	Tyrosine	35,3	mg/g	
(a)	Valine	60,6	mg/g	
FS04U	FS	Total Protein (Dumas) (For Most Products N x 6.25)	Method: AOAC 968.06 (mod.), AOAC 992.15 (mod.)	Dumas (TCD)
(a)	Protein	84,6	%	
FS05E	FS	Total Tryptophan by HPLC	Method: Agilent Publication, 2000. (Modified); Agilent Application Note 5990-4547, (2010); J of Chromatography, 431, pp 271-284; AOAC 988.15	LC-FLD
(a)	Tryptophan	12,7	mg/g	

SIGNATURE



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Sample code Nr. 493-2024-00069554**Date** 23.10.2024**Page** 2 / 2**Analytical Report Nr.** AR-24-FL-035703-02**EXPLANATORY NOTE**

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

FS - Eurofins Food Chemistry Testing US Madison, USA - ISO/IEC 17025:2017 A2LA 2918.01

Sample code Nr. 493-2024-00069530
Analytical Report Nr. AR-24-FL-035160-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd
Chris Landowski
Hämeentie 157
00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-7695
Sample described as: OnegoID-7695
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinijauhe

Results

JJ0EW	I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod.	IAC-LC-FLD
(a)	Aflatoxin B1	< 1,0 µg/kg		
(a)	Aflatoxin B2	< 1,0 µg/kg		
(a)	Aflatoxin G1	< 1,0 µg/kg		
(a)	Aflatoxin G2	< 1,0 µg/kg		
(a)	Sum of all positive Aflatoxins	not calculable µg/kg		
JJ613	I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding	Grinding
(a)	Sample preparation	done		
JJ0BG	I6	Fumonisin	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Fumonisin B1 (FB1)	< 20 µg/kg		
(a)	Fumonisin B2 (FB2)	< 20 µg/kg		
(a)	Fumonisin sum (B1+B2)	not calculable µg/kg		
JJ0FE	I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Deoxynivalenol (Vomitoxin)	< 20 µg/kg		
(a)	HT-2 Toxin	< 10 µg/kg		
(a)	sum T-2 HT-2 toxin	not calculable µg/kg		
(a)	T-2 Toxin	< 10 µg/kg		
(a)	Zearalenone (ZON)	< 10 µg/kg		
JJ0G5	I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod.	IAC-LC-FLD
(a)	Ochratoxin A	< 2,0 µg/kg		

SIGNATURE

Mira Mäntynen
Analytical Service Manager
+358 44 781 9026

EXPLANATORY NOTE

This report shall not be reproduced except in full. The results relate only to the sample analyzed. The laboratory is not responsible for the accuracy of the information received from the customer. Accredited methods of the analyzing laboratory have been evaluated by the national accreditation body of the country in question. Additional information on measurement uncertainties are available upon request. This report has been created electronically and has been verified and authorized. The results apply to the sample as received.

(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAKKS D-PL-14602-01-00

Sample code Nr. 493-2024-00069544
Analytical Report Nr. AR-24-FL-035162-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd
Chris Landowski
Hämeentie 157
00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-6309
Sample described as: OnegoID-6309
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results

JJ0EW	I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod.	IAC-LC-FLD
(a)	Aflatoxin B1	< 1,0 µg/kg		
(a)	Aflatoxin B2	< 1,0 µg/kg		
(a)	Aflatoxin G1	< 1,0 µg/kg		
(a)	Aflatoxin G2	< 1,0 µg/kg		
(a)	Sum of all positive Aflatoxins	not calculable µg/kg		
JJ613	I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding	Grinding
(a)	Sample preparation	done		
JJ0BG	I6	Fumonisin	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Fumonisin B1 (FB1)	< 20 µg/kg		
(a)	Fumonisin B2 (FB2)	< 20 µg/kg		
(a)	Fumonisin sum (B1+B2)	not calculable µg/kg		
JJ0FE	I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Deoxynivalenol (Vomitoxin)	< 20 µg/kg		
(a)	HT-2 Toxin	< 10 µg/kg		
(a)	sum T-2 HT-2 toxin	not calculable µg/kg		
(a)	T-2 Toxin	< 10 µg/kg		
(a)	Zearalenone (ZON)	< 10 µg/kg		
JJ0G5	I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod.	IAC-LC-FLD
(a)	Ochratoxin A	< 2,0 µg/kg		

SIGNATURE

Mira Mäntynen
Analytical Service Manager
+358 44 781 9026

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAKKS D-PL-14602-01-00

Sample code Nr. 493-2024-00069558
Analytical Report Nr. AR-24-FL-035164-01

Date 21.10.2024 Page 1 / 1



Onego Bio Ltd
Chris Landowski
Hämeentie 157
00560 HELSINKI
FINLAND

Client Code:: FL0003394

Copy to: Chris Landowski (chris@onego.bio), Elviira Kärkkäinen
(elviira.karkkainen@onego.bio)

Client Reference: OnegoID-8711
Sample described as: OnegoID-8711
Sample reception date: 03.10.2024 Analysis Starting Date: 03.10.2024
Batch comment: Ei-eläinperäinen ovalbumiinjauhe

Results

JJ0EW	I6	Aflatoxin B1, B2, G1, G2 (spices, special matrix)	Method: DIN EN 14123 (2008-03), mod.	IAC-LC-FLD
(a)	Aflatoxin B1	< 1,0 µg/kg		
(a)	Aflatoxin B2	< 1,0 µg/kg		
(a)	Aflatoxin G1	< 1,0 µg/kg		
(a)	Aflatoxin G2	< 1,0 µg/kg		
(a)	Sum of all positive Aflatoxins	not calculable µg/kg		
JJ613	I6	Dry-Homogenization, Dry milling for myco- and biotoxins	Method: Internal Method, Grinding	Grinding
(a)	Sample preparation	done		
JJ0BG	I6	Fumonisin	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Fumonisin B1 (FB1)	< 20 µg/kg		
(a)	Fumonisin B2 (FB2)	< 20 µg/kg		
(a)	Fumonisin sum (B1+B2)	not calculable µg/kg		
JJ0FE	I6	Fusarium toxins (DON, ZON, T2, HT2)	Method: Internal Method, LC-MS/MS	LC-MS/MS
(a)	Deoxynivalenol (Vomitoxin)	< 20 µg/kg		
(a)	HT-2 Toxin	< 10 µg/kg		
(a)	sum T-2 HT-2 toxin	not calculable µg/kg		
(a)	T-2 Toxin	< 10 µg/kg		
(a)	Zearalenone (ZON)	< 10 µg/kg		
JJ0G5	I6	Ochratoxin A (spices, special matrix)	Method: DIN EN 14132 (2009-09), mod.	IAC-LC-FLD
(a)	Ochratoxin A	< 2,0 µg/kg		

SIGNATURE

Mira Mäntynen
Analytical Service Manager
+358 44 781 9026

EXPLANATORY NOTE

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(a) = Accredited analysis

(MU) = Expanded measurement uncertainty (k=2)

I6 - Eurofins WEJ Contaminants GmbH (HH), GERMANY - DIN EN ISO/IEC 17025:2018 DAKKS D-PL-14602-01-00

From: [Diana Rueda](#)
To: [Wafula, Denis](#)
Subject: [EXTERNAL] Re: GRN 1249: Questions from the Technical Review Team
Date: Wednesday, June 11, 2025 9:29:42 AM
Attachments: [GRN 1249 Response to FDA RFI 1 11JUN25.pdf](#)
[GRN 1249 Ovalbumin from fermentation by T reesei NHANES food codes.pdf](#)

CAUTION: This email originated from outside of the organization. Do not click links or open attachments unless you recognize the sender and know the content is safe.

Dear Dr. Wafula,

I hope this email finds you well.

Please find attached Onego Bio's responses to the technical questions for GRN 1249, as included in your email dated 30 May 2025. I am submitting two files: (1) the main response letter addressing questions 1 through 9, and (2) a separate file listing the NHANES food codes referenced in Onego Bio's response to question 8.

Should you have any further questions or require additional clarification, we would be pleased to assist.

With kind regards,

Dr. Diana Rueda
Global Regulatory Affairs Manager

Based in Shanghai | Available until 12 PM CET
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Tel: [+86 150 2183 2447](tel:+8615021832447) | diana@atovaconsulting.com | www.atovaconsulting.com

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From: Diana Rueda <diana@atovaconsulting.com>

Date: Monday, 2 June 2025 at 10:55

To: Wafula, Denis <Denis.Wafula@fda.hhs.gov>

Subject: Re: GRN 1249: Questions from the Technical Review Team

Dear Dr. Wafula,

Thank you for your email. We acknowledge receipt of the questions and will review them carefully. We aim to provide a complete response within the requested timeframe of 10 business days.

With kind regards,

Dr. Diana Rueda

Global Regulatory Affairs Manager

Based in Shanghai | Available until 12 PM CET

Emails received after this time will be answered starting at 2 AM CET the next day



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From: Wafula, Denis <Denis.Wafula@fda.hhs.gov>

Date: Friday, 30 May 2025 at 22:46

To: Diana Rueda <diana@atovaconsulting.com>

Subject: GRN 1249: Questions from the Technical Review Team

Dear Diana Rueda,

During our review of GRN 1249, we noted several questions that need to be addressed and are attached to this email. Please provide a response within 10 business days. If you are unable to complete the response within that time frame, please contact me to discuss further options. Please do not include any confidential information in your responses. If you have any questions or need further clarification, please feel free to reach out to me.

Sincerely,

Denis Wafula

Denis Wafula, Ph.D.

Biologist

Office of Pre-Market Additive Safety



Onego Bio, Inc.
9215 Brown Deer Road, Suite C
San Diego, CA 92121
USA

June 11, 2025

Dr. Denis Wafula

Office of Pre-Market Additive Safety
Office of Food Chemical Safety, Dietary Supplements, and Innovation
Human Foods Program
US Food and Drug Administration
5001 Campus Drive
College Park, MD 20740

RE: GRAS Notice No. GRN 001249 – response to technical review questions from May 30, 2025

Dear Dr. Wafula,

This letter responds to your email dated May 30, 2025, regarding GRAS Notice No. GRN 001249. Included below are Onego Bio's responses to the technical questions outlined in your correspondence. For clarity, each question is restated as presented in the original communication, followed by the corresponding response.

We would appreciate confirmation of receipt of this response. If you have any questions or need additional information, please feel free to contact us – we will be happy to assist promptly.

Thank you for your attention.

Sincerely,



Maija Itkonen
CEO
Onego Bio, Inc.
+358 440 593313 (Finland) / +1 650 613 3712 (USA)
maija@onego.bio



GENERAL COMMENTS

Question 1: The following references contain errors, please address as appropriate;

- The reference “OECD 1993” on page 16 is not listed in the bibliography.
- Based on the bibliography, the references Lucey *et.al.*, 2029 on page 34 should be Lucey *et al.*, 2019, and Jasmin *et. al.*, 2121 on page 57 should be Jasmin *et. al.*, 2021.

Response 1

- The citation “OECD 1993” in page 16 contains a typographical error and should read “OECD 1998”; the reference is as listed in Section 7.1 – List of references.
- The entries contained typographical errors: “2029” on page 34 should have read “2019”, and “2121” on page 57 should have read “2021”. The correct entries are as follows:
 - o Citation on page 34: “(...). As in Lucey et al. (2019), no adverse effects were observed, further reinforcing the safety of egg protein hydrolysates in adults with metabolic syndrome.”
 - o Reference on page 57: Jasim, H.A., Misnan, R., Yadzir, Z.H.M., Abdullah, N., Bakhtiar, F., Arip, M., Ateshan, H.M., Keong, P.B. (2021). Identification of common and novel major crab allergens in *Scylla tranquebarica* and the allergen stability in untreated and vinegar-treated crab. *IJAAI*, 20(1), 76-87.

MICROBIOLOGY

Question 2: In Part 2.2.2. (page 17), the notifier states that “The approach included the selective modulation of endogenous transcription factors and the integration of exogenous transcription factors from non-pathogenic and non-toxicogenic sources.”

- Please clarify whether the DNA fragments encoding for the exogenous transcription factors are *de novo* synthesized or are cloned from the species of origin.
- Please clarify whether any undesirable secondary metabolites are produced from the production strain during fermentation and further co-purified with the ovalbumin.

Response 2:

- The DNA fragments encoding the exogenous transcription factors are *de novo* synthesized by GenScript, based on publicly available sequences. The transcription factors are plant-derived and sourced from non-pathogenic and non-toxicogenic organisms with a history of safe use in food or feed applications.
- No undesirable secondary metabolites are expected to be produced by *T. reesei* TR112 under the fermentation conditions employed by Onego Bio. As outlined in Table 15 of GRN 1249, a bioinformatics analysis was conducted on *T. reesei* TR112 to screen for genes and gene clusters associated with secondary metabolite biosynthesis. While the results suggest a theoretical potential for mycotoxin production, this is common to most fungal species and does not indicate active toxin synthesis under production conditions.

To verify this, analytical testing for mycotoxins was conducted on three non-consecutive batches of NOVA, as described in Section 2.5 of GRN 1249. No mycotoxins of food safety concern were detected (for the limit of detection (LOD) of the methods employed, please see our answer to Question 3 below).

Additionally, as noted in Section 6.3.1 of the GRAS notice, the parental strain *T. reesei* QM6a and its derivatives are eligible for exemption under the U.S. EPA's TSCA biotechnology regulations (40 CFR 725.420) when used in standard submerged fermentation, due to their non-pathogenic and non-toxicogenic nature. *T. reesei* has a well-established history of safe use in industrial fermentation, including in the production of GRAS substances, and qualifies for use



under the lowest containment level in accordance with Good Industrial Large-Scale Practice (GILSP) as defined by the OECD. As such, no other undesirable secondary metabolites are expected during the production of ovalbumin from fermentation with *T. reesei*.

CHEMISTRY

Question 3: Please provide the approximate molecular mass of the ingredient in kDa to support the claim of equivalence in molecular weight to native ovalbumin (OVA).

Answer 3:

The molecular weight of NOVA and OVA was determined to be approximately 42.75 kDa by intact mass spectrometry, consistent with the reported molecular weight of native chicken ovalbumin (Nisbet et al., 1981; UniProt, entry P01012). Both values reflect the unmodified amino acid sequence and do not include any post-translational modifications.

Nisbet, A.D., Saundry, R.H., Moir, A.J.G., Fothergill, L.A., Fothergill, J.E. (1981). The Complete Amino-Acid Sequence of Hen Ovalbumin. *European Journal of Biochemistry*, 115: 335-345.

Question 4: Please provide the Limit of Detection (LOD) for the LC-MS/MS test method used for mycotoxin analysis (as referenced in Table 7).

Answer 4:

Please find the methods of analysis for mycotoxins (as indicated in the CoAs included in the GRAS notice) with their respective limits of detection (LOD) and limits of quantification (LOQ) in table A1 below.

Table A1. Analytical methods used for mycotoxin detection in ovalbumin from fermentation by <i>T. reesei</i>			
Parameter	Method	LOD (µg/kg)	LOQ (µg/kg)
Aflatoxin B1	DIN EN 14123 (2008-03), mod. IAC-LC-FLD	0.33	1
Aflatoxin B2		0.33	1
Aflatoxin G1		0.33	1
Aflatoxin G2		0.33	1
Fumonisin B1	Internal method, LC-MS/MS	6.67	20
Fumonisin B2		6.67	20
Deoxynivalenol	Internal method, LC-MS/MS	6.67	20
HT2-toxin		3.33	10
T2-toxin		3.33	10
Zearalenone		3.33	10
Ochratoxin A	DIN EN 14132 (2009-09), mod. IAC-LC-FLD	0.67	2

DIN = Deutsches Institut für Normung (German Institute for Standardization); EN= European Norm; FLD = fluorescence detection; IAC = immunoaffinity column; LC = liquid chromatography; LOD = Limit of Detection; LOQ = Limit of Quantification

Question 5: Please provide a statement confirming that all raw materials and processing aids used in the manufacture of the ingredient are used in accordance with applicable U.S. regulations, are GRAS for their intended use, or are the subject of an effective food contact notification.

Answer 5:

We confirm that all raw materials and processing aids used in the manufacture of the ingredient are used in accordance with applicable US regulation, are GRAS for the intended use, or are the subject of an effective food contact notification.



Question 6: Please clarify if the intended food uses encompass broad food categories listed in the first column of Table 10 or are limited to the representative food categories listed in the second column of Table 10. If the intended food uses encompass the broad food categories in the first column of Table 1, please update your dietary exposure estimate accordingly to include all foods within each broad food category.

Answer 6:

The intended food uses encompass the broad food categories listed in the first column of Table 10. The dietary exposure assessment was conducted accordingly and includes all applicable and representative foods within each broad category; therefore, no update is required. The NHANES food codes selected for each intended food category are provided in a separate file (see Question 8 below).

Question 7: The intended food uses listed in Table 10 include alcoholic beverages. Please note that FDA's fortification policy does not support the addition of nutrients such as protein to alcoholic beverages. Please clarify the technical effect of your ingredient in alcoholic beverages.

Answer 7:

The reference to alcoholic beverages in Table 10 pertains specifically to products such as eggnog and egg-containing cocktails, where ovalbumin from fermentation with *T. reesei* is intended to serve as a functional substitute for egg. In this context, the ingredient is not added for nutritional fortification, but rather to replicate the technical and organoleptic properties of traditional egg-based formulations.

Question 8: Additionally, please provide the specific NHANES food codes selected as representative for each intended food category, along with the corresponding intended use levels of the ingredient in those categories.

Answer 8:

The specific NHANES food codes selected for the dietary exposure assessment of Onego's NOVA, as presented in Part 3 of GRN 1249, are presented in a separate file entitled GRN_1249_Ovalbumin_from_fermentation_by_T_reesei_NHANES_food_codes. The intended levels of use of the notified ingredient are provided for each food selected.

TOXICOLOGY

Question 9: Please perform an updated literature search and confirm whether you have retrieved/uncovered any data and information since you submitted the GRN that could contradict your GRAS conclusion.

Answer 9:

A literature search was conducted on June 6 – June 9, 2025, using Google Scholar and PubMed for peer-reviewed scientific literature, and Google for supplementary grey literature and regulatory updates. No new information was identified that contradicts Onego Bio's GRAS conclusion regarding the safety ovalbumin, *T. reesei*, or the raw materials used in manufacture.

NHANES food codes for the dietary exposure assessment of ovalbumin from fermentation by *T. reesei*

As described in Part 3 of GRN 1249, ovalbumin from *T. reesei* fermentation is intended as a general-purpose protein ingredient, used at up to 25 g per 100 g in a broad range of food categories listed in Table 10. These categories, defined in 21 CFR § 170.3, include baked goods, beverages (alcoholic and non-alcoholic), breakfast cereals, cheeses, confections, dairy analogs, egg products, fats and oils, frozen desserts, fruit ices, gelatins, grain products, gravies, milk products, plant proteins, snack foods (including bars), and soft candy.

The estimated daily intake (EDI), also detailed in Part 3, was calculated using food consumption data from the 2017–2020 pre-pandemic NHANES. The data sources, methods, and results are summarized in Part 3 of GRN 1249. The NHANES 2017–2020 food codes used in the assessment, along with the corresponding use levels of the ingredient applied in the EDI calculation, are listed below.

21 CFR § 170.3(n)(1) Baked goods and baking mixes

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
13252600	Tiramisu	6
14630200	Cheese souffle	8
51000100	Bread, NS as to major flour	3
51000110	Bread, NS as to major flour, toasted	3
51000180	Bread, made from home recipe or purchased at a bakery, NS as to major flour	3
51000190	Bread, made from home recipe or purchased at a bakery, toasted, NS as to major flour	3
51000200	Roll, NS as to major flour	3
51000300	Roll, hard, NS as to major flour	3
51000400	Roll, bran, NS as to type of bran	3
51101000	Bread, white	3
51101010	Bread, white, toasted	3
51101050	Bread, white, made from home recipe or purchased at a bakery	3
51101060	Bread, white, made from home recipe or purchased at a bakery, toasted	3
51102010	Bread, white with whole wheat swirl	3
51102020	Bread, white with whole wheat swirl, toasted	3
51105010	Bread, Cuban	3
51105040	Bread, Cuban, toasted	3
51106010	Bread, native, water, Puerto Rican style	3
51106020	Bread, native, water, toasted, Puerto Rican style	3
51106200	Bread, lard, Puerto Rican style	3
51106210	Bread, lard, toasted, Puerto Rican style	3
51106300	Bread, caressed, Puerto Rican style	3
51106310	Bread, caressed, toasted, Puerto Rican style	3
51107010	Bread, French or Vienna	3
51107040	Bread, French or Vienna, toasted	3
51108010	Focaccia, Italian, plain	3
51108100	Bread, naan	3
51109010	Bread, Italian, Grecian, Armenian	3
51109040	Bread, Italian, Grecian, Armenian, toasted	3
51109100	Bread, pita	3
51109110	Bread, pita, toasted	3
51109150	Bread, pita with fruit	3
51109200	Bread, pita with fruit, toasted	3
51111010	Bread, cheese	3
51111040	Bread, cheese, toasted	3
51113010	Bread, cinnamon	3
51113100	Bread, cinnamon, toasted	3
51115010	Bread, cornmeal and molasses	3
51115020	Bread, cornmeal and molasses, toasted	3
51119010	Bread, egg, Challah	3

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
51119040	Bread, egg, Challah, toasted	3
51121015	Garlic bread, NFS	3
51121025	Garlic bread, from fast food / restaurant	3
51121035	Garlic bread, from frozen	3
51121045	Garlic bread, with parmesan cheese, from fast food / restaurant	3
51121055	Garlic bread, with parmesan cheese, from frozen	3
51121065	Garlic bread, with melted cheese, from fast food / restaurant	3
51121075	Garlic bread, with melted cheese, from frozen	3
51121110	Bread, onion	3
51121120	Bread, onion, toasted	3
51122000	Bread, reduced calorie and/or high fiber, white or NFS	3
51122010	Bread, reduced calorie and/or high fiber, white or NFS, toasted	3
51122100	Bread, reduced calorie and/or high fiber, white or NFS, with fruit and/or nuts	3
51122110	Bread, reduced calorie and/or high fiber, white or NFS, with fruit and/or nuts, toasted	3
51122300	Bread, white, special formula, added fiber	3
51122310	Bread, white, special formula, added fiber, toasted	3
51123010	Bread, high protein	25
51123020	Bread, high protein, toasted	25
51127010	Bread, potato	3
51127020	Bread, potato, toasted	3
51129010	Bread, raisin	3
51129020	Bread, raisin, toasted	3
51130510	Bread, white, low sodium or no salt	3
51130520	Bread, white, low sodium or no salt, toasted	3
51133010	Bread, sour dough	3
51133020	Bread, sour dough, toasted	3
51134000	Bread, sweet potato	3
51134010	Bread, sweet potato, toasted	3
51135000	Bread, vegetable	3
51135010	Bread, vegetable, toasted	3
51140100	Bread, dough, fried	3
51150000	Roll, white, soft	3
51153000	Roll, white, hard	3
51154010	Roll, white, hot dog bun	3
51154100	Roll, white, hamburger bun	3
51154510	Roll, diet	3
51154550	Roll, egg bread	3
51154600	Roll, cheese	3
51155000	Roll, French or Vienna	3
51156500	Roll, garlic	3
51157000	Roll, white, hoagie, submarine	3
51158100	Roll, Mexican, bolillo	3
51159000	Roll, sour dough	3
51160000	Roll, sweet, no frosting	5
51160100	Roll, sweet, cinnamon bun, no frosting	5
51160110	Roll, sweet, cinnamon bun, frosted	5
51160200	Pan dulce, NFS	5
51161000	Pan Dulce, with fruit, no frosting	5
51161020	Roll, sweet, with fruit, frosted	5
51161030	Roll, sweet, with fruit, frosted, diet	5
51161050	Roll, sweet, frosted	5
51161250	Pan Dulce, no topping	5
51161270	Pan Dulce, with sugar topping	5
51161280	Pan Dulce, with raisins and icing	5
51165000	Coffee cake, yeast type	5
51166000	Croissant	5
51166100	Croissant, cheese	5
51166200	Croissant, chocolate	5
51166500	Croissant, fruit	5
51167000	Brioche	5
51168000	Bread, Spanish coffee	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
51180010	Bagel	3
51180030	Bagel, with raisins	3
51180080	Bagel, with fruit other than raisins	3
51182010	Bread stuffing	5
51182020	Bread stuffing made with egg	5
51183990	Breadsticks, NFS	3
51184200	Breadsticks, soft, NFS	3
51184210	Breadsticks, soft, fast food / restaurant	3
51184220	Breadsticks, soft, frozen	3
51184230	Breadsticks, soft, with parmesan cheese, fast food / restaurant	3
51184240	Breadsticks, soft, with parmesan cheese, from frozen	3
51184250	Breadsticks, soft, topped with melted cheese	3
51184260	Breadsticks, soft, stuffed or topped with melted cheese	3
51186010	Muffin, English	3
51186100	Muffin, English, with raisins	3
51186130	Muffin, English, cheese	3
51186160	Muffin, English, with fruit other than raisins	3
51187020	Anisette toast	3
51188100	Pannetone	5
51300050	Bread, whole grain white	3
51300060	Bread, whole grain white, toasted	3
51300100	Bagel, whole grain white	3
51300110	Bread, whole wheat	3
51300120	Bread, whole wheat, toasted	3
51300140	Bread, whole wheat, made from home recipe or purchased at bakery	3
51300150	Bread, whole wheat, made from home recipe or purchased at bakery, toasted	3
51300175	Bread, chappatti or roti	3
51300180	Bread, puri	3
51300185	Bread, paratha	3
51300210	Bread, whole wheat, with raisins	3
51300220	Bread, whole wheat, with raisins, toasted	3
51300300	Bread, sprouted wheat	3
51300310	Bread, sprouted wheat, toasted	3
51301010	Bread, wheat or cracked wheat	3
51301020	Bread, wheat or cracked wheat, toasted	3
51301040	Bread, wheat or cracked wheat, made from home recipe or purchased at bakery	3
51301050	Bread, wheat or cracked wheat, made from home recipe or purchased at bakery, toasted	3
51301120	Bread, wheat or cracked wheat, with raisins	3
51301130	Bread, wheat or cracked wheat, with raisins, toasted	3
51301510	Bread, wheat or cracked wheat, reduced calorie and/or high fiber	3
51301520	Bread, wheat or cracked wheat, reduced calorie and/or high fiber, toasted	3
51301540	Bread, French or Vienna, whole wheat	3
51301550	Bread, French or Vienna, whole wheat, toasted	3
51301600	Bread, pita, whole wheat	3
51301610	Bread, pita, whole wheat, toasted	3
51301620	Bread, pita, wheat or cracked wheat	3
51301630	Bread, pita, wheat or cracked wheat, toasted	3
51301700	Bagel, wheat	3
51301750	Bagel, whole wheat	3
51301800	Bagel, wheat, with raisins	3
51301805	Bagel, whole wheat, with raisins	3
51301820	Bagel, wheat, with fruit and nuts	3
51301900	Bagel, wheat bran	3
51302500	Muffin, English, wheat bran	3
51302520	Muffin, English, wheat bran, with raisins	3
51303010	Muffin, English, wheat or cracked wheat	3
51303030	Muffin, English, whole wheat	3
51303050	Muffin, English, wheat or cracked wheat, with raisins	3
51303070	Muffin, English, whole wheat, with raisins	3
51303100	Muffin, English, whole grain white	3

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
51320010	Roll, wheat or cracked wheat	3
51320060	Roll, wheat or cracked wheat, hot dog bun	3
51320070	Roll, wheat or cracked wheat, hamburger bun	3
51320500	Roll, whole wheat	3
51320550	Roll, whole wheat, hot dog bun	3
51320560	Roll, whole wheat, hamburger bun	3
51320700	Roll, whole grain white	3
51320710	Roll, whole grain white, hot dog bun	3
51320720	Roll, whole grain white, hamburger bun	3
51401010	Bread, rye	3
51401020	Bread, rye, toasted	3
51401030	Bread, marble rye and pumpernickel	3
51401040	Bread, marble rye and pumpernickel, toasted	3
51401200	Muffin, English, rye	3
51404010	Bread, pumpernickel	3
51404020	Bread, pumpernickel, toasted	3
51404500	Bagel, pumpernickel	3
51404550	Muffin, English, pumpernickel	3
51407010	Bread, black	3
51407020	Bread, black, toasted	3
51420000	Roll, rye	3
51421000	Roll, pumpernickel	3
51501010	Bread, oatmeal	3
51501020	Bread, oatmeal, toasted	3
51501040	Bread, oat bran	3
51501050	Bread, oat bran, toasted	3
51501080	Bagel, oat bran	3
51502010	Roll, oatmeal	3
51503000	Muffin, English, oat bran	3
51503040	Muffin, English, oat bran, with raisins	3
51601010	Bread, multigrain, toasted	3
51601020	Bread, multigrain	3
51601210	Bread, multigrain, with raisins	3
51601220	Bread, multigrain, with raisins, toasted	3
51602010	Bread, multigrain, reduced calorie and/or high fiber	3
51602020	Bread, multigrain, reduced calorie and/or high fiber, toasted	3
51620000	Roll, multigrain	3
51620020	Roll, multigrain, hot dog bun	3
51620030	Roll, multigrain, hamburger bun	3
51630000	Bagel, multigrain	3
51630100	Bagel, multigrain, with raisins	3
51630200	Muffin, English, multigrain	3
51801010	Bread, barley	3
51801020	Bread, barley, toasted	3
51804010	Bread, soy	3
51804020	Bread, soy, toasted	3
51805010	Bread, sunflower meal	3
51805020	Bread, sunflower meal, toasted	3
51806010	Bread, rice	3
51806020	Bread, rice, toasted	3
51807000	Injera, Ethiopian bread	3
51808000	Bread, gluten free	3
51808010	Bread, gluten free, toasted	3
51808100	Roll, gluten free	3
52101000	Biscuit, NFS	5
52101040	Crumpet	5
52102040	Biscuit, from refrigerated dough	5
52103000	Biscuit, from fast food / restaurant	5
52104010	Biscuit, home recipe	5
52104040	Biscuit, wheat	5
52104100	Biscuit, cheese	5
52104200	Biscuit with fruit	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
52105100	Scone	5
52105200	Scone, with fruit	5
52201000	Cornbread, prepared from mix	5
52202060	Cornbread, made from home recipe	5
52204000	Cornbread stuffing	5
52206010	Cornbread muffin, stick, round	5
52206060	Cornbread muffin, stick, round, made from home recipe	5
52207010	Corn flour patty or tart, fried	5
52208010	Corn pone, baked	5
52208020	Corn pone, fried	5
52209010	Hush puppy	5
52211010	Johnnycake	5
52213010	Spoonbread	5
52220110	Arepa Dominicana	5
52301000	Muffin, NFS	5
52302010	Muffin, fruit	5
52302020	Muffin, fruit, low fat	5
52302500	Muffin, chocolate chip	5
52302600	Muffin, chocolate	5
52303010	Muffin, whole wheat	5
52303500	Muffin, wheat	5
52304000	Muffin, whole grain	5
52304010	Muffin, wheat bran	5
52304040	Muffin, bran with fruit, lowfat	5
52304100	Muffin, oatmeal	5
52304150	Muffin, oat bran	5
52306010	Muffin, plain	5
52306300	Muffin, cheese	5
52306500	Muffin, pumpkin	5
52306550	Muffin, zucchini	5
52306700	Muffin, carrot	5
52311010	Popover	5
52401000	Bread, Boston Brown	5
52403000	Bread, nut	5
52404060	Bread, pumpkin	5
52405010	Bread, fruit	5
52407000	Bread, zucchini	5
52408000	Bread, Irish soda	5
53100050	Cake batter, raw, chocolate	6
53100070	Cake batter, raw, not chocolate	6
53100100	Cake or cupcake, NFS	6
53101100	Cake, angel food	6
53101200	Cake, angel food, with icing or filling	6
53101250	Cake, angel food, with fruit and icing or filling	6
53102100	Cake or cupcake, applesauce, without icing or filling	6
53102200	Cake or cupcake, apple	6
53102600	Cake or cupcake, banana, without icing or filling	6
53102700	Cake or cupcake, banana	6
53102800	Cake or cupcake, Black Forest	6
53103000	Cake, Boston cream pie	6
53104100	Cake or cupcake, carrot, without icing or filling	6
53104260	Cake or cupcake, carrot	6
53104300	Cake, carrot, diet	6
53104400	Cake or cupcake, coconut	6
53104500	Cheesecake, plain	6
53104550	Cheesecake, fruit	6
53104600	Cheesecake, chocolate	6
53105262	Cake or cupcake, chocolate with white icing, bakery	6
53105264	Cake or cupcake, chocolate with white icing, from mix	6
53105270	Cake or cupcake, chocolate with chocolate icing, bakery	6
53105272	Cake or cupcake, chocolate with chocolate icing, from mix	6
53105275	Cake or cupcake, chocolate, no icing	6

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53105300	Cake or cupcake, German chocolate	6
53105310	Cake or cupcake, gluten free	6
53105396	Cake, chocolate, flourless	6
53105500	Cake, chocolate, with icing, diet	6
53106500	Cake, cream	6
53108200	Snack cake, chocolate	6
53108220	Snack cake, chocolate, with icing or filling, reduced fat and calories	6
53109200	Snack cake, white	6
53109220	Snack cake, not chocolate, with icing or filling, reduced fat and calories	6
53109300	Cake, Dobos Torte	6
53110000	Cake, fruit cake	6
53111000	Cake or cupcake, gingerbread	6
53112100	Ice cream cake	6
53113000	Cake, jelly roll	6
53114000	Cake or cupcake, lemon, without icing or filling	6
53114100	Cake or cupcake, lemon	6
53115100	Cake or cupcake, marble, without icing or filling	6
53115200	Cake or cupcake, marble	6
53115310	Cake or cupcake, nut, without icing or filling	6
53115320	Cake or cupcake, nut, with icing or filling	6
53115410	Cake or cupcake, oatmeal	6
53115450	Cake or cupcake, peanut butter	6
53116000	Cake, pound	6
53116020	Cake, pound, with icing or filling	6
53116270	Cake, pound, chocolate	6
53116350	Cake, pound, Puerto Rican style	6
53116390	Cake, pound, reduced fat, cholesterol free	6
53116500	Cake or cupcake, pumpkin, without icing or filling	6
53116510	Cake or cupcake, pumpkin	6
53116520	Cake or cupcake, red velvet	6
53116550	Cake or cupcake, raisin-nut	6
53116570	Cake, Ravani	6
53116600	Cake, rice flour, without icing or filling	6
53116650	Cake, Quezadilla, El Salvadorian style	6
53117100	Cake or cupcake, spice, without icing or filling	6
53117200	Cake or cupcake, spice	6
53118100	Cake, sponge	6
53118110	Cake or cupcake, strawberry	6
53118200	Cake, sponge, with icing or filling	6
53118300	Cake, sponge, chocolate	6
53118410	Rum cake, without icing	6
53118500	Cake, torte	6
53118550	Cake, tres leche	6
53119000	Cake, pineapple, upside down	6
53120270	Cake or cupcake, white with white icing, bakery	6
53120272	Cake or cupcake, white with white icing, from mix	6
53120275	Cake or cupcake, white, without icing or filling	6
53121270	Cake or cupcake, white with chocolate icing, bakery	6
53121272	Cake or cupcake, white with chocolate icing, from mix	6
53121275	Cake or cupcake, white, no icing	6
53122070	Cake, shortcake, biscuit type, with whipped cream and fruit	6
53122080	Cake, shortcake, biscuit type, with fruit	6
53123070	Cake, strawberry shortcake	6
53123080	Cake, shortcake, sponge type, with fruit	6
53123500	Cake, shortcake, with whipped topping and fruit, diet	6
53124110	Cake or cupcake, zucchini	6
53200100	Cookie, batter or dough, raw	3
53201000	Cookie, NFS	3
53202000	Cookie, almond	3
53203000	Cookie, applesauce	3
53203500	Cookie, biscotti	3
53204000	Cookie, brownie, NS as to icing	3

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53204010	Cookie, brownie, without icing	3
53204100	Cookie, brownie, with icing or filling	3
53204840	Cookie, brownie, reduced fat, NS as to icing	3
53204860	Cookie, brownie, fat free, NS as to icing	3
53205250	Cookie, butterscotch, brownie	3
53205260	Cookie, bar, with chocolate	3
53206000	Cookie, chocolate chip	3
53206020	Cookie, chocolate chip, made from home recipe or purchased at a bakery	3
53206030	Cookie, chocolate chip, reduced fat	3
53206100	Cookie, chocolate chip sandwich	3
53206500	Cookie, chocolate, made with rice cereal	3
53206550	Cookie, chocolate, made with oatmeal and coconut, no bake	3
53207000	Cookie, chocolate or fudge	3
53207020	Cookie, chocolate or fudge, reduced fat	3
53207050	Cookie, chocolate, with chocolate filling or coating, fat free	3
53208000	Cookie, marshmallow, chocolate-covered	3
53208200	Cookie, marshmallow pie, chocolate covered	3
53209005	Cookie, chocolate, with icing or coating	3
53209010	Cookie, sugar wafer, chocolate-covered	3
53209015	Cookie, chocolate sandwich	3
53209020	Cookie, chocolate sandwich, reduced fat	3
53209100	Cookie, chocolate, sandwich, with extra filling	3
53209500	Cookie, chocolate and vanilla sandwich	3
53210000	Cookie, chocolate wafer	3
53210900	Cookie, graham cracker with chocolate and marshmallow	3
53211000	Cookie bar, with chocolate, nuts, and graham crackers	3
53215500	Cookie, coconut	3
53220000	Cookie, fruit-filled bar	3
53220010	Cookie, fruit-filled bar, fat free	3
53220030	Cookie, fig bar	3
53220040	Cookie, fig bar, fat free	3
53222010	Cookie, fortune	3
53222020	Cookie, cone shell, ice cream type, wafer or cake	3
53223000	Cookie, gingersnaps	3
53223100	Cookie, granola	3
53224000	Cookie, ladyfinger	3
53224250	Cookie, lemon bar	3
53225000	Cookie, macaroon	3
53226000	Cookie, marshmallow, with coconut	3
53226500	Cookie, marshmallow, with rice cereal, no bake	3
53226550	Cookie, marshmallow, with rice cereal and chocolate chips	3
53226600	Cookie, marshmallow and peanut butter, with oat cereal, no bake	3
53228000	Cookie, meringue	3
53230000	Cookie, molasses	3
53231000	Cookie, Lebkuchen	3
53231400	Cookie, multigrain, high fiber	3
53233000	Cookie, oatmeal	3
53233010	Cookie, oatmeal, with raisins	3
53233040	Cookie, oatmeal, reduced fat, NS as to raisins	3
53233050	Cookie, oatmeal sandwich, with creme filling	3
53233060	Cookie, oatmeal, with chocolate chips	3
53233080	Cookie, oatmeal sandwich, with peanut butter and jelly filling	3
53233100	Cookie, oatmeal, with chocolate and peanut butter, no bake	3
53234000	Cookie, peanut butter	3
53234100	Cookie, peanut butter, with chocolate	3
53234250	Cookie, peanut butter with rice cereal, no bake	3
53235000	Cookie, peanut butter sandwich	3
53235500	Cookie, with peanut butter filling, chocolate-coated	3
53235600	Cookie, Pfefferkuchen	3
53236000	Cookie, Pizzelle	3
53236100	Cookie, pumpkin	3
53237000	Cookie, raisin	3

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53237010	Cookie, raisin sandwich, cream-filled	3
53237500	Cookie, rum ball, no bake	3
53238000	Cookie, sandwich-type, not chocolate or vanilla	3
53239000	Cookie, shortbread	3
53239010	Cookie, shortbread, reduced fat	3
53239050	Cookie, shortbread, with icing or filling	3
53239100	Pocky	3
53240000	Cookie, animal	3
53240010	Cookie, animal, with frosting or icing	3
53241500	Cookie, butter or sugar	3
53241510	Marie biscuit	3
53241600	Cookie, butter or sugar, with fruit and/or nuts	3
53242000	Cookie, sugar wafer	3
53242500	Cookie, toffee bar	3
53243000	Cookie, vanilla sandwich	3
53243010	Cookie, vanilla sandwich, extra filling	3
53243050	Cookie, vanilla sandwich, reduced fat	3
53244010	Cookie, butter or sugar, with chocolate icing or filling	3
53244020	Cookie, butter or sugar, with icing or filling other than chocolate	3
53246000	Cookie, tea, Japanese	3
53247000	Cookie, vanilla wafer	3
53247050	Cookie, vanilla wafer, reduced fat	3
53247500	Cookie, vanilla with caramel, coconut, and chocolate coating	3
53251100	Cookie, rugelach	3
53260030	Cookie, chocolate chip, sugar free	3
53260200	Cookie, oatmeal, sugar free	3
53260300	Cookie, sandwich, sugar free	3
53260400	Cookie, sugar or plain, sugar free	3
53260500	Cookie, sugar wafer, sugar free	3
53260600	Cookie, peanut butter, sugar free	3
53261000	Cookie, gluten free	3
53270100	Cookies, Puerto Rican style	3
53300100	Pie, NFS	5
53300170	Pie, individual size or tart, NFS	5
53300180	Pie, fried, NFS	5
53300200	Tart, all types	5
53301000	Pie, apple	5
53301070	Pie, apple, fast food	5
53301080	Pie, apple, fried pie	5
53301500	Pie, apple, one crust	5
53302000	Pie, apricot, two crust	5
53302070	Pie, apricot, individual size or tart	5
53302080	Pie, apricot, fried pie	5
53303000	Pie, blackberry, two crust	5
53303070	Pie, blackberry, individual size or tart	5
53303500	Pie, berry	5
53303510	Pie, berry, not blackberry, blueberry, boysenberry, huckleberry, raspberry, or strawberry; one crust	5
53303570	Pie, berry, not blackberry, blueberry, boysenberry, huckleberry, raspberry, or strawberry, individual size or tart	5
53304000	Pie, blueberry	5
53304070	Pie, blueberry, individual size or tart	5
53305000	Pie, cherry	5
53305010	Pie, cherry, one crust	5
53305070	Pie, cherry, individual size or tart	5
53305080	Pie, cherry, fried pie	5
53305700	Pie, lemon	5
53305720	Pie, lemon, not cream or meringue, individual size or tart	5
53305750	Pie, lemon, fried pie	5
53305800	Pie, key lime	5
53307000	Pie, peach	5
53307050	Pie, peach, one crust	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53307070	Pie, peach, individual size or tart	5
53307080	Pie, peach, fried pie	5
53307500	Pie, pear, two crust	5
53307570	Pie, pear, individual size or tart	5
53308000	Pie, pineapple, two crust	5
53308070	Pie, pineapple, individual size or tart	5
53309000	Pie, raisin, two crust	5
53309070	Pie, raisin, individual size or tart	5
53310000	Pie, raspberry, one crust	5
53310050	Pie, raspberry, two crust	5
53311000	Pie, rhubarb, two crust	5
53312000	Pie, strawberry	5
53313000	Pie, strawberry-rhubarb, two crust	5
53314000	Pie, strawberry, individual size or tart	5
53340000	Pie, apple-sour cream	5
53340500	Pie, cherry, made with cream cheese and sour cream	5
53341000	Pie, banana cream	5
53341070	Pie, banana cream, individual size or tart	5
53341500	Pie, buttermilk	5
53341750	Pie, chess	5
53342000	Pie, chocolate cream	5
53342070	Pie, chocolate cream, individual size or tart	5
53343000	Pie, coconut cream	5
53343070	Pie, coconut cream, individual size or tart	5
53344000	Pie, custard	5
53344070	Pie, custard, individual size or tart	5
53344200	Mixed fruit tart filled with custard or cream cheese	5
53344300	Dessert pizza	5
53345000	Pie, lemon cream	5
53345070	Pie, lemon cream, individual size or tart	5
53346000	Pie, peanut butter cream	5
53346500	Pie, pineapple cream	5
53347000	Pie, pumpkin	5
53347070	Pie, pumpkin, individual size or tart	5
53347500	Pie, sour cream, raisin	5
53347600	Pie, squash	5
53348000	Pie, strawberry cream	5
53348070	Pie, strawberry cream, individual size or tart	5
53360000	Pie, sweet potato	5
53365000	Pie, vanilla cream	5
53370000	Pie, chiffon, not chocolate	5
53371000	Pie, chiffon, chocolate	5
53373000	Pie, black bottom	5
53381000	Pie, lemon meringue	5
53381070	Pie, lemon meringue, individual size or tart	5
53382000	Pie, chocolate-marshmallow	5
53385000	Pie, pecan	5
53385070	Pie, pecan, individual size or tart	5
53385500	Pie, oatmeal	5
53386000	Pie, pudding, flavors other than chocolate	5
53387000	Pie, Toll house chocolate chip	5
53390000	Pie, shoo-fly	5
53390100	Pie, tofu with fruit	5
53391000	Pie shell	5
53391100	Pie shell, graham cracker	5
53391150	Pie shell, chocolate	5
53391200	Vanilla wafer dessert base	5
53400200	Blintz, cheese-filled	5
53400300	Blintz, fruit-filled	5
53410100	Cobbler, apple	5
53410200	Cobbler, apricot	5
53410300	Cobbler, berry	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53410500	Cobbler, cherry	5
53410800	Cobbler, peach	5
53410850	Cobbler, pear	5
53410880	Cobbler, plum	5
53410900	Cobbler, rhubarb	5
53415100	Crisp, apple	5
53415110	Fritter, plain	5
53415120	Fritter, fruit	5
53415200	Fritter, banana	5
53415220	Fritter, berry	5
53415300	Crisp, berry	5
53415400	Crisp, cherry	5
53415500	Crisp, peach	5
53420000	Cream puff, eclair, custard or cream filled, NS as to icing	5
53420100	Cream puff, eclair, custard or cream filled, not iced	5
53420200	Cream puff, eclair, custard or cream filled, iced	5
53420210	Cream puff, eclair, custard or cream filled, iced, reduced fat	5
53420250	Cream puff, no filling or icing	5
53420300	Air filled fritter or fried puff, without syrup, Puerto Rican style	5
53420310	Wheat flour fritter, without syrup	5
53420400	Sopaipilla, without syrup or honey	5
53420410	Sopaipilla with syrup or honey	5
53430000	Crepe, NS as to filling	5
53430100	Crepe, chocolate filled	5
53430200	Crepe, fruit filled	5
53430700	Tamale, sweet	5
53430750	Tamale, sweet, with fruit	5
53440000	Strudel, apple	5
53440300	Strudel, berry	5
53440500	Strudel, cherry	5
53440600	Strudel, cheese	5
53440700	Strudel, peach	5
53440800	Strudel, cheese and fruit	5
53441110	Baklava	5
53441210	Basbousa	5
53450000	Turnover, fruit	5
53450300	Turnover or dumpling, berry	5
53450500	Turnover or dumpling, cherry	5
53450800	Turnover or dumpling, lemon	5
53451000	Turnover or dumpling, peach	5
53451500	Turnover, guava	5
53451750	Turnover, pumpkin	5
53452100	Pastry, fruit-filled	5
53452120	Pastry, made with bean or lotus seed paste filling, baked	5
53452130	Pastry, made with bean paste and salted egg yolk filling, baked	5
53452150	Pastry, Chinese, made with rice flour	5
53452170	Pastry, cookie type, fried	5
53452200	Pastry, Italian, with cheese	5
53452400	Pastry, puff	5
53452420	Pastry, puff, custard or cream filled, iced or not iced	5
53452450	Cheese pastry puffs	5
53452500	Pastry, mainly flour and water, fried	5
53453150	Empanada, fruit	5
53453170	Empanada, Mexican turnover, pumpkin	5
53500100	Breakfast pastry, NFS	5
53510000	Danish pastry, plain or spice	5
53510100	Danish pastry, with fruit	5
53511000	Danish pastry, with cheese	5
53520000	Doughnut, NFS	5
53520100	Doughnut, cake type, plain	5
53520120	Doughnut, chocolate	5
53520130	Doughnut, cake type, powdered sugar	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53520135	Doughnut, cake type, with icing	5
53520140	Doughnut, cake type, chocolate icing	5
53520160	Doughnut, chocolate, with chocolate icing	5
53520170	Doughnut holes	5
53520200	Churros	5
53520510	Beignet	5
53521110	Doughnut, yeast type	5
53521130	Doughnut, yeast type, with chocolate icing	5
53521140	Doughnut, jelly	5
53521210	Doughnut, custard-filled	5
53521230	Doughnut, custard-filled, with icing	5
53530000	Breakfast tart	5
53530010	Breakfast tart, lowfat	5
53610100	Coffee cake, crumb or quick-bread type	5
53610170	Coffee cake, crumb or quick-bread type, with fruit	5
53610200	Coffee cake, crumb or quick-bread type, cheese-filled	5
54102010	Graham crackers	3
54102015	Graham crackers (Teddy Grahams)	3
54102020	Graham crackers, chocolate covered	3
54102060	Crackers, Cuban	3
54102100	Graham crackers, reduced fat	3
54102200	Graham crackers, sandwich, with filling	3
54103000	Crackers, breakfast biscuit	3
55100005	Pancakes, NFS	5
55100010	Pancakes, plain, frozen	5
55100015	Pancakes, plain, reduced fat, from frozen	5
55100020	Pancakes, fruit, frozen	5
55100025	Pancakes, chocolate, frozen	5
55100030	Pancakes, whole grain, frozen	5
55100035	Pancakes, whole grain, reduced fat, from frozen	5
55100040	Pancakes, gluten free, from frozen	5
55100050	Pancakes, plain, fast food / restaurant	5
55100055	Pancakes, fruit, fast food / restaurant	5
55100060	Pancakes, chocolate, fast food / restaurant	5
55100065	Pancakes, whole grain, fast food / restaurant	5
55100070	Pancakes, whole grain and nuts, from fast food / restaurant	5
55100080	Pancakes, school	5
55101000	Pancakes, plain	5
55101015	Pancakes, plain, reduced fat	5
55103000	Pancakes, fruit	5
55103020	Pancakes, pumpkin	5
55103100	Pancakes, chocolate	5
55105000	Pancakes, buckwheat	5
55105100	Pancakes, cornmeal	5
55105200	Pancakes, whole grain	5
55105205	Pancakes, whole grain, reduced fat	5
55106000	Pancakes, gluten free	5
55200010	Waffle, NFS	5
55200020	Waffle, plain, frozen	5
55200030	Waffle, plain, reduced fat	5
55200040	Waffle, fruit, frozen	5
55200050	Waffle, chocolate, frozen	5
55200060	Waffle, whole grain, frozen	5
55200070	Waffle, whole grain, reduced fat	5
55200080	Waffle, whole grain, fruit, frozen	5
55200090	Waffle, gluten free, from frozen	5
55200100	Waffle, plain, fast food / restaurant	5
55200110	Waffle, chocolate, fast food / restaurant	5
55200120	Waffle, fruit, fast food / restaurant	5
55200130	Waffle, whole grain, fast food / restaurant	5
55200200	Waffle, school	5
55201000	Waffle, plain	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
55203000	Waffle, fruit	5
55203600	Waffle, chocolate	5
55203700	Waffle, cinnamon	5
55204000	Waffle, cornmeal	5
55205000	Waffle, whole grain	5
55208000	Waffle, gluten free	5
55211050	Waffle, plain, reduced fat	5
55212000	Waffle, whole grain, reduced fat	5
55300010	French toast, NFS	5
55300020	French toast, frozen	5
55300030	French toast, whole grain, from frozen	5
55300040	French toast, gluten free, from frozen	5
55300050	French toast, fast food / restaurant	5
55300055	French toast, whole grain, from fast food / restaurant	5
55300060	French toast, school	5
55301000	French toast, plain	5
55301010	French toast, plain, reduced fat	5
55301015	French toast, whole grain	5
55301020	French toast, whole grain, reduced fat	5
55301025	French toast, gluten free	5
55301030	French toast sticks, NFS	5
55301031	French toast sticks	5
55301040	French toast sticks, fast food / restaurant	5
55301048	French toast sticks, school	5
55301050	French toast sticks, plain	5
55301055	French toast sticks, whole grain	5
55310100	Fried bread, Puerto Rican style	5
55400010	Crepe, NFS	5
55401000	Crepe, plain	5
55501000	Chinese pancake	5
55702100	Dosa, plain	5
55801000	Funnel cake with sugar	5
58116100	Empanada, NFS	5
58116115	Empanada, no meat	5
58116310	Cheese turnover, Puerto Rican style	5
58117110	Cornmeal fritter, Puerto Rican style	5
58117210	Cornmeal stick, Puerto Rican style	5
58121610	Pierogi	5
58122250	Kishke, stuffed derma	5
58122310	Knish, potato	5
58122320	Knish	5
58124210	Pastry, cheese-filled	5
58124220	Pastry, egg and cheese filled	5
58124250	Spanakopita	5
58124500	Samosa	5
58125110	Quiche with meat, poultry or fish	12
58125120	Spinach quiche, meatless	12
58125180	Cheese quiche, meatless	12
58126100	Turnover or hot pocket, NFS	5
58126105	Turnover or hot pocket, meatless	5
58126160	Turnover or hot pocket, pizza pocket	5
58126400	Turnover or breakfast pocket, egg	5
58127110	Vegetables in pastry	5
58127150	Vegetables and cheese in pastry	5
58174100	Dosa, with filling	5
58175000	Vada	5
58175110	Tabbouleh	5
71930200	Casabe, cassava bread	5
71945020	Yam buns; Puerto Rican style	5
72125240	Spinach souffle	8
73305020	Squash, winter, souffle	8
75411020	Fritter, corn	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
75418060	Squash, summer, souffle	8
91550100	Coconut cream cake, Puerto Rican style	6
99995000	Breading or batter as ingredient in food	3

21 CFR § 170.3(n)(2) Beverages, alcoholic

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11531000	Eggnog	4
93301000	Cocktail, NFS	4
93301160	Whiskey sour	4
93301230	Sloe gin fizz	4
93301250	White Russian	4
93301450	Liqueur with cream	4
93301550	Eggnog, alcoholic	4
93301600	Gin fizz	4

21 CFR § 170.3(n)(3) Beverages and beverage bases, nonalcoholic

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
64134015	Fruit smoothie, with whole fruit, no dairy	12
64134020	Fruit smoothie, with whole fruit, no dairy, added protein	25
64134025	Fruit smoothie, with whole fruit, non-dairy	12
64134030	Fruit smoothie juice drink, no dairy	12
64134100	Fruit smoothie, light	12
64134200	Fruit smoothie, bottled	12
78101100	Fruit and vegetable smoothie, with dairy	12
78101110	Fruit and vegetable smoothie, added protein	25
78101115	Fruit and vegetable smoothie, non-dairy	12
78101118	Fruit and vegetable smoothie, non-dairy, added protein	25
78101120	Fruit and vegetable smoothie, bottled	12
78101125	Fruit and vegetable smoothie, no dairy	12
78101130	Vegetable smoothie	12
95101000	Nutritional drink or shake, ready-to-drink (Boost)	25
95101010	Nutritional drink or shake, ready-to-drink (Boost Plus)	25
95102000	Nutritional drink or shake, ready-to-drink (Carnation Instant Breakfast)	25
95103000	Nutritional drink or shake, ready-to-drink (Ensure)	25
95103010	Nutritional drink or shake, ready-to-drink (Ensure Plus)	25
95104000	Nutritional drink or shake, ready-to-drink, sugar free (Glucerna)	25
95105000	Nutritional drink or shake, ready-to-drink (Kellogg's Special K Protein)	25
95106000	Nutritional drink or shake, ready-to-drink (Muscle Milk)	25
95106010	Nutritional drink or shake, ready-to-drink, light (Muscle Milk)	25
95110000	Nutritional drink or shake, ready-to-drink (Slim Fast)	25
95110010	Nutritional drink or shake, ready-to-drink, sugar free (Slim Fast)	25
95110020	Nutritional drink or shake, high protein, ready-to-drink (Slim Fast)	25
95120000	Nutritional drink or shake, ready-to-drink, NFS	25
95120010	Nutritional drink or shake, high protein, ready-to-drink, NFS	25
95120020	Nutritional drink or shake, high protein, light, ready-to-drink, NFS	25
95120050	Nutritional drink or shake, liquid, soy-based	25
95201000	Nutritional powder mix (Carnation Instant Breakfast)	25
95201010	Nutritional powder mix, sugar free (Carnation Instant Breakfast)	25
95201200	Nutritional powder mix (EAS Whey Protein Powder)	25
95201300	Nutritional powder mix (EAS Soy Protein Powder)	25
95201500	Nutritional powder mix, high protein (Herbalife)	25
95201600	Nutritional powder mix (Isopure)	25
95201700	Nutritional powder mix (Kellogg's Special K20 Protein Water)	25
95202000	Nutritional powder mix (Muscle Milk)	25

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
95202010	Nutritional powder mix, light (Muscle Milk)	25
95210000	Nutritional powder mix (Slim Fast)	25
95210010	Nutritional powder mix, sugar free (Slim Fast)	25
95210020	Nutritional powder mix, high protein (Slim Fast)	25
95220000	Nutritional powder mix, NFS	25
95220010	Nutritional powder mix, high protein, NFS	25
95230000	Nutritional powder mix, whey based, NFS	25
95230010	Nutritional powder mix, protein, soy based, NFS	25
95230020	Nutritional powder mix, protein, light, NFS	25
95230030	Nutritional powder mix, protein, NFS	25

21 CFR § 170.3(n)(4) Breakfast cereals

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
57000100	Cereal, oat, NFS	10
57100100	Cereal, ready-to-eat, NFS	10
57101000	Cereal (Kellogg's All-Bran)	10
57103000	Cereal (Post Alpha-Bits)	10
57103100	Cereal, O's, flavored	10
57104000	Cereal (Kellogg's Apple Jacks)	10
57106050	Cereal (Post Great Grains Banana Nut Crunch)	10
57106060	Cereal (General Mills Cheerios Banana Nut)	10
57106260	Cereal (General Mills Cheerios Berry Burst)	10
57117000	Cereal (Quaker Cap'n Crunch)	10
57117500	Cereal (Quaker Christmas Crunch)	10
57119000	Cereal, crunch	10
57120000	Cereal (Quaker Cap'n Crunch's Peanut Butter Crunch)	10
57123000	Cereal, O's, plain	10
57124030	Cereal (General Mills Chex Chocolate)	10
57124050	Cereal (General Mills Chex Cinnamon)	10
57124100	Cereal (General Mills Cheerios Chocolate)	10
57124200	Cereal, chocolate puffs	10
57124300	Cereal (General Mills Lucky Charms Chocolate)	10
57125000	Cereal, cinnamon toast	10
57125010	Cereal (General Mills 25% Less Sugar Cinnamon Toast Crunch)	10
57125900	Cereal (General Mills Honey Nut Clusters)	10
57126000	Cereal, chocolate crispy	10
57127000	Cereal (Post Cocoa Pebbles)	10
57128000	Cereal (General Mills Cocoa Puffs)	10
57130000	Cereal (General Mills Cookie Crisp)	10
57132000	Cereal, corn squares	10
57134000	Cereal, corn flakes, plain	10
57135000	Cereal (Kellogg's Corn Flakes)	10
57137000	Cereal, corn puffs	10
57139000	Cereal (General Mills Count Chocula)	10
57143000	Cereal (Kellogg's Cracklin' Oat Bran)	10
57143500	Cereal (Post Great Grains, Cranberry Almond Crunch)	10
57148000	Cereal (Kellogg's Crispix)	10
57151000	Cereal, rice crispy, plain	10
57206700	Cereal (General Mills Fiber One)	10
57206710	Cereal (General Mills Fiber One Honey Clusters)	10
57206715	Cereal (General Mills Fiber One Raisin Bran Clusters)	10
57207000	Cereal, bran flakes, plain	10
57208000	Cereal (Kellogg's All-Bran Complete Wheat Flakes)	10
57209000	Cereal (Post Bran Flakes)	10
57211000	Cereal (General Mills Frankenberry)	10
57213000	Cereal (Kellogg's Froot Loops)	10
57213010	Cereal (Kellogg's Froot Loops Marshmallow)	10
57213850	Cereal (General Mills Cheerios Frosted)	10

57214000	Cereal, shredded wheat, flavored	10
57216000	Cereal, rice crispy, flavored	10
57221700	Cereal, fruit rings	10
57221810	Cereal (General Mills Cheerios Fruity)	10
57223000	Cereal, fruit crispy	10
57224000	Cereal (General Mills Golden Grahams)	10
57227000	Cereal, granola	10
57228000	Granola, homemade	10
57229000	Cereal (Kellogg's Low Fat Granola)	10
57230000	Cereal (Post Grape-Nuts)	10
57231200	Cereal (Post Great Grains Raisins, Dates, and Pecans)	10
57237100	Cereal, oat bunches	10
57237200	Cereal (Post Honey Bunches of Oats with Vanilla Bunches)	10
57237300	Cereal (Post Honey Bunches of Oats with Almonds)	10
57238000	Cereal (Post Honeycomb)	10
57240100	Cereal, corn squares, flavored	10
57241000	Cereal, O's, honey nut	10
57241200	Cereal (Post Shredded Wheat Honey Nut)	10
57243000	Cereal (Kellogg's Honey Smacks)	10
57301500	Cereal (Kashi 7 Whole Grain Puffs)	10
57301505	Cereal (Kashi Autumn Wheat)	10
57301510	Cereal (Kashi GOLEAN)	10
57301511	Cereal (Kashi GOLEAN Crunch)	10
57301512	Cereal (Kashi GOLEAN Crunch Honey Almond Flax)	10
57301530	Cereal (Kashi Heart to Heart Honey Toasted Oat)	10
57301600	Cereal, multigrain	10
57303100	Cereal (General Mills Kix)	10
57303105	Cereal (General Mills Honey Kix)	10
57303200	Cereal (Kellogg's Krave)	10
57304100	Cereal, oat squares	10
57305100	Cereal (General Mills Lucky Charms)	10
57305150	Cereal, frosted oats with marshmallows	10
57305160	Cereal (Malt-O-Meal Blueberry Muffin Tops)	10
57305165	Cereal (Malt-O-Meal Cinnamon Toasters)	10
57305170	Cereal (Malt-O-Meal Coco-Roos)	10
57305174	Cereal (Malt-O-Meal Colossal Crunch)	10
57305175	Cereal (Malt-O-Meal Cocoa Dyno-Bites)	10
57305180	Cereal (Malt-O-Meal Corn Bursts)	10
57305210	Cereal (Malt-O-Meal Frosted Flakes)	10
57305300	Cereal (Malt-O-Meal Fruity Dyno-Bites)	10
57305400	Cereal (Malt-O-Meal Honey Graham Squares)	10
57305500	Cereal (Malt-O-Meal Honey Nut Toasty O's)	10
57305600	Cereal (Malt-O-Meal Marshmallow Mateys)	10
57306500	Cereal (Malt-O-Meal Golden Puffs)	10
57306700	Cereal (Malt-O-Meal Toasted Oat Cereal)	10
57306800	Cereal (Malt-O-Meal Tootie Fruities)	10
57308190	Cereal, muesli	10
57308400	Cereal, O's, multigrain	10
57309100	Cereal (Nature Valley Granola)	10
57316380	Cereal (General Mills Cheerios Oat Cluster Crunch)	10
57316385	Cereal (General Mills Cheerios Protein)	10
57316450	Cereal (General Mills Oatmeal Crisp with Almonds)	10
57316710	Cereal (Quaker Honey Graham Oh's)	10
57320500	Cereal (Quaker Granola with Oats, Honey, and Raisins)	10
57321900	Cereal (Nature's Path Organic Flax Plus)	10
57326000	Cereal (Barbara's Puffins)	10
57327450	Cereal (Quaker Toasted Oat Bran)	10
57327500	Cereal (Quaker Oatmeal Squares)	10
57329000	Cereal, bran flakes, flavored	10
57330000	Cereal (Kellogg's Raisin Bran)	10
57330010	Cereal (Kellogg's Raisin Bran Crunch)	10
57331000	Cereal (Post Raisin Bran)	10
57332100	Cereal (General Mills Raisin Nut Bran)	10
57335550	Cereal (General Mills Reese's Puffs)	10
57336000	Cereal, rice squares	10
57337000	Cereal, rice flakes	10

57339000	Cereal (Kellogg's Rice Krispies)	10
57339500	Cereal (Kellogg's Rice Krispies Treats Cereal)	10
57340000	Cereal, puffed rice	10
57341200	Cereal (Kellogg's Smart Start Strong)	10
57341300	Cereal (Kellogg's Smorz)	10
57344000	Cereal, K's, plain	10
57344001	Cereal (Kellogg's Special K Blueberry)	10
57344005	Cereal (Kellogg's Special K Chocolatey Delight)	10
57344010	Cereal, K's, flavored	10
57344015	Cereal (Kellogg's Special K Fruit & Yogurt)	10
57344020	Cereal (Kellogg's Special K Vanilla Almond)	10
57344025	Cereal (Kellogg's Special K Cinnamon Pecan)	10
57347000	Cereal, flavored puffs	10
57348000	Cereal, corn flakes, flavored	10
57349000	Cereal (Kellogg's Frosted Flakes)	10
57355000	Cereal (Post Golden Crisp)	10
57401100	Cereal, O's, NFS	10
57407100	Cereal (General Mills Trix)	10
57408100	Cereal (Uncle Sam)	10
57411000	Cereal, wheat squares	10
57416000	Cereal, plain puffs	10
57416010	Cereal, puffed wheat, sweetened	10
57417000	Cereal, shredded wheat, plain	10
57418000	Cereal, wheat flakes	10
57420100	Cereal, other, NFS	10
57420110	Cereal, other, plain	10
57420120	Cereal, other, fruit flavored	10
57420130	Cereal, other, chocolate	10
57420140	Cereal, other, peanut butter	10
57420150	Cereal, other, honey	10
57420160	Cereal, reduced sugar	10

21 CFR § 170.3(n)(5) Cheeses

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
14660200	Mozzarella sticks, breaded, baked, or fried	6

21 CFR § 170.3(n)(9) Confections and frostings

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
91304040	Topping, marshmallow	3
91723000	Candy, marshmallow	3
91723010	Marshmallow, chocolate covered	3
91723020	Marshmallow, candy-coated	3

21 CFR § 170.3(n)(10) Dairy product analogs

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11300100	Non-dairy milk, NFS	10
11320000	Soy milk	10
11320100	Soy milk, light	10
11320200	Soy milk, nonfat	10
11321000	Soy milk, chocolate	10
11321100	Soy milk, light, chocolate	10
11321200	Soy milk, nonfat, chocolate	10
11350000	Almond milk, sweetened	10

11350010	Almond milk, sweetened, chocolate	10
11350020	Almond milk, unsweetened	10
11350030	Almond milk, unsweetened, chocolate	10
11360000	Rice milk	10
11370000	Coconut milk	10
42401010	Coconut milk, used in cooking	10
42402010	Coconut cream, canned, sweetened	10

21 CFR § 170.3(n)(11) Egg analogs

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
31101010	Egg, whole, raw	12
31102000	Egg, whole, cooked, NS as to cooking method	12
31103010	Egg, whole, boiled or poached	12
31105005	Egg, whole, fried, NS as to fat	12
31105010	Egg, whole, fried no added fat	12
31105020	Egg, whole, fried with margarine	12
31105030	Egg, whole, fried with oil	12
31105040	Egg, whole, fried with butter	12
31105060	Egg, whole, fried with animal fat or meat drippings	12
31105080	Egg, whole, fried with cooking spray	12
31105085	Egg, whole, fried, NS as to fat type	12
31105090	Egg, whole, fried, from fast food / restaurant	12
31106000	Egg, whole, baked, NS as to fat	12
31106010	Egg, whole, baked, no added fat	12
31106020	Egg, whole, baked, fat added	12
31107000	Egg, whole, pickled	12
31108010	Egg, white only, raw	12
31108100	Egg, white, cooked, NS as to fat	12
31108110	Egg, white, cooked, no added fat	12
31108120	Egg, white, cooked, fat added	12
31110010	Egg, yolk only, raw	12
31111000	Egg, yolk only, cooked, NS as to fat	12
31111010	Egg, yolk only, cooked, no added fat	12
31111020	Egg, yolk only, cooked, fat added	12
31201000	Duck egg, cooked	12
31202000	Goose egg, cooked	12
31203000	Quail egg, canned	12
32101000	Egg, creamed	12
32101500	Egg, Benedict	12
32102000	Egg, deviled	12
32103000	Egg salad, made with mayonnaise	12
32103015	Egg salad, made with light mayonnaise	12
32103020	Egg salad, made with mayonnaise-type salad dressing	12
32103025	Egg salad, made with light mayonnaise-type salad dressing	12
32103030	Egg salad, made with creamy dressing	12
32103035	Egg salad, made with light creamy dressing	12
32103040	Egg salad, made with Italian dressing	12
32103045	Egg salad, made with light Italian dressing	12
32103050	Egg Salad, made with any type of fat free dressing	12
32105180	Huevos rancheros	12
32105190	Egg casserole with bread, cheese, milk and meat	12
32105310	Ripe plantain omelet, Puerto Rican style	12
32105330	Scrambled eggs with jerked beef, Puerto Rican style	12
32110100	Eggs a la Malaguena, Puerto Rican style	12
32110150	Shrimp-egg patty	12
32129990	Egg omelet or scrambled egg, NS as to fat	12
32130000	Egg omelet or scrambled egg, made with margarine	12
32130010	Egg omelet or scrambled egg, made with oil	12
32130020	Egg omelet or scrambled egg, made with butter	12
32130040	Egg omelet or scrambled egg, made with animal fat or meat drippings	12
32130060	Egg omelet or scrambled egg, made with cooking spray	12
32130065	Egg omelet or scrambled egg, NS as to fat type	12

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
32130070	Egg omelet or scrambled egg, no added fat	12
32130080	Egg omelet or scrambled egg, from fast food / restaurant	12
32130100	Egg omelet or scrambled egg, with cheese, made with margarine	12
32130110	Egg omelet or scrambled egg, with cheese, made with oil	12
32130120	Egg omelet or scrambled egg, with cheese, made with butter	12
32130140	Egg omelet or scrambled egg, with cheese, made with animal fat or meat drippings	12
32130160	Egg omelet or scrambled egg, with cheese, made with cooking spray	12
32130170	Egg omelet or scrambled egg, with cheese, no added fat	12
32130190	Egg omelet or scrambled egg, with meat, NS as to fat	12
32130200	Egg omelet or scrambled egg, with meat, made with margarine	12
32130210	Egg omelet or scrambled egg, with meat, made with oil	12
32130220	Egg omelet or scrambled egg, with meat, made with butter	12
32130240	Egg omelet or scrambled egg, with meat, made with animal fat or meat drippings	12
32130260	Egg omelet or scrambled egg, with meat, made with cooking spray	12
32130265	Egg omelet or scrambled egg, with meat, NS as to fat type	12
32130270	Egg omelet or scrambled egg, with meat, no added fat	12
32130290	Egg omelet or scrambled egg, with cheese and meat, NS as to fat	12
32130300	Egg omelet or scrambled egg, with cheese and meat, made with margarine	12
32130310	Egg omelet or scrambled egg, with cheese and meat, made with oil	12
32130320	Egg omelet or scrambled egg, with cheese and meat, made with butter	12
32130340	Egg omelet or scrambled egg, with cheese and meat, made with animal fat or meat drippings	12
32130360	Egg omelet or scrambled egg, with cheese and meat, made with cooking spray	12
32130365	Egg omelet or scrambled egg, with cheese and meat, NS as to fat type	12
32130370	Egg omelet or scrambled egg, with cheese and meat, no added fat	12
32130400	Egg omelet or scrambled egg, with tomatoes, fat added	12
32130410	Egg omelet or scrambled egg, with tomatoes, no added fat	12
32130420	Egg omelet or scrambled egg, with tomatoes, NS as to fat	12
32130430	Egg omelet or scrambled egg, with dark-green vegetables, fat added	12
32130440	Egg omelet or scrambled egg, with dark-green vegetables, no added fat	12
32130450	Egg omelet or scrambled egg, with dark-green vegetables, NS as to fat	12
32130460	Egg omelet or scrambled egg, with tomatoes and dark-green vegetables, fat added	12
32130470	Egg omelet or scrambled egg, with tomatoes and dark-green vegetables, no fat added	12
32130480	Egg omelet or scrambled egg, with tomatoes and dark-green vegetables, NS as to fat	12
32130490	Egg omelet or scrambled egg, with vegetables other than dark green and/or tomatoes, fat added	12
32130500	Egg omelet or scrambled egg, with vegetables other than dark green and/or tomatoes, no added fat	12
32130510	Egg omelet or scrambled egg, with vegetables other than dark green and/or tomatoes, NS as to fat	12
32130600	Egg omelet or scrambled egg, with cheese and tomatoes, fat added	12
32130610	Egg omelet or scrambled egg, with cheese and tomatoes, no added fat	12
32130620	Egg omelet or scrambled egg, with cheese and tomatoes, NS as to fat	12
32130630	Egg omelet or scrambled egg, with cheese and dark-green vegetables, fat added	12
32130640	Egg omelet or scrambled egg, with cheese and dark-green vegetables, no added fat	12
32130650	Egg omelet or scrambled egg, with cheese and dark-green vegetables, NS as to fat	12
32130660	Egg omelet or scrambled egg, with cheese, tomatoes, and dark-green vegetables, fat added	12
32130670	Egg omelet or scrambled egg, with cheese, tomatoes, and dark-green vegetables, no added fat	12
32130680	Egg omelet or scrambled egg, with cheese, tomatoes, and dark-green vegetables, NS as to fat	12
32130690	Egg omelet or scrambled egg, with cheese and vegetables other than dark green and/or tomatoes, fat added	12

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
32130700	Egg omelet or scrambled egg, with cheese and vegetables other than dark green and/or tomatoes, no added fat	12
32130710	Egg omelet or scrambled egg, with cheese and vegetables other than dark green and/or tomatoes, NS as to fat	12
32130800	Egg omelet or scrambled egg, with meat and tomatoes, fat added	12
32130810	Egg omelet or scrambled egg, with meat and tomatoes, no added fat	12
32130820	Egg omelet or scrambled egg, with meat and tomatoes, NS as to fat	12
32130830	Egg omelet or scrambled egg, with meat and dark-green vegetables, fat added	12
32130840	Egg omelet or scrambled egg, with meat and dark-green vegetables, no added fat	12
32130850	Egg omelet or scrambled egg, with meat and dark-green vegetables, NS as to fat	12
32130860	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, fat added	12
32130870	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, no added fat	12
32130880	Egg omelet or scrambled egg, with meat, tomatoes, and dark-green vegetables, NS as to fat	12
32130890	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, fat added	12
32130900	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, no added fat	12
32130910	Egg omelet or scrambled egg, with meat and vegetables other than dark-green and/or tomatoes, NS as to fat	12
32131000	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, fat added	12
32131010	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, no added fat	12
32131020	Egg omelet or scrambled egg, with cheese, meat, and tomatoes, NS as to fat	12
32131030	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, fat added	12
32131040	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, no added fat	12
32131050	Egg omelet or scrambled egg, with cheese, meat, and dark-green vegetables, NS as to fat	12
32131060	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, fat added	12
32131070	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, no added fat	12
32131080	Egg omelet or scrambled egg, with cheese, meat, tomatoes, and dark-green vegetables, NS as to fat	12
32131090	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, fat added	12
32131100	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, no added fat	12
32131110	Egg omelet or scrambled egg, with cheese, meat, and vegetables other than dark-green and/or tomatoes, NS as to fat	12
32131200	Egg omelet or scrambled egg, with potatoes and/or onions, fat added	12
32131210	Egg omelet or scrambled egg, with potatoes and/or onions, no added fat	12
32131220	Egg omelet or scrambled egg, with potatoes and/or onions, NS as to fat	12
32400055	Egg white omelet, scrambled, or fried, NS as to fat	12
32400060	Egg white omelet, scrambled, or fried, made with margarine	12
32400065	Egg white omelet, scrambled, or fried, made with oil	12
32400070	Egg white omelet, scrambled, or fried, made with butter	12
32400075	Egg white omelet, scrambled, or fried, made with cooking spray	12
32400078	Egg white omelet, scrambled, or fried, NS as to fat type	12
32400080	Egg white omelet, scrambled, or fried, no added fat	12
32400100	Egg white, omelet, scrambled, or fried, with cheese	12
32400200	Egg white, omelet, scrambled, or fried, with meat	12
32400300	Egg white, omelet, scrambled, or fried, with vegetables	12
32400400	Egg white, omelet, scrambled, or fried, with cheese and meat	12
32400500	Egg white, omelet, scrambled, or fried, with cheese and vegetables	12
32400600	Egg white, omelet, scrambled, or fried, with meat and vegetables	12
32400700	Egg white, omelet, scrambled, or fried, with cheese, meat, and vegetables	12
33001010	Egg substitute, omelet, scrambled, or fried, fat added	12

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
33001050	Egg substitute, omelet, scrambled, or fried, no added fat	12
33401000	Egg substitute, omelet, scrambled, or fried, with cheese	12
33401100	Egg substitute, omelet, scrambled, or fried, with meat	12
33401200	Egg substitute, omelet, scrambled, or fried, with vegetables	12
33401300	Egg substitute, omelet, scrambled, or fried, with cheese and meat	12
33401400	Egg substitute, omelet, scrambled, or fried, with cheese and vegetables	12
33401500	Egg substitute, omelet, scrambled, or fried, with meat and vegetables	12
33401600	Egg substitute, omelet, scrambled, or fried, with cheese, meat, and vegetables	12

21 CFR § 170.3(n)(12) Fats and oils

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
81302040	Sandwich spread	10
81302050	Tartar sauce	10
83100100	Salad dressing, NFS, for salads	10
83100200	Salad dressing, NFS, for sandwiches	10
83101000	Blue or roquefort cheese dressing	10
83101600	Bacon and tomato dressing	10
83102000	Caesar dressing	10
83103000	Coleslaw dressing	10
83104000	French or Catalina dressing	10
83105500	Honey mustard dressing	10
83106000	Italian dressing, made with vinegar and oil	10
83107000	Mayonnaise, regular	10
83108000	Vegan mayonnaise	10
83109000	Russian dressing	10
83110000	Mayonnaise-type salad dressing	10
83112000	Avocado dressing	10
83112400	Creamy Italian dressing	10
83112500	Creamy dressing	10
83112950	Poppy seed dressing	10
83112990	Sesame dressing	10
83113500	Ranch dressing	10
83114000	Thousand Island dressing	10
83115000	Yogurt dressing	10
83200100	Salad dressing, light, NFS	10
83201000	Blue or roquefort cheese dressing, light	10
83201400	Coleslaw dressing, light	10
83201500	Creamy Italian dressing, light	10
83202020	French or Catalina dressing, light	10
83203000	Caesar dressing, light	10
83204000	Mayonnaise, light	10
83204030	Mayonnaise, reduced fat, with olive oil	10
83204050	Mayonnaise-type salad dressing, light	10
83204500	Honey mustard dressing, light	10
83205450	Italian dressing, light	10
83205560	Ranch dressing, light	10
83206000	Russian dressing, light	10
83206500	Sesame dressing, light	10
83207000	Thousand Island dressing, light	10
83210100	Creamy dressing, light	10
83300100	Blue or roquefort cheese dressing, fat free	10
83300200	Caesar dressing, fat free	10
83300250	Creamy Italian dressing, fat free	10
83300300	Creamy dressing, fat free	10
83300400	French or Catalina dressing, fat free	10
83300500	Honey mustard dressing, fat free	10
83300600	Italian dressing, fat free	10
83300700	Mayonnaise, fat free	10

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
83300750	Ranch dressing, fat free	10
83300800	Russian dressing, fat free	10
83300900	Salad dressing, fat free, NFS	10
83301000	Thousand Island dressing, fat free	10

21 CFR § 170.3(n)(20) Frozen dairy desserts and mixes

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11459990	Frozen yogurt, NFS	8
11460000	Frozen yogurt, vanilla	8
11460100	Frozen yogurt, chocolate	8
11460500	Frozen yogurt, soft serve, vanilla	8
11460510	Frozen yogurt, soft serve, chocolate	8
11461200	Frozen yogurt sandwich	8
11461210	Frozen yogurt bar, vanilla	8
11461220	Frozen yogurt bar, chocolate	8
11461250	Frozen yogurt cone, chocolate	8
11461260	Frozen yogurt cone, vanilla	8
11461300	Frozen yogurt cone, vanilla, waffle cone	8
11461320	Frozen yogurt cone, chocolate, waffle cone	8
11459990	Frozen yogurt, NFS	8
13110000	Ice cream, NFS	8
13110100	Ice cream, vanilla	8
13110102	Ice cream, vanilla, with additional ingredients	8
13110110	Ice cream, chocolate	8
13110112	Ice cream, chocolate, with additional ingredients	8
13110200	Ice cream, soft serve, vanilla	8
13110210	Ice cream, soft serve, chocolate	8
13110460	Gelato, vanilla	8
13110470	Gelato, chocolate	8
13120050	Ice cream bar, vanilla	8
13120100	Ice cream bar, vanilla, chocolate coated	8
13120110	Ice cream candy bar	8
13120140	Ice cream bar, chocolate	8
13120500	Ice cream sandwich, vanilla	8
13120510	Ice cream sandwich, chocolate	8
13120550	Ice cream cookie sandwich	8
13120730	Ice cream cone, scooped, vanilla	8
13120735	Ice cream cone, scooped, vanilla, waffle cone	8
13120740	Ice cream cone, NFS	8
13120770	Ice cream cone, scooped, chocolate	8
13120775	Ice cream cone, scooped, chocolate, waffle cone	8
13120782	Ice cream cone, soft serve, vanilla	8
13120784	Ice cream cone, soft serve, chocolate	8
13120786	Ice cream cone, soft serve, vanilla, waffle cone	8
13120788	Ice cream cone, soft serve, chocolate, waffle cone	8
13120790	Ice cream cone, vanilla, prepackaged	8
13120792	Ice cream cone, chocolate, prepackaged	8
13121000	Ice cream sundae, NFS	8
13121100	Ice cream sundae, fruit topping	8
13121120	Banana split	8
13121300	Ice cream sundae, hot fudge topping	8
13121400	Ice cream sundae, caramel topping	8
13126000	Ice cream, fried	8
13130100	Light ice cream, NFS	8
13130300	Light ice cream, vanilla	8
13130310	Light ice cream, chocolate	8
13130700	Soft serve, blended with candy or cookies, from fast food	8
13135000	Light ice cream sandwich, vanilla	8

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
13135010	Light ice cream sandwich, chocolate	8
13140000	Light ice cream bar, vanilla	8
13140100	Light ice cream bar, vanilla, chocolate coated	8
13140115	Light ice cream bar, chocolate	8
13140700	Creamsicle	8
13140710	Creamsicle, light	8
13140900	Fudgesicle	8
13142100	Light ice cream cone, vanilla, prepackaged	8
13142110	Light ice cream cone, chocolate, prepackaged	8
13150000	Sherbet, all flavors	8
13161600	Fudgesicle, light	8

21 CFR § 170.3(n)(21) Fruit and water ices

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
63420105	Frozen fruit juice bar	8
63420205	Frozen fruit juice bar, no sugar added	8
63430150	Sorbet	8
91601000	Italian Ice	8
91601010	Italian Ice, no sugar added	8
91610900	Popsicle, NFS	8
91611000	Popsicle	8
91611100	Popsicle, no sugar added	8
91612000	Freezer pop	8
91621000	Snow cone	8
91621050	Snow cone, no sugar added	8

21 CFR § 170.3(n)(22) Gelatins, puddings, and fillings

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
13200110	Pudding, chocolate, NFS	5
13210110	Pudding, bread	5
13210280	Pudding, flavors other than chocolate, NFS	5
13210300	Custard	5
13210350	Flan	5
13210370	Creme brulee	5
13210410	Pudding, rice	5
13210450	Firni, Indian pudding	5
13210520	Pudding, tapioca, made from dry mix	5
13220110	Pudding, flavors other than chocolate, made from dry mix	5
13220120	Pudding, chocolate, made from dry mix	5
13220210	Pudding, flavors other than chocolate, made from dry mix, sugar free	5
13220220	Pudding, chocolate, made from dry mix, sugar free	5
13230110	Pudding, flavors other than chocolate, ready-to-eat	5
13230120	Pudding, flavors other than chocolate, ready-to-eat, sugar free	5
13230130	Pudding, chocolate, ready-to-eat	5
13230140	Pudding, chocolate, ready-to-eat, sugar free	5
13230500	Pudding, tapioca, ready-to-eat	5
13241000	Banana pudding	5
13250000	Mousse	5
13252200	Dulce de leche	5
13252500	Barfi or Burfi, Indian dessert	5
13252590	Trifle	5
32401000	Meringues	5
58118210	Cornmeal coconut dessert, Puerto Rican style	5
63403150	Lime souffle	5

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
91501010	Gelatin dessert	5
91501020	Gelatin dessert with fruit	5
91501100	Gelatin salad with vegetables	5
91511010	Gelatin dessert, sugar free	5
91511020	Gelatin dessert, sugar free, with fruit	5
91560100	Haupia	5

21 CFR § 170.3(n)(23) Grain products and pastas

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
55610300	Dumpling, no meat	6
56104000	Pasta, vegetable, cooked	10
56112000	Noodles, cooked	10
56113000	Noodles, whole grain, cooked	10
56113990	Noodles, vegetable, cooked	10
56116990	Long rice noodles, made from mung beans, cooked	10
56117090	Rice noodles, cooked	10
56130000	Pasta, cooked	10
56132990	Pasta, whole grain, cooked	10
56140100	Pasta, gluten free	10
56201550	Cornmeal dumpling	6
58110110	Egg roll, meatless	6
58110120	Egg roll, with shrimp	6
58110130	Egg roll, with beef and/or pork	6
58110170	Egg roll, with chicken or turkey	6
58110200	Roll with meat and/or shrimp, vegetables and rice paper, not fried	6
58111110	Wonton, dumpling or pot sticker, fried	6
58111120	Wonton, dumpling or pot sticker, fried, no meat	6
58112510	Wonton, dumpling or pot sticker, steamed	6
58121620	Dumpling, vegetable	6
58123110	Bao bun	6
58123120	Bao bun, no meat	6

21 CFR § 170.3(n)(24) Gravies and sauces

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11440010	Chipotle dip, yogurt based	10
11440020	Dill dip, yogurt based	10
11440030	Onion dip, yogurt based	10
11440040	Ranch dip, yogurt based	10
11440050	Spinach dip, yogurt based	10
11440060	Tzatziki dip	10
11440070	Vegetable dip, yogurt based	10
12350010	Dip, NFS	10
12350200	Chipotle dip, regular	10
12350205	Chipotle dip, light	10
12350210	Dill dip, regular	10
12350215	Dill dip, light	10
12350220	Onion dip, regular	10
12350225	Onion dip, light	10
12350230	Ranch dip, regular	10
12350235	Ranch dip, light	10
12350240	Spinach dip, regular	10
12350245	Spinach dip, light	10
12350250	Vegetable dip, regular	10
12350255	Vegetable dip, light	10

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
13411000	White sauce or gravy	10
14620110	Artichoke dip	10
14620115	Spinach and artichoke dip	10
14620150	Cheese dip with chili pepper	10
14620200	Cheese dip	10
42204100	Gravy, vegetarian	10
81302010	Hollandaise sauce	10

21 CFR § 170.3(n)(31) Milk products

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11511000	Chocolate milk, NFS	10
11511100	Chocolate milk, ready to drink, whole	10
11511200	Chocolate milk, ready to drink, reduced fat	10
11511300	Chocolate milk, ready to drink, fat free	10
11511400	Chocolate milk, ready to drink, low fat	10
11511550	Chocolate milk, ready to drink, reduced sugar, NS as to milk	10
11511600	Chocolate milk, ready to drink, low fat (Nesquik)	10
11511610	Chocolate milk, ready to drink, fat free (Nesquik)	10
11511700	Chocolate milk, ready to drink, low fat, no sugar added (Nesquik)	10
11512010	Hot chocolate / Cocoa, ready to drink	10
11512020	Hot chocolate / Cocoa, ready to drink, made with nonfat milk	10
11512030	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk	10
11512100	Hot chocolate / Cocoa, ready to drink, with whipped cream	10
11512110	Hot chocolate / Cocoa, ready to drink, made with nonfat milk and whipped cream	10
11512120	Hot chocolate / Cocoa, ready to drink, made with non-dairy milk and whipped cream	10
11513000	Chocolate milk, made from dry mix, NS as to type of milk	10
11513100	Chocolate milk, made from dry mix with whole milk	10
11513150	Chocolate milk, made from dry mix with reduced fat milk	10
11513200	Chocolate milk, made from dry mix with low fat milk	10
11513300	Chocolate milk, made from dry mix with fat free milk	10
11513310	Chocolate milk, made from dry mix with non-dairy milk	10
11513350	Chocolate milk, made from reduced sugar mix, NS as to type of milk	10
11513355	Chocolate milk, made from reduced sugar mix with whole milk	10
11513360	Chocolate milk, made from reduced sugar mix with reduced fat milk	10
11513365	Chocolate milk, made from reduced sugar mix with low fat milk	10
11513370	Chocolate milk, made from reduced sugar mix with fat free milk	10
11513375	Chocolate milk, made from reduced sugar mix with non-dairy milk	10
11513380	Chocolate milk, made from dry mix, NS as to type of milk (Nesquik)	10
11513381	Chocolate milk, made from dry mix with whole milk (Nesquik)	10
11513382	Chocolate milk, made from dry mix with reduced fat milk (Nesquik)	10
11513383	Chocolate milk, made from dry mix with low fat milk (Nesquik)	10
11513384	Chocolate milk, made from dry mix with fat free milk (Nesquik)	10
11513385	Chocolate milk, made from dry mix with non-dairy milk (Nesquik)	10
11513390	Chocolate milk, made from no sugar added dry mix, NS as to type of milk (Nesquik)	10
11513391	Chocolate milk, made from no sugar added dry mix with whole milk (Nesquik)	10
11513392	Chocolate milk, made from no sugar added dry mix with reduced fat milk (Nesquik)	10
11513393	Chocolate milk, made from no sugar added dry mix with low fat milk (Nesquik)	10
11513394	Chocolate milk, made from no sugar added dry mix with fat free milk (Nesquik)	10
11513395	Chocolate milk, made from no sugar added dry mix with non-dairy milk (Nesquik)	10
11513400	Chocolate milk, made from syrup, NS as to type of milk	10
11513500	Chocolate milk, made from syrup with whole milk	10
11513550	Chocolate milk, made from syrup with reduced fat milk	10
11513600	Chocolate milk, made from syrup with low fat milk	10

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
11513700	Chocolate milk, made from syrup with fat free milk	10
11513750	Chocolate milk, made from syrup with non-dairy milk	10
11513800	Chocolate milk, made from light syrup, NS as to type of milk	10
11513801	Chocolate milk, made from light syrup with whole milk	10
11513802	Chocolate milk, made from light syrup with reduced fat milk	10
11513803	Chocolate milk, made from light syrup with low fat milk	10
11513804	Chocolate milk, made from light syrup with fat free milk	10
11513805	Chocolate milk, made from light syrup with non-dairy milk	10
11513850	Chocolate milk, made from sugar free syrup, NS as to type of milk	10
11513851	Chocolate milk, made from sugar free syrup with whole milk	10
11513852	Chocolate milk, made from sugar free syrup with reduced fat milk	10
11513853	Chocolate milk, made from sugar free syrup with low fat milk	10
11513854	Chocolate milk, made from sugar free syrup with fat free milk	10
11513855	Chocolate milk, made from sugar free syrup with non-dairy milk	10
11514100	Hot chocolate / Cocoa, made with dry mix and water	10
11514110	Hot chocolate / Cocoa, made with dry mix and whole milk	10
11514120	Hot chocolate / Cocoa, made with dry mix and reduced fat milk	10
11514130	Hot chocolate / Cocoa, made with dry mix and low fat milk	10
11514140	Hot chocolate / Cocoa, made with dry mix and fat free milk	10
11514150	Hot chocolate / Cocoa, made with dry mix and non-dairy milk	10
11514310	Hot chocolate / Cocoa, made with no sugar added dry mix and water	10
11514320	Hot chocolate / Cocoa, made with no sugar added dry mix and whole milk	10
11514330	Hot chocolate / Cocoa, made with no sugar added dry mix and reduced fat milk	10
11514340	Hot chocolate / Cocoa, made with no sugar added dry mix and low fat milk	10
11514350	Hot chocolate / Cocoa, made with no sugar added dry mix and fat free milk	10
11514360	Hot chocolate / Cocoa, made with no sugar added dry mix and non-dairy milk	10
11519040	Strawberry milk, NFS	10
11519050	Strawberry milk, whole	10
11519105	Strawberry milk, reduced fat	10
11519200	Strawberry milk, low fat	10
11519205	Strawberry milk, fat free	10
11519210	Strawberry milk, reduced sugar	10
11519215	Strawberry milk, non-dairy	10
11526000	Milk, malted	10
11541110	Milk shake, home recipe, chocolate	10
11541120	Milk shake, home recipe, flavors other than chocolate	10
11541130	Milk shake, home recipe, chocolate, light	10
11541135	Milk shake, home recipe, flavors other than chocolate, light	10
11541400	Milk shake with malt	10
11542100	Milk shake, fast food, chocolate	10
11542200	Milk shake, fast food, flavors other than chocolate	10
11543000	Milk shake, bottled, chocolate	10
11543010	Milk shake, bottled, flavors other than chocolate	10
11553100	Fruit smoothie, NFS	12
11553110	Fruit smoothie, with whole fruit and dairy	12
11553120	Fruit smoothie, with whole fruit and dairy, added protein	25
11553130	Fruit smoothie juice drink, with dairy	12
11560000	Chocolate milk drink	10
13120800	Ice cream soda, flavors other than chocolate	10
13120810	Ice cream soda, chocolate	10

21 CFR § 170.3(n)(33) Plant protein products

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
27564200	Hot dog sandwich, meat and poultry, on wheat bread	10
27564420	Hot dog sandwich, vegetarian, on bun	10
27564430	Hot dog sandwich, vegetarian, on bread	10
27564560	Frankfurter or hot dog sandwich, meatless, on bun, with meatless chili	10

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
27564570	Frankfurter or hot dog sandwich, meatless, on bread, with meatless chili	10
41810200	Bacon strip, meatless	10
41810400	Breakfast link, pattie, or slice, meatless	10
41810600	Chicken, meatless, NFS	10
41810610	Chicken, meatless, breaded, fried	10
41811400	Hot dog, vegetarian	10
41811600	Luncheon slice, meatless-beef, chicken, salami or turkey	10
41811800	Meatball, meatless	10
41811890	Veggie burger patty, no bun	10
41811950	Swiss steak, with gravy, meatless	10
41812000	Sandwich spread, meat substitute type	10
41812450	Vegetarian chili, made with meat substitute	10
41812600	Vegetarian, fillet	10
41812800	Vegetarian stew	10
41812850	Vegetarian stroganoff	10
41901010	Veggie burger, on bun	10
41901020	Veggie burger, on bun, with cheese	10
59003000	Meat substitute, cereal- and vegetable protein-based, fried	10

21 CFR § 170.3(n)(37) Snack foods – protein and snack bars

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53710400	Cereal or granola bar (General Mills Fiber One Chewy Bar)	10
53710500	Cereal or granola bar (Kellogg's Nutri-Grain Cereal Bar)	10
53710502	Cereal or granola bar (Kellogg's Nutri-Grain Yogurt Bar)	10
53710504	Cereal or granola bar (Kellogg's Nutri-Grain Fruit and Nut Bar)	10
53710600	Milk 'n Cereal bar	10
53710700	Cereal or granola bar (Kellogg's Special K bar)	10
53710800	Cereal or granola bar (Kashi Chewy)	10
53710802	Cereal or granola bar (Kashi Crunchy)	10
53710810	Cereal or granola bar (KIND Fruit and Nut Bar)	10
53710900	Cereal or granola bar (General Mills Nature Valley Chewy Trail Mix)	10
53710902	Cereal or granola bar, with yogurt coating (General Mills Nature Valley Chewy Granola Bar)	10
53710904	Cereal or granola bar (General Mills Nature Valley Sweet and Salty Granola Bar)	10
53710906	Cereal or granola bar (General Mills Nature Valley Crunchy Granola Bar)	10
53711000	Cereal or granola bar (Quaker Chewy Granola Bar)	10
53711002	Cereal or granola bar (Quaker Chewy 90 Calorie Granola Bar)	10
53711004	Cereal or granola bar (Quaker Chewy 25% Less Sugar Granola Bar)	10
53711006	Cereal or granola bar (Quaker Chewy Dippys Granola Bar)	10
53711100	Cereal or granola bar (Quaker Granola Bites)	10
53712000	Snack bar, oatmeal	10
53712100	Cereal or Granola bar, NFS	10
53712200	Cereal or granola bar, lowfat, NFS	10
53712210	Cereal or granola bar, nonfat	10
53713000	Cereal or granola bar, reduced sugar, NFS	10
53713010	Cereal or granola bar, fruit and nut	10
53713100	Cereal or granola bar, peanuts , oats, sugar, wheat germ	10
53714200	Cereal or granola bar, chocolate coated, NFS	10
53714210	Cereal or granola bar, with coconut, chocolate coated	10
53714220	Cereal or granola bar with nuts, chocolate coated	10
53714230	Cereal or granola bar, oats, nuts, coated with non-chocolate coating	10
53714250	Cereal or granola bar, coated with non-chocolate coating	10
53714300	Cereal or granola bar, high fiber, coated with non-chocolate yogurt coating	10
53714400	Cereal or granola bar, with rice cereal	10
53714500	Breakfast bar, NFS	10
53714510	Breakfast bar, date, with yogurt coating	10
53714520	Breakfast bar, cereal crust with fruit filling, lowfat	10

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
53720100	Nutrition bar (Balance Original Bar)	25
53720200	Nutrition bar (Clif Bar)	25
53720210	Nutrition bar (Clif Kids Organic Zbar)	25
53720300	Nutrition bar (PowerBar)	25
53720400	Nutrition bar (Slim Fast Original Meal Bar)	25
53720500	Nutrition bar (Snickers Marathon Protein Bar)	25
53720600	Nutrition bar (South Beach Living Meal Bar)	25
53720610	Nutrition bar (South Beach Living High Protein Bar)	25
53720700	Nutrition bar (Tiger's Milk)	25
53720800	Nutrition bar (Zone Perfect Classic Crunch)	25
53729000	Nutrition bar or meal replacement bar, NFS	25

21 CFR § 170.3(n)(38) Soft candy

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
91702010	Butterscotch morsels	3
91703010	Caramel, chocolate-flavored roll	3
91703020	Caramel, flavor other than chocolate	3
91703030	Caramel, with nuts	3
91703040	Caramel candy, chocolate covered	3
91703050	Caramel with nuts and cereal, chocolate covered	3
91703060	Caramel with nuts, chocolate covered	3
91703070	Rolo	3
91703080	Caramel, all flavors, sugar free	3
91703150	Toblerone, milk chocolate with honey and almond nougat	3
91703200	TWIX Caramel Cookie Bars	3
91703250	TWIX Chocolate Fudge Cookie Bars	3
91703300	TWIX Peanut Butter Cookie Bars	3
91703400	Whatchamacallit	3
91703600	Espresso coffee beans, chocolate-covered	3
91705005	Chocolate candy, other, NFS	3
91705010	Chocolate candy	3
91705012	Chocolate candy with nuts, other, NFS	3
91705015	Chocolate candy with nuts	3
91705020	Chocolate candy with cereal	3
91705030	Kit Kat	3
91705040	Chocolate, milk, with nuts, not almond or peanuts	3
91705050	Milk chocolate candy, with fruit and nuts	3
91705060	Milk chocolate candy, with almonds	3
91705070	Chocolate, milk, with peanuts	3
91705080	Chocolate candy, cookie filled	3
91705090	Chocolate candy with fondant and caramel	3
91705200	Chocolate chips	3
91705290	Dark chocolate candy, other, NFS	3
91705300	Dark chocolate candy	3
91705310	Chocolate, sweet or dark, with almonds	3
91705312	Dark chocolate candy with nuts, other, NFS	3
91705315	Dark chocolate candy with nuts	3
91705440	Chocolate candy, fudge	3
91705450	Chocolate candy, caramel filled	3
91705460	Chocolate candy, caramel filled with nuts	3
91705470	Chocolate candy, coconut filled	3
91705480	Chocolate candy, cream filled	3
91705500	Mexican chocolate, tablet	3
91705510	Chocolate candy, nougat filled	3
91705520	Chocolate candy, nougat filled with nuts	3
91705530	Chocolate candy, peanut butter filled	3
91705550	Chocolate candy with dried fruit	3
91706000	Coconut candy, chocolate covered	3

Food Code	Food description	Maximum use level (g per 100 g food as consumed)
91706010	Chocolate candy, sugar free	3
91707000	Fondant	3
91707010	Fondant, chocolate covered	3
91708150	Yogurt covered fruit snacks candy, with added vitamin C	3
91708160	Yogurt covered fruit snacks candy rolls, with high vitamin C	3
91709000	Gumdrops, chocolate covered	3
91713010	Fudge, chocolate, chocolate-coated	3
91713020	Fudge, chocolate, chocolate-coated, with nuts	3
91713030	Fudge, chocolate	3
91713040	Fudge, chocolate, with nuts	3
91713050	Fudge, peanut butter	3
91713060	Fudge, peanut butter, with nuts	3
91713070	Fudge, vanilla	3
91713080	Fudge, vanilla, with nuts	3
91713090	Fudge, divinity	3
91715000	Fudge, caramel and nut, chocolate-coated candy	3
91715100	SNICKERS Bar	3
91715200	Baby Ruth	3
91715300	100 GRAND Bar	3
91716110	Halvah, chocolate covered	3
91718050	Honey-combed hard candy with peanut butter, chocolate covered	3
91718100	Butterfinger	3
91718110	Butterfinger Crisp	3
91718200	Chocolate-flavored sprinkles	3
91723030	Candy, caramel	3
91726000	Nougat, plain	3
91726110	Nougat, with caramel, chocolate covered	3
91726130	MILKY WAY Bar	3
91726140	MILKY WAY MIDNIGHT Bar	3
91726150	MARS Almond Bar	3
91726410	Nougat, chocolate covered	3
91726420	3 MUSKETEERS Bar	3
91726425	3 Musketeers Truffle Crisp Bar	3
91727010	Nuts, chocolate covered, not almonds or peanuts	3
91728000	Candy, nougat with nuts	3
91731000	Peanuts, chocolate covered candy	3
91731010	M&M's Peanut Chocolate Candies	3
91731060	M&M's Peanut Butter Chocolate Candies	3
91731150	Peanuts, yogurt covered	3
91733200	Peanut Bar, chocolate covered candy	3
91734000	Peanut butter, chocolate covered	3
91734100	Reese's Peanut Butter Cup	3
91734300	Reese's Sticks	3
91734400	Reese's Fast Break	3
91734450	Reese's Crispy Crunchy Bar	3
91735000	Pralines	3
91739010	Raisins, chocolate covered	3
91739600	Raisins, yogurt covered	3
91745000	Candy, mint	3
91745010	Candy, gummy	3
91746010	Sugar-coated chocolate discs	3
91746100	Chocolate candy, candy shell	3
91746110	Chocolate candy, candy shell with nuts	3
91746120	Sixlets	3
91746150	Easter egg, candy coated chocolate	3
91746200	M&M's Pretzel Chocolate Candies	3
91750000	Candy, taffy	3
91760500	Truffles	3

From: [Diana Rueda](#)
To: [Wafula, Denis](#)
Subject: [EXTERNAL] Re: GRN 1249: Further Questions from the Technical Review Team
Date: Thursday, July 3, 2025 10:52:12 PM
Attachments: [GRN 1249 Response to FDA RFI 2 \(04JUL25\).pdf](#)

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Dear Dr. Wafula,

I hope this email finds you well.

Attached, please find Onego Bio's response to your latest technical question.

Please let me know if you have any further questions or if anything requires clarification.

With kind regards,

Dr. Diana Rueda
Global Regulatory Affairs Manager

Based in Shanghai | Available until 12 PM CET
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From: Diana Rueda <diana@atovaconsulting.com>
Date: Monday, 30 June 2025 at 10:52
To: Wafula, Denis <Denis.Wafula@fda.hhs.gov>
Subject: Re: GRN 1249: Further Questions from the Technical Review Team

Dear Dr. Wafula,

Many thanks for the helpful discussion. We appreciate you and your team taking the time to meet with us.

We will review the follow-up question and respond accordingly in the coming days.

With kind regards,

Dr. Diana Rueda
Global Regulatory Affairs Manager

Based in Shanghai | Available until 12 PM CET
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Onego Bio, Inc.
9215 Brown Deer Road, Suite C
San Diego, CA 92121
USA

July 04, 2025

Dr. Denis Wafula

Office of Pre-Market Additive Safety
Office of Food Chemical Safety, Dietary Supplements, and Innovation
Human Foods Program
US Food and Drug Administration
5001 Campus Drive
College Park, MD 20740

RE: GRAS Notice No. GRN 001249 – response to technical review question from July 27, 2025

Dear Dr. Wafula,

This letter responds to your technical question dated June 27, 2025, regarding GRAS Notice No. GRN 001249. To maintain consistency with your earlier correspondence, which included Questions 1 through 9, we have designated this new inquiry as **Question 10**.

For clarity, the question is restated below as it appeared in your original message, followed by Onego Bio's corresponding response.

We kindly ask that you confirm receipt. Should you have any questions or require further information, please do not hesitate to reach out – we would be glad to provide any additional details.

Thank you for your attention.

Sincerely,



Maija Itkonen
CEO
Onego Bio, Inc.
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Question 10: In response to Q6, you note that the intended food uses encompass the “broad” food categories listed in the first column of Table 10 (pages 26-27). We consider that if you want to include all foods in this “broad” category, you should also include food codes to account for all broad categories. Please further clarify the intended food categories other than those specified in Table A1 below. Accordingly, please revise the dietary exposure assessment to reflect the intended use categories.

Table A1. Food categories and intended use levels	
Food category	Use level (%)
Baked goods and baking mixes	3-25
Beverages and bases, non-alcoholic (smoothies, powdered shake mixes)	4-25
Beverages, alcoholic (eggnog, egg white cocktails)	4
Breakfast cereals, ready-to-eat (RTE)	10
Cheeses (breaded mozzarella sticks)	6
Confections and frostings (marshmallows)	3
Dairy product analogs (milk substitute beverages)	10
Egg products (egg analogs)	12
Fats and oils (dressing, vegetable oils, mayonnaise)	10
Frozen dairy desserts (ice cream, frozen yogurt, sherbert)	8
Fruit and water ices (frozen, flavored, and sweetened)	8
Gelatins, puddings, fillings	5
Grain products and pastas (pastas, egg rolls, dumplings)	6-10
Gravies sauces (sauces, dips)	6
Milk products	10
Plant protein products (meat analogs)	10
Protein and snack bars (cereal or granola, protein bars)	10-25
Soft candy (snack bars)	3

Answer 10:

The NHANES food codes used in the dietary exposure assessment for NOVA (as provided in response to Question 8) are representative of the full range of intended uses for the ingredient. The exposure estimate presented in Part 3 of the notice is based on these codes and captures the complete scope of intended use.

The intended uses of NOVA are detailed in Table 10 of the GRAS notice, which references the food categories defined in 21 CFR § 170.3. For clarity, a revised table (Table A2) is provided below, outlining the intended uses based on the specific NHANES food codes used in the assessment. As the current estimate already reflects these uses, no update to the dietary exposure assessment is necessary.

Table A2. Revised food categories and intended use levels of NOVA		
Food category	Specific subcategories covering the intended uses of NOVA ¹	Use level ² (%)
Baked goods and baking mixes	Baked goods and baking mixes	3-25
Beverages and beverage bases, nonalcoholic	Smoothies; powdered shake mixes; ready-to-drink shakes or shake substitutes	12-25
Beverages, alcoholic	Eggnog; egg white cocktails	4
Breakfast cereals	Ready-to-eat cereals	10
Cheeses	Coated or breaded processed cheese products	6
Confections and frostings	Marshmallows	3
Dairy product analogs	Milk substitute beverages	10
Egg products	Egg analogs	12
Fats and oils	Salad dressings; vegetable oils; mayonnaise; spreads	10
Frozen dairy desserts and mixes	Ice cream; frozen yogurt; sherbet	8
Fruit and water ices	Frozen, flavored and sweetened ices and pops	8
Gelatins, puddings, and fillings	Gelatins; puddings; custards	5
Grain products and pastas	Plain pasta and noodles; filled grain-based products (e.g., dumplings, egg rolls); grain-based prepared foods (non-pasta, e.g., steamed buns)	6-10
Gravies and sauces	Sauces; dips	10



Table A2. Revised food categories and intended use levels of NOVA

Food category	Specific subcategories covering the intended uses of NOVA¹	Use level² (%)
Milk products	Flavored milk and milk-based drinks	10
Plant protein products	Meat analogs	10
Protein and snack bars	Cereal bars; breakfast bars; granola bars; protein bars	10-25
Soft candy	Snack bars; chocolate-based and caramel-based candy; fudge and fondant-type candy	3

¹ These subcategories represent the full scope of NOVA's intended uses. The NHANES food codes used in the dietary exposure assessment (see response to Question 8) are aligned with and fully cover these subcategories.

² The use levels shown correspond to the maximum intended levels, as listed in Table 10 of the GRAS notice. For food categories that include multiple subcategories with different maximum use levels, a range is provided to reflect this variation. The specific maximum level for each subcategory can be found in Table 10.